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1.1 Background of Study

- Brief description of the current situation or problem

Groove Monkey Pinoy Korean Bistro experiences problems regarding manual order which causes errors and delays, limited online presence and no Digital Ordering Options, and inaccurate Manual Inventory Tracking which leads to food waste. Many businesses that rely on manual ordering face inefficiencies such as slow processing, increased risk of human error, and difficulty in handling large volumes of orders. These issues often lead to poor customer satisfaction. Additionally, having a limited online presence restricts the business’s ability to reach a wider market, reducing potential sales and competitiveness in an increasingly digital marketplace. Furthermore, inaccurate manual inventory tracking makes it challenging to monitor stock levels in real time, resulting in overstocking or stock shortages, which can disrupt operations and affect overall profitability.

Another significant challenge faced by Groove Monkey Pinoy Korean Bistro is the difficulty in maintaining accurate records of customer orders during peak business hours. Since orders are processed manually, employees may unintentionally misread, miswrite, or overlook customer requests, leading to incorrect orders and dissatisfaction among customers. Such mistakes can negatively impact the restaurant's reputation and may result in the loss of repeat customers who expect fast and accurate service.

The absence of a digital ordering system also limits convenience for customers who prefer using online platforms to place their orders. In today's technology-driven environment, many consumers rely on mobile devices and online applications for food purchases. Without these options, potential customers may choose competitors that offer online ordering, delivery integration, and digital payment methods, thereby reducing the bistro's ability to maximize sales opportunities and expand its customer base.

Moreover, manual processes require additional time and effort from employees, reducing overall operational efficiency. Staff members must spend considerable time recording orders, checking inventory levels, and preparing reports manually. This not only increases workload but also diverts attention from other important tasks such as customer service, food quality assurance, and business development activities. As a result, productivity may decline, affecting the restaurant's overall performance.

Inventory management is another critical area that suffers from the lack of automation. Manual inventory tracking often results in delayed updates and inaccurate stock records, making it difficult to determine the exact quantity of ingredients available. Consequently, management may encounter unexpected shortages of essential supplies or purchase excessive quantities that exceed demand. Both situations can lead to increased operational costs, wasted resources, and disruptions in daily restaurant operations.

To remain competitive in the food service industry, Groove Monkey Pinoy Korean Bistro must adapt to the growing demand for digital solutions. Implementing an integrated ordering and inventory management system can streamline business processes, improve order accuracy, provide real-time inventory monitoring, and strengthen the restaurant's online presence. These improvements can enhance customer satisfaction, increase operational efficiency, and support sustainable business growth in the long term.

The lack of real-time data availability also hinders management's ability to make informed business decisions. Since records are maintained manually, gathering and analyzing information regarding sales trends, best-selling menu items, and customer preferences becomes time-consuming and prone to inaccuracies. Without reliable data, management may find it difficult to develop effective marketing strategies, forecast demand, and identify opportunities for business improvement.

Customer experience is greatly affected when service processes are inefficient. Long waiting times caused by manual order taking and processing can create frustration among customers, especially during busy periods. In a highly competitive food industry, customers value speed, convenience, and accuracy. Delays in service may encourage customers to seek alternative dining establishments that offer more efficient and technology-driven services.

In addition, manual record-keeping increases the risk of data loss and documentation errors. Physical order slips, inventory logs, and sales records can be misplaced, damaged, or lost over time. Such incidents can create difficulties in tracking transactions, resolving customer concerns, and generating accurate financial reports.

Maintaining organized and secure records becomes increasingly challenging as the business grows and handles a larger volume of transactions.

The limited online presence of the bistro also restricts customer engagement and brand visibility. Modern consumers frequently discover restaurants through social media platforms, online food directories, and digital marketing campaigns. Without a strong digital presence, Groove Monkey Pinoy Korean Bistro may struggle to attract new customers, promote special offers, and build lasting relationships with its target market. This limitation can affect the business's ability to compete effectively with restaurants that actively utilize digital marketing channels.

Furthermore, the absence of an integrated system reduces the business's capacity to adapt to changing customer demands and market trends. As consumer expectations continue to evolve toward digital convenience, businesses that fail to embrace technological solutions risk falling behind competitors. By adopting a modern ordering and inventory management system, Groove Monkey Pinoy Korean Bistro can improve operational control, enhance customer satisfaction, increase revenue opportunities, and establish a stronger foundation for future expansion and long-term success.

Another issue arising from the current manual system is the difficulty in monitoring employee performance and accountability. Since transactions and inventory updates are recorded manually, it becomes challenging for management to accurately track employee activities, identify errors, and determine areas that require improvement. This lack of transparency can affect operational control and make it harder to implement effective performance evaluation measures.

The manual ordering process can also lead to communication gaps between front-of-house and kitchen staff. Handwritten orders may be difficult to read or incomplete, resulting in misunderstandings during food preparation. Such communication errors can cause incorrect orders, longer preparation times, and increased food wastage. Consequently, both customer satisfaction and operational efficiency may be negatively affected.

Financial management is another area that can be impacted by the absence of automated systems. Manual computation of daily sales, expenses, and inventory costs requires significant time and effort while increasing the likelihood of calculation errors. Inaccurate financial records may affect budgeting, profit analysis, and decisionmaking processes, making it difficult for management to assess the true financial performance of the business.

As the business continues to grow, the limitations of manual operations become more evident. A higher volume of customers and transactions increases the complexity of managing orders, inventory, and sales records. Without technological support, employees may struggle to handle the increasing workload efficiently, which can lead to slower service, operational bottlenecks, and reduced productivity during peak periods.

Implementing a digital management solution would provide a more scalable and sustainable approach to business operations. Features such as automated order processing, real-time inventory updates, digital reporting, and customer data management can significantly improve accuracy and efficiency. By embracing technology, Groove Monkey Pinoy Korean Bistro can better meet customer expectations, optimize resource utilization, and strengthen its position within the highly competitive restaurant industry.

The current manual system limits the restaurant's ability to provide consistent and efficient customer service. As customer volume increases, employees may struggle to keep up with order processing and record management. This can result in delayed responses to customer inquiries and slower service delivery, which may negatively influence the overall dining experience.

Another challenge is the inability to quickly identify sales patterns and customer preferences. Since information is manually recorded, management must spend considerable time reviewing records to determine which menu items are performing well and which are not. This delay in obtaining valuable business insights can hinder the development of effective promotional campaigns and menu improvement strategies.

The absence of a centralized database makes information management more difficult. Customer orders, inventory records, and sales reports are often stored in separate documents or logbooks, increasing the risk of inconsistencies and duplication of information.

A fragmented record-keeping system can create confusion and reduce the accuracy of business data.

Manual inventory management can also affect supplier coordination. Without accurate and up-to-date inventory information, management may find it difficult to determine when ingredients need to be replenished. This can lead to emergency purchases, delayed deliveries, and higher procurement costs, which may reduce the restaurant's profitability.

Customer retention can become a challenge when businesses lack digital engagement tools. Modern customers often appreciate features such as online ordering, loyalty programs, digital promotions, and personalized offers. Without these capabilities, Groove Monkey Pinoy Korean Bistro may miss opportunities to strengthen customer relationships and encourage repeat business.

The restaurant's limited digital presence may also reduce its visibility among tourists, working professionals, and younger consumers who frequently rely on online platforms to discover dining options. As more consumers search for restaurants through social media and search engines, businesses with minimal online exposure may struggle to attract new customers and expand their market reach.

Operational delays caused by manual procedures can increase employee stress and workload. Staff members may need to handle multiple tasks simultaneously, including taking orders, updating inventory records, and processing payments. This additional burden can lead to fatigue, decreased productivity, and a higher likelihood of errors during busy operating hours.

Furthermore, the lack of automated reporting tools makes it difficult for management to generate timely business reports. Preparing sales summaries, inventory reports, and performance analyses manually requires significant effort and may delay important management decisions. Access to real-time reports would allow managers to respond more quickly to operational issues and market changes.

Data security is another concern associated with manual recordkeeping. Physical documents can be lost, damaged, or accessed by unauthorized individuals. The absence of secure digital storage increases the risk of data loss and limits the restaurant's ability to maintain accurate historical records for future reference and business planning.

Overall, the combination of manual ordering, limited online presence, and inefficient inventory tracking creates significant operational challenges for Groove Monkey Pinoy Korean Bistro. Addressing these issues through the implementation of an integrated digital ordering and inventory management system would enhance efficiency, improve customer satisfaction, support data-driven decision-making, and contribute to the long-term growth and sustainability of the business.

The reliance on manual operations can significantly affect the restaurant’s ability to adapt to rapidly changing market conditions. As customer preferences evolve and new dining trends emerge, businesses require accurate and timely information to make strategic decisions. Without digital tools, management may experience delays in identifying market opportunities and responding to customer demands.

Manual order processing can also create difficulties in handling special requests and customized orders. Customers may request modifications to menu items due to dietary restrictions, allergies, or personal preferences. When such requests are recorded manually, there is a greater possibility of miscommunication, which can lead to customer dissatisfaction and potential health risks in cases involving food allergies.

Another concern is the lack of efficient coordination between different operational functions within the restaurant. Ordering, inventory management, sales recording, and customer service activities often operate independently when handled manually. This lack of integration can result in inconsistencies, delays, and challenges in monitoring overall business performance.

The restaurant may also encounter difficulties in managing peak hours and seasonal demand fluctuations. During weekends, holidays, and special events, the volume of customer orders may increase substantially. Manual systems often struggle to accommodate these surges efficiently, causing bottlenecks in service delivery and reducing the restaurant’s ability to maximize revenue opportunities.

Employee training and onboarding can become more complicated when processes rely heavily on manual procedures. New staff members may require extensive training to learn various recording methods, documentation procedures, and inventory tracking practices. The complexity of these manual processes can increase training costs and extend the time needed for employees to become fully productive.

In addition, the absence of digital customer records limits the restaurant’s ability to understand customer behavior. Valuable information such as purchase history, favorite menu items, and frequency of visits cannot be easily tracked or analyzed. As a result, the business misses opportunities to implement personalized marketing strategies that could improve customer engagement and loyalty.

The restaurant’s current operational setup may also hinder business expansion efforts. As the customer base grows and additional services are introduced, managing larger volumes of transactions manually becomes increasingly difficult. The lack of scalable systems may prevent the business from efficiently supporting future growth and development initiatives.

Furthermore, manual inventory monitoring often makes it challenging to detect inventory discrepancies, spoilage, and theft promptly. Since stock counts are updated periodically rather than in real time, losses may go unnoticed until physical inventory checks are conducted. This can contribute to increased operational costs and reduced profitability over time.

The growing popularity of food delivery services and online marketplaces presents another challenge for businesses with limited digital capabilities. Customers increasingly expect restaurants to offer convenient online ordering and delivery options. Without integration with digital platforms, Groove Monkey Pinoy Korean Bistro may lose potential customers to competitors that provide more accessible and user-friendly ordering experiences.

Ultimately, the continued use of manual systems places limitations on the restaurant’s efficiency, competitiveness, and growth potential. Implementing a comprehensive digital solution would enable the business to streamline operations, improve communication, enhance inventory control, strengthen customer relationships, and establish a more sustainable and technology-driven foundation for future success.

One of the major concerns associated with the manual ordering process is the difficulty in maintaining consistency in service quality. As employees manually record and process customer orders, variations in work practices may occur, leading to inconsistent customer experiences. Maintaining a standardized process becomes more challenging, especially when different staff members handle transactions in different ways.

The lack of automation can also result in slower transaction processing at the point of sale. Customers often expect quick and efficient service when dining in or placing takeout orders. Delays in order entry, payment processing, and order confirmation may discourage customers from returning and may negatively affect the restaurant’s reputation within the local community.

Another challenge is the limited ability to monitor business performance across different periods. Manual records require significant effort to compile and compare, making it difficult to evaluate monthly, quarterly, or annual performance trends. Without access to comprehensive business analytics, management may struggle to identify strengths, weaknesses, and opportunities for improvement.

The restaurant may also experience challenges in managing menu availability. Since inventory records are not updated in real time, staff may unknowingly accept orders for menu items that are already unavailable. This situation can lead to customer disappointment, order cancellations, and unnecessary delays in service, ultimately affecting customer satisfaction.

Manual documentation procedures can increase administrative workload for both employees and management. Staff members must spend valuable time completing paperwork, updating records, and organizing documents. This reduces the amount of time available for customer interaction, food preparation, quality control, and other essential operational activities.

In today's digital age, customers often expect businesses to provide convenient communication channels. The limited online presence of Groove Monkey Pinoy Korean Bistro may make it difficult for customers to access information about menu offerings, operating hours, promotions, and special events. As a result, potential customers may choose competitors with stronger digital visibility and accessibility.

Another limitation of manual systems is the lack of immediate access to operational information. Managers may need to wait until the end of the day or week to review sales and inventory records. This delay can prevent timely decision-making and reduce the business’s ability to respond quickly to operational challenges or emerging opportunities.

The absence of automated inventory alerts further contributes to inventory management problems. Without notifications for low-stock items or expiring ingredients, employees may overlook critical inventory issues. This can result in stock shortages that disrupt service or excess inventory that leads to spoilage and financial losses.

Moreover, manual processes may hinder collaboration among employees. Information recorded on paper may not be readily accessible to all staff members, causing communication gaps and misunderstandings. Improved information sharing through a digital system can help ensure that employees have access to accurate and updated operational data whenever needed.

As competition within the food service industry continues to increase, businesses must continuously innovate and improve their operations to remain competitive. The implementation of an integrated digital ordering and inventory management system would allow Groove Monkey Pinoy Korean Bistro to modernize its operations, enhance service efficiency, improve customer satisfaction, reduce operational costs, and position itself for long-term growth and success in an increasingly technology-driven market.

The use of manual systems can also affect the accuracy of sales tracking and revenue monitoring. Since transactions are recorded by hand, there is a greater possibility of missing entries, duplicate records, or incorrect calculations. These inaccuracies can make it difficult for management to obtain a clear picture of the restaurant's financial performance and overall business health.

Another issue is the limited capability to monitor customer feedback effectively. In a manual environment, customer comments and complaints are often recorded informally or not documented at all. Without a structured system for collecting and analyzing customer feedback, the business may miss valuable insights that could help improve products, services, and customer satisfaction.

The restaurant may also encounter challenges in implementing promotional activities and discount programs. Tracking the effectiveness of promotions manually can be time-consuming and prone to errors. As a result, management may find it difficult to determine which marketing campaigns generate the highest returns and which strategies require improvement.

The lack of digital records can create difficulties during audits and financial reviews. Retrieving historical data from paper-based files requires significant effort and may consume valuable time. In some cases, incomplete or damaged records may prevent the business from producing accurate reports, affecting transparency and accountability.

Manual inventory systems may also contribute to inefficient resource allocation. Without precise information regarding stock movement and consumption rates, management may struggle to forecast future inventory requirements. This can lead to unnecessary expenditures on excess inventory or operational disruptions caused by insufficient supplies.

Furthermore, the absence of data integration between different business functions can reduce operational efficiency. Sales information, inventory records, and purchasing activities are often managed separately, making it difficult to establish a seamless flow of information. This lack of coordination can lead to delays in decision-making and reduced productivity across various departments.

Customer expectations continue to evolve as technology becomes more integrated into daily life. Many consumers now prefer cashless payments, online reservations, and digital ordering systems because of their convenience and accessibility. Without these features, Groove Monkey Pinoy Korean Bistro may find it challenging to meet modern customer expectations and maintain a competitive advantage.

The restaurant's limited digital capabilities may also restrict its ability to participate in partnerships with food delivery platforms and online marketplaces. These platforms provide opportunities to reach a broader audience and generate additional revenue streams. Failure to integrate with such services may limit business growth and reduce market exposure.

In addition, the manual management of operational data can affect strategic planning efforts. Reliable and timely information is essential for setting business goals, evaluating performance, and identifying growth opportunities. When data is incomplete or difficult to access, management may face challenges in making informed decisions that support long-term development.

Overall, the continued reliance on manual ordering, inventory management, and record-keeping processes creates numerous operational inefficiencies that can impact customer satisfaction, employee productivity, and business profitability. Addressing these challenges through the adoption of a digital ordering and inventory management system would provide Groove Monkey Pinoy Korean Bistro with the tools necessary to improve operational effectiveness, enhance customer experiences, and achieve sustainable growth in a competitive business environment.

- Industry or community context

This study is situated within the food service industry which is characterized by small to medium-sized dining establishments that offer a casual yet refined dining experience. Bistros typically focus on providing affordable meals, unique menu offerings, and a comfortable ambiance to attract customers.

In recent years, the food industry has experienced significant changes due to increasing competition, evolving customer preferences, and the growing demand for convenience through digital platforms. Many bistros face challenges such as manual ordering, limited online presence, and inaccurate inventory tracking, which can lead to inefficiencies and reduced customer satisfaction.

The rapid advancement of technology has significantly transformed the way food service establishments operate and interact with customers. Digital solutions such as online ordering systems, mobile applications, digital payment methods, and automated inventory management have become essential tools for improving operational efficiency and enhancing customer convenience. Businesses that fail to adopt these technologies may find it increasingly difficult to compete in a highly dynamic and technology-driven marketplace.

Within the local community, restaurants and bistros play an important role in providing employment opportunities, supporting local suppliers, and contributing to economic growth. As customer expectations continue to evolve, establishments are expected to deliver not only quality food but also fast, reliable, and convenient services. The ability to meet these expectations has become a key factor in maintaining customer loyalty and achieving long-term business success.

The growing popularity of social media and online review platforms has also influenced consumer behavior in the food service industry. Customers often rely on digital channels to search for restaurants, compare menu offerings, read reviews, and share their dining experiences. As a result, maintaining a strong online presence has become increasingly important for attracting new customers, building brand awareness, and strengthening customer engagement within the community.

Moreover, effective inventory management has become a critical component of restaurant operations due to rising food costs and increasing concerns regarding sustainability. Inaccurate inventory tracking can lead to excessive food waste, unnecessary expenses, and reduced profitability. Implementing modern inventory management practices enables businesses to optimize resource utilization, minimize losses, and ensure the consistent availability of menu items for customers.

Given these industry trends and challenges, there is a growing need for food service establishments such as Groove Monkey Pinoy Korean Bistro to adopt innovative technological solutions that enhance operational efficiency and customer satisfaction. By integrating digital ordering and inventory management systems, businesses can improve service quality, streamline daily operations, and strengthen their competitive position within the food service industry while meeting the evolving needs of the community they serve.

The food service industry is recognized as one of the most dynamic and competitive sectors of the economy. Restaurants and bistros must continuously adapt to changing consumer preferences, technological advancements, and market trends to remain relevant and profitable. Businesses that embrace innovation are more likely to achieve operational efficiency and maintain a strong competitive advantage within the industry.

In recent years, customers have become increasingly reliant on digital technologies when making purchasing decisions. Many consumers use smartphones and online platforms to search for restaurants, view menus, compare prices, and place orders. This shift in consumer behavior has encouraged food establishments to invest in digital solutions that improve accessibility and convenience for their customers.

The increasing demand for fast and efficient service has also reshaped operational practices within the food service industry. Customers expect shorter waiting times, accurate order fulfillment, and seamless transactions. Businesses that continue to rely solely on traditional manual methods may face difficulties meeting these expectations, potentially affecting customer satisfaction and retention.

Another significant trend in the industry is the growing popularity of food delivery and takeout services. The expansion of online delivery platforms has created new opportunities for restaurants to reach a wider customer base beyond their physical locations. Establishments that lack digital ordering capabilities may be unable to fully benefit from these emerging market opportunities.

Competition among food establishments has intensified as new restaurants, cafes, and specialty dining concepts continue to enter the market. To attract and retain customers, businesses must differentiate themselves not only through food quality but also through service efficiency, convenience, and customer engagement. Technological integration has become an important factor in achieving these objectives.

From a community perspective, local restaurants contribute significantly to economic development by generating employment opportunities and supporting local suppliers. Their success has a direct impact on the livelihoods of employees, farmers, distributors, and other stakeholders involved in the food supply chain. Therefore, improving operational efficiency can have positive effects beyond the business itself.

Sustainability has also become an important concern within the food service industry. Consumers are becoming more conscious of environmental issues and expect businesses to minimize waste and use resources responsibly. Accurate inventory management systems can help reduce food spoilage and waste, supporting both environmental sustainability and cost efficiency.

The increasing use of data analytics in restaurant management has further transformed industry practices. Businesses can now collect and analyze data related to customer preferences, sales performance, and inventory consumption. These insights enable managers to make informed decisions regarding menu planning, pricing strategies, and marketing initiatives, ultimately improving business performance.

Furthermore, government regulations and food safety standards require restaurants to maintain accurate records and implement proper inventory controls. Digital systems can support compliance efforts by providing organized documentation, improving traceability, and facilitating the preparation of reports required by regulatory authorities.

Given these industry developments, the adoption of technology is no longer merely an option but a necessity for many food service establishments. For Groove Monkey Pinoy Korean Bistro, implementing a digital ordering and inventory management system aligns with current industry trends and community needs. Such a system can improve operational efficiency, enhance customer experiences, strengthen business competitiveness, and contribute to sustainable growth within the local food service sector.

The modernization of the food service industry has encouraged businesses to transition from traditional operational methods to technology-based solutions. Digital transformation has become a key strategy for improving efficiency, reducing operational costs, and enhancing overall customer experiences. Restaurants that successfully adopt these innovations are often better positioned to respond to market demands and sustain long-term growth.

Consumer expectations have changed significantly as technology becomes increasingly integrated into everyday life. Customers now expect restaurants to offer convenient services such as online reservations, digital menus, contactless payments, and real-time order tracking. These features not only improve customer satisfaction but also influence purchasing decisions and brand loyalty.

The rise of social media marketing has further impacted the food service industry. Platforms such as Facebook, Instagram, and TikTok allow businesses to promote their products, engage with customers, and expand their market reach. Restaurants with limited online visibility may struggle to attract new customers, particularly younger consumers who frequently rely on digital platforms for dining recommendations and reviews.

Community engagement has also become an important aspect of restaurant operations. Many local dining establishments contribute to community development by supporting local events, sourcing ingredients from local suppliers, and providing employment opportunities to residents. Strengthening operational efficiency through technology can enable businesses to allocate more resources toward communityfocused initiatives and customer engagement activities.

Another notable trend in the industry is the increasing emphasis on customer relationship management. Businesses are recognizing the importance of understanding customer preferences and purchasing behaviors. Digital systems can help collect and analyze customer data, allowing restaurants to personalize promotions, improve service quality, and develop stronger relationships with their patrons.

The integration of technology has also improved communication between various departments within food establishments. Automated systems facilitate the smooth flow of information between cashiers, kitchen staff, inventory managers, and business owners. Improved communication minimizes misunderstandings, reduces errors, and ensures that operations run more efficiently throughout the day.

As labor costs continue to rise, many food service establishments seek solutions that maximize productivity without compromising service quality. Automated ordering and inventory systems can reduce repetitive administrative tasks, enabling employees to focus on customer service, food preparation, and other value-adding activities that contribute directly to customer satisfaction.

The growing importance of business resilience has become evident in recent years. Unexpected disruptions, such as economic fluctuations, supply chain challenges, and changes in consumer behavior, require businesses to be adaptable and responsive. Access to accurate real-time information through digital systems can help restaurant managers make timely decisions and maintain operational stability during challenging situations.

In addition, technological advancements have made digital solutions more accessible and affordable for small and medium-sized enterprises. Modern cloud-based systems, mobile applications, and integrated management platforms allow businesses to implement efficient operational tools without requiring extensive financial investments. This accessibility encourages more establishments to adopt digital innovations as part of their growth strategies.

Considering the ongoing developments within the food service industry and the evolving needs of the community, the implementation of a digital ordering and inventory management system represents a practical and strategic solution for Groove Monkey Pinoy Korean Bistro. Such a system can support operational excellence, improve customer satisfaction, enhance competitiveness, and ensure that the business remains aligned with current industry standards and future technological advancements.

The food service industry continues to evolve as businesses strive to meet the increasing expectations of modern consumers. Customers are no longer concerned solely with food quality; they also value convenience, speed, accessibility, and personalized service. As a result, restaurants and bistros are under constant pressure to improve their operational processes and customer engagement strategies.

The widespread use of mobile devices has significantly influenced dining habits and purchasing behavior. Consumers frequently use smartphones to search for nearby restaurants, browse menus, compare prices, and read customer reviews before making dining decisions. Businesses that maintain an active digital presence are more likely to attract potential customers and remain competitive within the industry.

Another important development in the food service sector is the increasing reliance on digital payment systems. Cashless transactions have become more common due to their convenience, speed, and security. Restaurants that integrate digital payment options can provide a smoother customer experience while reducing the time required to process transactions and manage cash handling procedures.

The growing availability of cloud-based technologies has also transformed restaurant management practices. These technologies allow business owners and managers to access operational information remotely, monitor sales performance in real time, and make informed decisions based on accurate data. Such capabilities are particularly valuable in ensuring efficient business management and continuous operational improvement.

Customer reviews and online ratings now play a significant role in shaping the reputation of food establishments. Positive online feedback can attract new customers, while negative reviews can quickly influence public perception. Therefore, restaurants must consistently provide high-quality service and maintain efficient operations to build a strong reputation within both the local community and digital marketplace.

The increasing diversity of customer preferences has created additional challenges for food service providers. Customers often seek unique dining experiences, menu customization, and flexible ordering options. Businesses that leverage digital systems can better accommodate these demands by streamlining order management processes and improving communication between customers and staff.

Inventory management remains a critical concern throughout the industry due to rising food costs and supply chain uncertainties. Effective inventory control helps businesses maintain product availability, reduce waste, and manage expenses more efficiently. Digital inventory systems provide real-time tracking and reporting capabilities that support more accurate forecasting and purchasing decisions.

The competitive nature of the restaurant industry has encouraged businesses to adopt innovative solutions that improve operational efficiency. Establishments that continue to rely heavily on manual systems may face challenges in keeping pace with competitors that utilize automated technologies to optimize their services and reduce operational bottlenecks.

Within the community, successful restaurants contribute to local economic growth by generating employment, supporting local suppliers, and attracting visitors. Enhancing operational efficiency through technology can strengthen these contributions by improving business sustainability and enabling establishments to expand their services and customer reach.

As the food service industry becomes increasingly technologydriven, businesses must continuously adapt to changing market conditions and customer expectations. The implementation of digital ordering and inventory management solutions provides an opportunity for Groove Monkey Pinoy Korean Bistro to align with industry best practices, improve operational performance, and strengthen its role as a valued contributor to the local community and economy.

The food service industry has become increasingly customercentered, with businesses focusing on delivering exceptional dining experiences alongside high-quality food products. Customers today expect restaurants to provide efficient service, accurate order processing, and convenient access to information. These expectations have encouraged establishments to adopt technological innovations that improve customer interactions and operational performance.

Technological advancements have enabled restaurants to streamline many aspects of their daily operations. From digital ordering systems to automated inventory tracking, modern tools help businesses reduce manual workloads and improve overall efficiency. As these technologies become more common within the industry, businesses that do not adapt may face difficulties maintaining their competitiveness.

The expansion of internet connectivity has also changed how customers discover and interact with food establishments. Online platforms allow consumers to explore menu offerings, compare prices, and read customer reviews before making purchasing decisions.

Maintaining a strong digital presence has therefore become essential for attracting new customers and increasing brand awareness.

In many communities, local restaurants serve as social gathering places where individuals and families come together to enjoy meals and build relationships. As such, the quality and efficiency of restaurant services directly influence customer experiences and community satisfaction. Businesses that consistently provide reliable service are more likely to develop strong relationships with their customers and earn community trust.

The increasing popularity of online food delivery services has further transformed the industry landscape. Customers appreciate the convenience of ordering meals from their homes or workplaces, creating new opportunities for restaurants to expand their market reach. Establishments that integrate digital ordering capabilities are better positioned to capitalize on this growing demand and increase revenue streams.

Operational efficiency has become a critical factor in ensuring long-term sustainability within the food service sector. Efficient processes help businesses minimize costs, optimize resource utilization, and improve service quality. By reducing manual procedures and adopting automated systems, restaurants can enhance productivity while maintaining high standards of customer service.

The use of digital technologies also supports better decisionmaking by providing access to accurate and timely information. Managers can monitor sales trends, inventory levels, and customer preferences in real time, enabling them to make informed decisions regarding menu planning, staffing, purchasing, and marketing activities. Data-driven management practices contribute significantly to business success in a competitive environment.

Environmental sustainability has emerged as an important concern within the industry. Restaurants are increasingly seeking ways to reduce food waste, conserve resources, and implement environmentally responsible practices. Effective inventory management systems can help achieve these goals by ensuring accurate stock monitoring and reducing unnecessary spoilage of ingredients.

Furthermore, the development of smart business solutions has created opportunities for small and medium-sized enterprises to compete more effectively with larger establishments. Affordable software applications and cloud-based systems allow businesses to access advanced management tools without requiring substantial investments. This technological accessibility promotes innovation and growth throughout the industry.

Considering these industry and community developments, it is evident that technology plays a vital role in shaping the future of food service operations. The implementation of a digital ordering and inventory management system at Groove Monkey Pinoy Korean Bistro would not only address existing operational challenges but also align the business with current industry trends, enhance customer satisfaction, strengthen community engagement, and support sustainable business growth.

- Why the study/system is needed

The study is needed to address the operational challenges commonly experienced by food services, particularly those relying on manual processes. Many bistros still use traditional methods in handling orders, managing inventory, and promoting their products, which often result in delays, human errors, and inefficiencies.

With the increasing demand for faster and more efficient service in the food industry, there is a need for a system that can automate ordering, improve inventory accuracy, and enhance online visibility. Implementing such a system will not only streamline operations but also improve customer experience, increase productivity, and help the business stay competitive.

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The continuous growth of the food service industry has encouraged businesses to seek innovative methods for improving customer satisfaction and operational effectiveness. As customer expectations become more sophisticated, restaurants must adopt modern technologies and management practices to remain competitive and relevant in the marketplace.

Digital transformation has become a major trend across various industries, including food service. Many restaurants have implemented integrated systems that combine order management, inventory tracking, customer relationship management, and sales reporting into a single platform. These technologies help reduce manual work and improve the accuracy of business operations.

The increasing use of e-commerce and online food ordering platforms has changed the way customers interact with restaurants. Consumers now expect the convenience of placing orders online, tracking their purchases, and making secure digital payments. Businesses that provide these services are often more successful in attracting and retaining customers compared to those that rely solely on traditional ordering methods.

In addition, demographic changes have influenced dining preferences and consumer behavior. Younger generations, particularly those who are highly familiar with technology, often prefer businesses that offer digital convenience and seamless service experiences. This shift highlights the importance of adopting technological solutions that cater to the needs of modern consumers.

The food service industry also faces challenges related to rising operational costs, including food prices, labor expenses, and utility costs. To remain profitable, businesses must identify ways to optimize resources and minimize waste. Automated inventory systems can contribute to cost reduction by providing accurate stock monitoring and helping managers make informed purchasing decisions.

Customer loyalty has become increasingly important in a highly competitive market. Restaurants must consistently deliver positive experiences to encourage repeat visits and generate positive wordofmouth recommendations. Digital systems can support customer retention efforts by enabling personalized promotions, loyalty programs, and efficient service delivery.

The role of technology extends beyond operational efficiency and also contributes to business transparency and accountability. Digital records provide accurate documentation of transactions, inventory movements, and financial activities. These records can assist management in conducting audits, evaluating performance, and ensuring compliance with industry standards and regulatory requirements.

Another significant trend is the growing emphasis on datadriven decision-making. Businesses that collect and analyze operational data are better equipped to identify market opportunities, understand customer preferences, and develop effective business strategies. Access to reliable information allows managers to make decisions based on evidence rather than assumptions.

Community development is closely linked to the success of local businesses such as restaurants and bistros. Thriving food establishments contribute to local employment, stimulate economic activity, and create opportunities for collaboration with local suppliers and service providers. Enhancing operational efficiency can strengthen these contributions and support broader community growth.

Given the increasing importance of technology, efficiency, and customer satisfaction within the food service industry, businesses must continuously adapt to changing market conditions. The adoption of a digital ordering and inventory management system can help Groove Monkey Pinoy Korean Bistro address industry challenges, improve service quality, optimize resource management, and establish a strong foundation for sustainable growth and long-term success.

The globalization of food trends has significantly influenced consumer expectations within the restaurant industry. Customers are increasingly exposed to diverse cuisines, dining concepts, and service innovations through social media, travel, and digital platforms. As a result, food establishments are expected to continuously improve not only their menu offerings but also the quality and efficiency of their services to remain competitive.

The emergence of smart technologies has created new opportunities for restaurants to optimize their operations. Modern restaurant management systems allow businesses to automate routine tasks, reduce administrative burdens, and improve workflow efficiency. These innovations help employees focus more on customer service and food quality rather than spending excessive time on manual documentation and record-keeping.

In many communities, food establishments contribute to cultural exchange and social interaction by offering unique dining experiences. Restaurants that provide specialty cuisines, such as Korean and Filipino fusion dishes, help introduce customers to different culinary traditions while fostering a sense of community engagement. Maintaining efficient operations ensures that these experiences remain enjoyable and accessible to customers.

The increasing demand for transparency in food service operations has also influenced industry practices. Customers are becoming more interested in food quality, ingredient sourcing, and service standards. Digital systems can support transparency by providing accurate records, improving inventory management, and helping businesses maintain consistent service quality and food safety standards.

Supply chain management has become a critical aspect of restaurant operations due to fluctuations in ingredient availability and pricing. Businesses that lack effective inventory control may experience difficulties in maintaining adequate stock levels and managing supplier relationships. Technology-driven inventory systems help address these concerns by providing accurate forecasts and facilitating timely purchasing decisions.

The growing popularity of customer feedback platforms has increased the importance of maintaining high service standards.

Online reviews and ratings can significantly influence a restaurant's reputation and purchasing decisions among potential customers. Efficient order processing, accurate service delivery, and consistent product quality contribute to positive customer experiences and favorable reviews.

Workforce management is another area that benefits from technological integration. Restaurant managers must coordinate employee schedules, monitor productivity, and ensure adequate staffing levels during peak hours. Digital systems provide valuable operational data that can assist management in allocating resources effectively and improving overall workforce efficiency.

The adoption of technology can also support business continuity and risk management efforts. Automated systems reduce dependence on paper-based records, which are vulnerable to loss, damage, or human error. Secure digital storage helps preserve important business information and ensures that operational data remains accessible when needed.

As urbanization and population growth continue to influence consumer demand, restaurants must be prepared to serve larger and more diverse customer groups. Scalable digital solutions enable businesses to manage increasing transaction volumes without compromising service quality, thereby supporting long-term expansion and growth objectives.

Considering these industry developments, it is evident that technological innovation has become an essential component of successful restaurant management. For Groove Monkey Pinoy Korean Bistro, embracing digital ordering and inventory management solutions can enhance operational efficiency, improve customer satisfaction, strengthen competitiveness, and contribute to the continued development of both the business and the community it serves.

The rapid pace of technological innovation continues to reshape the food service industry, creating new opportunities for businesses to improve efficiency and customer engagement. Restaurants that adopt modern technologies are often able to respond more effectively to market demands and maintain a competitive position within the industry. As digital solutions become increasingly accessible, their implementation has become an important factor in achieving operational success.

Consumer lifestyles have also evolved significantly over the years. Busy work schedules, increased urbanization, and the growing preference for convenience have influenced how customers purchase food and dining services. Many consumers now favor establishments that offer quick ordering processes, multiple payment options, and accessible online services that simplify their overall dining experience.

The integration of digital tools in restaurant operations has proven beneficial in reducing human errors and improving service accuracy. Automated systems can assist in order processing, inventory monitoring, and sales reporting, thereby minimizing the risks associated with manual data entry and paper-based record management. This contributes to more reliable operations and improved customer satisfaction.

Furthermore, restaurants are increasingly utilizing technology to gain a deeper understanding of customer preferences and behavior. Information collected through digital platforms can help businesses identify purchasing patterns, popular menu items, and customer feedback trends. These insights support the development of targeted marketing strategies and more informed business decisions.

The demand for efficiency is particularly important in small and medium-sized food establishments, where resources may be limited. Business owners must carefully manage labor, inventory, and operational costs to maintain profitability. Implementing technology-driven solutions can help optimize resource allocation and improve overall business performance without significantly increasing operational expenses.

The food service industry also plays a vital role in supporting local economic activity. Restaurants create employment opportunities, generate tax revenues, and contribute to the growth of related industries such as agriculture, transportation, and food distribution. Enhancing the operational capabilities of local establishments can therefore produce positive economic impacts within the community.

As environmental awareness continues to grow, many businesses are adopting sustainable practices to reduce waste and improve resource efficiency. Inventory management technologies can assist restaurants in monitoring stock levels accurately, reducing overordering, and minimizing food spoilage. These practices not only support environmental sustainability but also contribute to cost savings and improved profitability.

Another significant development within the industry is the increasing importance of customer convenience. Features such as digital menus, online reservations, mobile ordering, and contactless payments have become common expectations among consumers. Businesses that provide these conveniences are often better positioned to attract and retain customers in a competitive marketplace.

The availability of real-time operational data has become a valuable asset for restaurant management. Access to timely information enables managers to monitor performance indicators, identify operational issues, and implement corrective actions promptly. This proactive approach helps improve efficiency, maintain service quality, and support strategic planning efforts.

In light of these industry trends and community needs, the adoption of an integrated digital ordering and inventory management system represents a practical and forward-looking solution for Groove Monkey Pinoy Korean Bistro. By embracing technological innovation, the business can strengthen its operational capabilities, improve customer experiences, support sustainable practices, and contribute more effectively to the growth and development of the local food service industry.

- Existing issues or inefficiencies

Bistro businesses commonly experience several operational inefficiencies due to reliance on manual processes, including errors in order-taking that lead to delays and miscommunication, limited online presence that restricts market reach and customer convenience, and inaccurate inventory tracking that results in stock shortages or overstocking. These issues contribute to reduced productivity, slower service, and lower customer satisfaction, ultimately affecting the overall performance and competitiveness of the business.

One significant inefficiency associated with manual order-taking is the increased likelihood of human error during busy operating hours. Employees may incorrectly record customer orders, overlook special requests, or enter inaccurate information, resulting in order discrepancies and customer complaints. Such mistakes can negatively impact customer trust and reduce the likelihood of repeat business.

Another challenge is the time-consuming nature of manual administrative tasks. Staff members often need to spend considerable time recording transactions, updating inventory logs, and preparing operational reports. This additional workload reduces the time available for customer service and other critical activities, potentially affecting both employee productivity and service quality.

The absence of a strong online presence also limits the business's ability to engage with current and potential customers. In an era where consumers frequently rely on digital platforms to discover dining options and place orders, businesses with limited online visibility may struggle to attract new customers. This can restrict market growth and reduce opportunities to increase sales and brand recognition.

Inaccurate inventory tracking can create operational disruptions that affect daily business activities. When stock levels are not monitored effectively, essential ingredients may run out unexpectedly, causing delays in food preparation and limiting menu availability. Conversely, excessive inventory purchases can lead to spoilage and unnecessary expenses, reducing the overall profitability of the business.

Furthermore, the lack of integrated systems makes it difficult for management to access accurate and timely operational information. Without real-time data on sales, inventory, and customer transactions, decision-making becomes more challenging and less efficient. As a result, management may struggle to identify operational issues promptly and implement effective solutions that support business growth and customer satisfaction.

The reliance on manual systems often creates communication gaps between employees and different operational departments. Information regarding customer orders, inventory updates, and special requests may not be communicated accurately or promptly, resulting in misunderstandings that can affect service quality and operational efficiency.

Manual order processing can also lead to longer customer waiting times, especially during peak business hours. As employees handle multiple tasks simultaneously, delays in taking orders, processing payments, and communicating requests to the kitchen may occur. These delays can negatively impact the dining experience and reduce customer satisfaction.

Another inefficiency arises from the difficulty of maintaining accurate sales records. Handwritten records and manual calculations are more susceptible to errors, omissions, and inconsistencies. Inaccurate sales information can make it challenging for management to assess business performance and identify opportunities for improvement.

The lack of automated reporting tools further limits the ability of managers to monitor key performance indicators. Generating sales summaries, inventory reports, and operational analyses manually requires significant time and effort. Consequently, management may not have immediate access to the information needed to make timely and informed decisions.

Inventory discrepancies are also common in businesses that rely on manual stock monitoring. Differences between recorded inventory levels and actual stock quantities may occur due to human error, unrecorded usage, or delayed updates. These discrepancies can result in inefficient purchasing decisions and increased operational costs.

In addition, manual systems provide limited support for forecasting future demand. Without accurate historical data and analytical tools, managers may find it difficult to predict customer purchasing patterns and inventory requirements. This can lead to either insufficient stock levels or excess inventory that may eventually go to waste.

The absence of digital customer engagement channels can also affect customer retention. Modern consumers often expect businesses to provide online ordering, social media interaction, and digital communication options. Businesses that fail to offer these conveniences may struggle to build lasting relationships with customers and maintain loyalty.

Employee productivity may be affected by repetitive administrative tasks associated with manual operations. Staff members spend valuable time completing paperwork, updating records, and verifying information instead of focusing on customer service and other revenue-generating activities. This can reduce overall workplace efficiency and employee satisfaction.

The business may also face challenges in responding quickly to operational issues due to limited access to real-time information. Problems such as low inventory levels, sudden increases in customer demand, or order processing errors may not be identified immediately, resulting in delayed corrective actions and potential service disruptions.

Overall, these inefficiencies highlight the limitations of traditional manual processes within the food service industry. Addressing these challenges through the implementation of an integrated digital ordering and inventory management system can significantly improve operational efficiency, enhance customer satisfaction, reduce errors, and support the long-term success and competitiveness of the business.

Another inefficiency commonly observed in manual business operations is the difficulty in tracking daily transactions accurately. As the number of customer orders increases, maintaining organized and complete records becomes more challenging. Missing or inaccurate transaction records can affect financial reporting and create discrepancies in sales monitoring.

The lack of automation can also hinder the restaurant’s ability to manage customer expectations effectively. Customers often expect prompt service and accurate order fulfillment, but manual processes may result in delays, forgotten requests, or incorrect orders. These issues can negatively affect customer perceptions and decrease overall satisfaction with the dining experience.

Manual inventory monitoring often requires frequent physical stock counts, which consume valuable time and labor resources. Employees must manually verify inventory levels, record stock movements, and update logs. This process can be inefficient and may still fail to provide completely accurate inventory information due to human error.

Another concern is the inability to obtain real-time updates regarding stock availability. Since inventory records are updated manually, management may not immediately know when ingredients are running low or when replenishment is necessary. This delay can disrupt operations and affect the restaurant’s ability to consistently meet customer demands.

The absence of centralized data storage can create difficulties in accessing important business information. Records may be stored in different notebooks, spreadsheets, or paper files, making it difficult to retrieve information quickly when needed. This fragmented approach can slow down decision-making and increase the likelihood of data inconsistencies.

Businesses that rely heavily on manual processes may also face challenges in maintaining service consistency. Variations in employee performance, record-keeping practices, and communication methods can result in inconsistent customer experiences. Standardized digital systems help minimize these variations by providing structured procedures and automated workflows.

Limited technological integration can reduce the effectiveness of promotional and marketing efforts. Without digital tools to track customer behavior and purchasing patterns, businesses may struggle to identify which promotions are successful and which customer segments are most responsive to marketing campaigns. This can limit the return on investment of marketing activities.

Furthermore, manual systems often make it difficult to identify operational bottlenecks. Since data collection and analysis require significant effort, management may not easily recognize areas where delays, inefficiencies, or resource wastage occur. This lack of visibility can prevent the implementation of timely process improvements.

Employee workload can increase significantly when administrative tasks are performed manually. Staff members may become overwhelmed by responsibilities such as recording transactions, checking inventory, preparing reports, and processing customer orders simultaneously. Excessive workload can contribute to stress, fatigue, and a greater likelihood of mistakes during daily operations.

These operational inefficiencies collectively impact the restaurant's productivity, profitability, and customer satisfaction. Without effective technological solutions, the business may continue to face challenges in optimizing resources, improving service delivery, and maintaining competitiveness. Implementing an integrated digital ordering and inventory management system can help address these issues by streamlining processes, improving data accuracy, and enhancing overall business performance.

The dependence on paper-based records can significantly slow down the flow of information within the restaurant. Employees often need to manually transfer data from one record to another, increasing the possibility of transcription errors and delays. This process can become particularly problematic during busy periods when quick access to accurate information is essential for maintaining efficient operations.

Another inefficiency is the difficulty in monitoring order status throughout the service process. Without a digital tracking system, staff members may have limited visibility regarding whether an order is pending, being prepared, or ready for serving. This can lead to confusion among employees, delayed deliveries, and unnecessary customer waiting times.

Manual systems also make it challenging to manage multiple customer orders simultaneously. As the volume of transactions increases, employees may struggle to organize and prioritize orders effectively. This can result in misplaced order slips, duplicated orders, or missed transactions that negatively impact customer satisfaction and operational efficiency.

The lack of automated notifications and alerts further contributes to operational challenges. Management may not receive timely warnings regarding low inventory levels, delayed supplier deliveries, or unusual sales patterns. Consequently, important issues may remain unnoticed until they begin to affect daily operations and customer service quality.

Businesses that rely on manual processes often encounter difficulties in evaluating employee performance objectively. Since operational data is not automatically recorded and analyzed, managers may have limited information regarding employee productivity, order accuracy, and task completion rates. This can make performance assessments less reliable and more time-consuming.

Another concern is the limited ability to monitor food wastage accurately. Without detailed inventory tracking and consumption records, management may find it difficult to identify ingredients that are frequently wasted or spoiled. This lack of visibility can result in higher operational costs and reduced profitability over time.

Manual operations may also affect the restaurant's responsiveness to customer feedback and complaints. When information is not systematically documented, recurring service issues may go unnoticed, preventing management from implementing appropriate corrective measures. As a result, customer concerns may persist and negatively influence the business's reputation.

The absence of integrated operational systems can create barriers to business scalability. As the restaurant grows and customer demand increases, manual procedures may become increasingly difficult to manage. The business may experience operational bottlenecks that limit its ability to expand services, accommodate more customers, or introduce new product offerings efficiently.

Financial planning and budgeting can also be affected by inaccurate or incomplete operational data. Reliable financial decisions depend on accurate information regarding sales performance, inventory costs, and operational expenses. Manual record-keeping may compromise the quality of this information, making it more difficult for management to develop effective financial strategies.

Ultimately, the persistence of these inefficiencies can hinder the restaurant's ability to achieve its business objectives and maintain a competitive position within the food service industry. Addressing these challenges through digital transformation can improve operational control, enhance decision-making, reduce resource wastage, and provide a more efficient and customer-focused service environment.

The absence of a centralized management system often results in fragmented information across different areas of the business. Sales records, inventory data, customer information, and supplier details may be stored separately, making it difficult for management to obtain a complete overview of operations. This lack of integration can lead to inefficiencies and delays in decision-making.

Manual order-taking procedures may also increase the likelihood of customer dissatisfaction due to inconsistent service experiences. Employees may unintentionally omit specific customer requests, incorrectly record menu selections, or fail to communicate important details to kitchen staff. Such errors can affect food quality, service reliability, and overall customer perception of the restaurant.

Another operational challenge is the inability to monitor inventory movement accurately. Without automated tracking, management may struggle to determine how quickly ingredients are being consumed and which items require replenishment. This can lead to poor inventory planning and disruptions in food preparation activities.

The restaurant may also experience difficulties in identifying high-demand and low-demand menu items. Manual sales records often require extensive review and analysis before useful insights can be obtained. As a result, management may miss opportunities to optimize menu offerings, eliminate underperforming products, or focus on items that generate higher profits.

In addition, the lack of digital records can make it challenging to maintain historical business data. Paper-based documents may become damaged, misplaced, or difficult to retrieve over time. Losing valuable operational information can affect future planning, trend analysis, and business evaluation efforts.

Manual processes can also contribute to increased labor costs. Employees spend considerable time performing repetitive administrative tasks such as recording transactions, updating inventory logs, and preparing reports. These tasks consume resources that could otherwise be directed toward improving customer service and enhancing business operations.

Another inefficiency involves the management of supplier transactions and purchasing activities. Without automated monitoring systems, it can be difficult to track purchase orders, delivery schedules, and supplier performance. This may lead to delayed restocking, procurement errors, and challenges in maintaining consistent inventory levels.

The limited availability of real-time business information may prevent management from responding quickly to operational issues. Problems such as declining sales, inventory shortages, or sudden increases in customer demand require prompt action. Manual systems often delay the identification of these issues, reducing the effectiveness of corrective measures.

Furthermore, businesses that lack digital capabilities may find it difficult to remain competitive in an increasingly technologydriven market. Competitors that utilize automated systems can often provide faster service, more accurate transactions, and greater customer convenience. This competitive disadvantage may affect customer retention and long-term business growth.

Overall, the continued reliance on manual operational procedures creates numerous challenges that impact efficiency, accuracy, and customer satisfaction. Implementing an integrated digital ordering and inventory management system can address these inefficiencies by streamlining workflows, improving data management, enhancing operational visibility, and supporting sustainable business development.

1.2 Statement of the Problem

- General problem statement

This study aims to address the operational inefficiencies experienced by food services, particularly in terms of manual order processing, limited online presence, and inaccurate inventory tracking. These challenges result in delays, errors, reduced customer satisfaction, and ineffective resource management, which hinder the overall performance and competitiveness of the business. Therefore, there is a need to develop a system that can improve operational efficiency, enhance customer service, and support better business management.

One of the primary concerns addressed by this study is the inefficiency associated with traditional manual processes in restaurant operations. Manual order-taking and record management require significant time and effort from employees, increasing the likelihood of human errors and reducing overall productivity. These inefficiencies can negatively affect the quality of service provided to customers and limit the business's ability to operate effectively during peak hours.

Another issue that this study seeks to address is the growing demand for digital accessibility among consumers. Modern customers increasingly prefer businesses that offer convenient online services, such as digital ordering, online reservations, and electronic payment options. The absence of these features may prevent businesses from meeting customer expectations and limit their ability to attract and retain a larger customer base.

The study also recognizes the importance of accurate inventory management in ensuring smooth business operations. Inaccurate stock monitoring can lead to shortages of essential ingredients, disruptions in food preparation, and unnecessary inventory costs resulting from overstocking. These challenges not only affect operational efficiency but also have a direct impact on customer satisfaction and business profitability.

Furthermore, the lack of real-time operational data makes it difficult for managers to make informed decisions regarding sales, inventory control, and resource allocation. Without reliable information, business owners may struggle to identify operational issues, monitor performance, and implement strategies that support growth and continuous improvement. An integrated system can provide timely and accurate information to support effective decision-making.

By addressing these operational challenges, this study aims to contribute to the development of a more efficient and technologydriven business environment. The proposed system is expected to streamline restaurant operations, improve service delivery, enhance inventory accuracy, and strengthen customer engagement. Ultimately, these improvements can help businesses achieve greater productivity, profitability, and long-term sustainability in the highly competitive food service industry.

The increasing complexity of restaurant operations has made traditional management methods less effective in meeting the demands of modern business environments. As customer expectations continue to evolve, food service establishments must manage larger volumes of transactions, inventory records, and customer interactions. Manual systems often struggle to handle these growing operational requirements efficiently.

A significant concern addressed by this study is the impact of operational inefficiencies on customer satisfaction. Delays in order processing, inaccurate order fulfillment, and inconsistent service quality can negatively affect the customer experience. Since customer satisfaction plays a crucial role in business success, addressing these issues is essential for maintaining customer loyalty and attracting new patrons.

The study also focuses on the challenges associated with limited business visibility in digital platforms. In an increasingly connected world, customers rely on online channels to search for restaurants, view menus, and place orders. Businesses that lack an effective digital presence may experience difficulties reaching potential customers and competing with establishments that utilize modern online marketing and ordering platforms.

Another problem identified in this study is the inefficient utilization of resources caused by inaccurate inventory management. Poor inventory control can result in unnecessary expenditures, food wastage, and stock shortages that disrupt daily operations. These issues can reduce profitability and hinder the restaurant’s ability to provide consistent service to customers.

The absence of automated systems can also affect communication and coordination among employees. Information related to customer orders, stock availability, and operational updates may not be communicated effectively through manual methods. This can lead to misunderstandings, service delays, and operational errors that affect overall business performance.

Furthermore, the study recognizes the challenges faced by managers in monitoring business activities and evaluating operational performance. Manual reporting processes often require substantial time and effort, making it difficult to obtain timely and accurate information. Without access to reliable data, management may encounter difficulties in identifying problems and implementing effective solutions.

The increasing reliance on technology within the food service industry has created new standards for operational efficiency and customer convenience. Businesses that continue to depend heavily on manual procedures may face disadvantages in terms of speed, accuracy, and service quality. This study seeks to explore solutions that align with current industry practices and technological advancements.

Another area of concern is the inability of traditional systems to support business growth and expansion. As customer demand increases, manual processes may become more difficult to manage, resulting in operational bottlenecks and reduced efficiency. Implementing a scalable digital solution can help businesses accommodate future growth while maintaining service quality and operational control.

The study also addresses the need for improved data management and record-keeping practices. Accurate and organized information is essential for financial reporting, inventory monitoring, and strategic planning. Digital systems provide a more secure and efficient method of storing, accessing, and analyzing business data compared to traditional paper-based records.

Ultimately, this study seeks to provide a practical solution to the operational challenges commonly experienced by food service establishments. By developing and implementing a digital ordering and inventory management system, the study aims to improve efficiency, enhance customer satisfaction, strengthen decision-making capabilities, and support the long-term competitiveness and sustainability of the business in an evolving industry landscape.

The food service industry operates in a highly competitive environment where efficiency and customer satisfaction are critical factors for success. Businesses that fail to streamline their operations often encounter difficulties in meeting customer expectations and maintaining profitability. This study addresses the need for improved operational practices that can help restaurants remain competitive in an increasingly demanding market.

One of the major problems identified is the dependence on manual procedures that consume valuable time and resources. Employees are often required to perform repetitive administrative tasks such as recording orders, updating inventory records, and preparing reports. These activities can reduce productivity and limit the amount of attention given to customer service and operational improvement.

The study also highlights the challenges associated with maintaining service consistency. Human errors in order-taking and inventory management can result in varying service experiences for customers. Inconsistent service quality may negatively affect customer perceptions and reduce the likelihood of repeat visits, ultimately impacting business growth.

Another important issue involves the lack of immediate access to critical business information. Managers often rely on manually compiled reports that may not accurately reflect current operational conditions. The absence of real-time data limits the ability to make timely decisions regarding inventory replenishment, staffing requirements, and customer service improvements.

The growing digitalization of the business environment has increased customer expectations regarding convenience and accessibility. Consumers now expect businesses to provide online ordering options, digital payment methods, and accessible information through online platforms. Businesses that do not offer these features may struggle to attract and retain customers in a technology-driven marketplace.

In addition, inefficient inventory management can have significant financial consequences for the business. Overstocking can lead to increased storage costs and food spoilage, while understocking can result in menu limitations and lost sales opportunities. Effective inventory control is therefore essential for maximizing profitability and ensuring operational continuity.

The study further addresses the issue of limited operational transparency. Manual systems often make it difficult to track transactions, monitor stock movements, and evaluate employee performance accurately. This lack of visibility can hinder management's ability to identify inefficiencies and implement appropriate corrective actions.

Another concern is the difficulty of adapting to changing customer preferences and market trends. Without access to accurate sales and customer data, businesses may struggle to identify emerging opportunities and make strategic adjustments. A modern information system can provide valuable insights that support innovation and continuous improvement.

As the volume of business transactions increases, manual processes become more vulnerable to delays, inaccuracies, and operational bottlenecks. These challenges can affect the restaurant’s ability to provide efficient service and maintain customer satisfaction during periods of high demand. Addressing these limitations is essential for supporting future growth and expansion.

Therefore, this study emphasizes the importance of developing an integrated digital ordering and inventory management system that addresses existing operational challenges. By improving process efficiency, enhancing data accuracy, strengthening customer engagement, and supporting informed decision-making, the proposed system can contribute significantly to the overall success, sustainability, and competitiveness of the business.

The current business environment requires food service establishments to operate with greater efficiency, accuracy, and responsiveness. As customer expectations continue to rise, businesses must ensure that their operational processes are capable of delivering consistent and high-quality service. The persistence of manual procedures can create obstacles that limit the restaurant’s ability to achieve these objectives effectively.

One major concern addressed by this study is the inefficiency of traditional record-keeping methods. Manual documentation often requires substantial effort and time to maintain, organize, and retrieve information. As business transactions increase, these methods become more difficult to manage and may contribute to inaccuracies that affect daily operations and management decisions.

The study also examines the impact of operational inefficiencies on employee performance and productivity. Staff members who spend excessive time performing repetitive administrative tasks may have fewer opportunities to focus on customer service and operational excellence. This can affect workplace efficiency and reduce the overall quality of service provided to customers.

Another problem involves the lack of integration among various business functions. Ordering, inventory management, sales tracking, and reporting activities are often performed independently in manual systems. This separation of information can create inconsistencies and delays that hinder effective communication and coordination throughout the organization.

The increasing importance of customer convenience has made digital accessibility a necessity for modern businesses. Customers prefer services that offer quick ordering processes, flexible payment options, and easy access to information. The absence of these features can place a business at a disadvantage compared to competitors that have adopted technology-driven solutions.

Inventory-related challenges remain a significant concern within the food service industry. Inaccurate stock records can lead to difficulties in maintaining adequate ingredient supplies, resulting in service interruptions and customer dissatisfaction. At the same time, excessive inventory purchases can increase storage costs and contribute to food waste, reducing overall profitability.

The study further recognizes the need for accurate and timely business information. Effective decision-making depends on reliable data regarding sales performance, inventory levels, and customer demand. Manual systems often limit the availability of such information, making it difficult for management to identify trends, evaluate performance, and implement strategic improvements.

Another challenge addressed by this study is the restaurant’s ability to adapt to changing industry conditions. The food service sector continues to evolve due to technological advancements, changing consumer behavior, and increasing competition. Businesses that rely solely on traditional methods may struggle to respond efficiently to these changes and maintain their competitive position.

Furthermore, operational inefficiencies can affect the restaurant’s long-term growth and sustainability. Delays, errors, and resource wastage may accumulate over time, leading to reduced profitability and lower customer retention rates. Addressing these issues through technological innovation can help create a more stable and efficient business environment.

In response to these challenges, this study proposes the development of a digital ordering and inventory management system designed to streamline operations, improve service quality, and enhance business performance. By providing automated processes, realtime information, and improved operational control, the system aims to address existing inefficiencies and support the restaurant’s longterm success in an increasingly competitive industry.

- Specific problems (listed in numbered or bullet form)

1. How does the current manual order processing system affect the speed and accuracy of service in the bistro?
2. What challenges are encountered due to the bistro's limited online presence in terms of customer reach and accessibility?
3. How does manual inventory tracking impact the accuracy of stock management and overall business operations?
4. What inefficiencies arise in the workflow and productivity of staff due to lack of an automated system?
5. How can a proposed system improve order processing, inventory management, and online visibility of the bistro?

The first problem regarding manual order processing is significant because it directly affects the speed of service delivery. When orders are taken manually, there is a higher chance of delays, miscommunication, and human error, which can lead to incorrect orders being served and reduced customer satisfaction. This study further examines how these inefficiencies influence overall customer experience.

In addition, manual order processing often requires staff to repeatedly verify and record orders, which increases workload and slows down service. This situation becomes more problematic during peak hours when customer demand is high, resulting in longer waiting times and decreased operational efficiency within the bistro.

The second problem focuses on the limited online presence of the bistro, which restricts its ability to reach a wider audience. In today's digital environment, many customers rely on online platforms to discover food establishments, view menus, and place orders. Without a strong online presence, the bistro may miss potential customers and sales opportunities.

Furthermore, limited accessibility through digital platforms reduces customer convenience. Potential customers may find it difficult to check menu offerings, prices, or availability, which may discourage them from choosing the bistro over competitors that offer online ordering and digital engagement.

The third problem involves manual inventory tracking, which often leads to inaccuracies in stock management. Without real-time updates, staff may struggle to determine the exact availability of ingredients and supplies, resulting in either overstocking or shortages that disrupt daily operations.

Inaccurate inventory tracking can also lead to financial inefficiencies, such as unnecessary purchasing or food waste. These issues not only increase operational costs but also affect the overall profitability of the business, making inventory management a critical area of concern.

The fourth problem highlights inefficiencies in staff workflow and productivity due to the absence of an automated system. Employees are required to perform repetitive tasks manually, such as recording orders, updating inventory logs, and calculating bills, which consumes valuable time and effort.

These manual processes often lead to task duplication and communication gaps among staff members. As a result, employees may experience confusion or delays in task completion, which ultimately affects the overall efficiency of business operations and service delivery.

The fifth problem addresses the need for a proposed system that can integrate and improve key business processes. A web-based management system has the potential to streamline order processing, automate inventory updates, and enhance online visibility, thereby addressing the identified operational challenges.

Such a system can also centralize all business data, allowing for real-time monitoring and easier access to information. By integrating multiple functions into a single platform, the bistro can improve coordination among staff, reduce errors, and enhance overall decision-making.

Overall, these specific problems highlight the importance of transitioning from manual processes to a digital system. Addressing these issues through a web-based solution will significantly improve operational efficiency, customer satisfaction, and business performance for Groove Monkey Pinoy Korean Bistro.

Another important concern in manual order processing is the increased risk of miscommunication between staff and kitchen personnel. When orders are relayed verbally or written on paper, details such as modifications, add-ons, or special instructions may be missed, leading to incorrect food preparation and customer dissatisfaction.

Manual processes also make it difficult to maintain a clear and organized record of transactions. Paper-based orders can easily be lost, damaged, or misplaced, which creates challenges in tracking previous sales and verifying past customer orders when needed for reference or dispute resolution.

The limited online presence of the bistro also affects its competitiveness in the food service industry. Many customers now prefer establishments that offer online menus, social media visibility, and digital ordering options. Without these, the bistro may struggle to compete with more digitally active competitors.

In addition, the absence of digital marketing channels reduces the bistro’s ability to promote new products, discounts, or special offers effectively. This limits customer engagement and reduces opportunities for attracting repeat customers or new market segments.

Manual inventory tracking also makes it difficult to identify usage patterns and consumption trends. Without automated records, management cannot easily determine which ingredients are frequently used or which items are underutilized, leading to inefficient purchasing decisions.

This lack of accurate inventory data may also result in unexpected stockouts during peak hours. When ingredients run out without proper forecasting, the bistro may be unable to fulfill customer orders, negatively affecting service quality and customer trust.

Staff productivity is further affected by the lack of integration between different operational tasks. Employees often need to transfer information manually between ordering, billing, and inventory logs, which increases workload and reduces the time available for direct customer service.

Another inefficiency arises from the absence of real-time updates across departments. When changes occur in orders or inventory, not all staff may be immediately informed, leading to inconsistencies in operations and potential errors in service delivery.

The lack of an automated system also limits the ability to generate timely and accurate business reports. Manual report preparation is time-consuming and prone to errors, making it difficult for management to quickly assess business performance and make informed decisions.

Another issue arising from manual order processing is the difficulty in handling multiple orders simultaneously. When customer volume increases, staff may struggle to manage several transactions at once, leading to order delays and increased chances of mistakes in recording or preparation.

Manual systems also make it challenging to track order status in real time. Without an automated system, staff cannot easily determine whether an order is pending, in preparation, or completed, which may result in confusion and slower service coordination.

The limited online presence of the bistro also reduces its ability to respond quickly to customer inquiries. Customers who want information about menu items, pricing, or availability may experience delays, which can discourage engagement and reduce potential sales opportunities.

Another problem is the lack of digital customer feedback mechanisms. Without an online platform, it becomes difficult for the bistro to collect and analyze customer reviews, which are essential for improving service quality and understanding customer preferences.

Manual inventory tracking also creates challenges in identifying expired or near-expiry products. Without proper monitoring tools, ingredients may be used past their optimal condition or wasted due to poor tracking, affecting both food quality and cost efficiency.

This lack of automation in inventory management also makes it difficult to implement efficient stock replenishment strategies. Management may rely on estimation rather than accurate data, which can lead to overordering or underordering of supplies.

Staff workflow inefficiency is further worsened by the duplication of tasks. Employees may need to input the same information in multiple records, increasing workload and raising the possibility of inconsistencies between different logs.

Another inefficiency is the lack of standardized procedures in manual operations. Different employees may follow different methods when recording orders or updating inventory, resulting in inconsistent data and confusion in daily operations.

The absence of an integrated system also limits data accessibility for decision-making. Important business information is scattered across different records, making it difficult for management to quickly retrieve and analyze operational data when needed.

Overall, these additional issues emphasize the critical need for a centralized and automated system. Addressing these problems will significantly improve accuracy, efficiency, communication, and competitiveness, ultimately enhancing the overall performance of Groove Monkey Pinoy Korean Bistro.

1.3 Objectives of the Study

1.3.1 General Objective

The main goal of this study is to create a web-based management system for Groove Monkey Pinoy Korean Bistro. This system aims to improve operational efficiency, provide better customer service, and ensure accurate data management.

The proposed web-based management system is designed to address the operational challenges currently experienced by Groove Monkey Pinoy Korean Bistro. By integrating various business functions into a single platform, the system seeks to streamline daily operations and reduce the inefficiencies associated with manual processes. This integration will enable the business to manage orders, inventory, and records more effectively and accurately.

One of the primary objectives of the system is to improve the speed and accuracy of order processing. Through digital order management, customer requests can be recorded, transmitted, and monitored efficiently, reducing the likelihood of errors and ensuring that orders are fulfilled promptly. This improvement is expected to enhance the overall dining experience and increase customer satisfaction.

The system also aims to strengthen inventory management by providing real-time monitoring of stock levels and inventory movements. With accurate and up-to-date inventory information, management can make informed decisions regarding purchasing, stock replenishment, and resource allocation. This capability can help minimize food waste, prevent stock shortages, and optimize operational costs.

Another important objective is to improve data accuracy and reliability. Manual record-keeping methods are often prone to human error, resulting in incomplete or inaccurate information. The proposed system will automate data collection and storage processes, ensuring that business records are accurate, organized, and readily accessible whenever needed.

The web-based nature of the system will provide greater accessibility and convenience for both management and staff. Authorized users will be able to access operational information through internet-enabled devices, allowing them to monitor business activities and perform administrative tasks more efficiently regardless of their location.

The study also aims to enhance customer service by providing a more organized and responsive operational environment. Faster order processing, improved communication among staff, and accurate inventory information can contribute to a smoother service experience, helping the restaurant meet customer expectations and build long-term customer loyalty.

Furthermore, the proposed system seeks to support managerial decision-making through the generation of comprehensive reports and analytical information. Managers will have access to detailed records of sales, inventory status, and operational performance, enabling them to identify trends, evaluate business activities, and implement datadriven strategies for improvement.

Another objective is to improve operational transparency and accountability within the organization. The system will maintain detailed records of transactions, inventory updates, and user activities, making it easier to track business operations and monitor employee responsibilities. This feature can contribute to better internal control and operational management.

The implementation of a web-based management system is also intended to support the future growth and expansion of the business. As customer demand increases, the system can accommodate larger transaction volumes and more complex operational requirements without significantly increasing administrative workload. This scalability makes the system a valuable long-term investment for the restaurant.

Ultimately, the development of the web-based management system aims to provide Groove Monkey Pinoy Korean Bistro with a modern and efficient technological solution that enhances operational performance, improves customer satisfaction, strengthens business management, and supports sustainable growth. Through the integration of digital technologies, the restaurant can better position itself to meet the demands of a competitive and rapidly evolving food service industry.

The development of the web-based management system seeks to modernize the operational processes of Groove Monkey Pinoy Korean Bistro by replacing traditional manual methods with automated and technologydriven solutions. Through digitalization, the business can improve efficiency, minimize errors, and create a more organized workflow that supports daily operations.

A key objective of the study is to establish a centralized platform where essential business information can be stored, managed, and accessed efficiently. By consolidating order records, inventory data, sales information, and customer transactions into a single system, the restaurant can improve information management and reduce the risk of data inconsistencies.

The proposed system aims to enhance communication among employees by providing a more efficient method of sharing operational information. Orders can be transmitted directly between service staff and kitchen personnel, reducing communication gaps and ensuring that customer requests are accurately processed and fulfilled.

Another important goal is to reduce the time required to perform routine administrative tasks. Automated processes for recording transactions, updating inventory levels, and generating reports can significantly decrease the workload of employees and management. This allows staff to focus more on customer service and other essential operational responsibilities.

The system is also intended to improve operational monitoring and control. Through real-time tracking features, managers can oversee business activities more effectively, identify potential issues promptly, and implement corrective measures before problems escalate. This capability contributes to smoother and more efficient restaurant operations.

In addition, the study aims to improve customer convenience by supporting a more efficient ordering process. Customers can benefit from faster service, reduced waiting times, and improved order accuracy. These enhancements can contribute to a more positive dining experience and strengthen customer loyalty toward the restaurant.

The proposed web-based solution also seeks to improve resource utilization by ensuring that inventory and operational resources are managed effectively. Accurate inventory monitoring enables the business to optimize purchasing decisions, avoid unnecessary expenses, and reduce losses associated with food spoilage and overstocking.

Another objective is to strengthen business adaptability in a rapidly evolving technological environment. By adopting digital tools and automated systems, Groove Monkey Pinoy Korean Bistro can better respond to changing customer preferences, emerging industry trends, and increasing market competition. This adaptability is essential for long-term business sustainability.

The study further aims to support strategic planning and performance evaluation through reliable data collection and analysis. The system can generate accurate reports and performance metrics that help management assess operational efficiency, monitor business growth, and identify opportunities for continuous improvement.

Ultimately, the web-based management system is intended to serve as a comprehensive solution that integrates multiple business functions into a cohesive and efficient platform. Through improved operational processes, enhanced customer service, accurate information management, and better decision-making capabilities, the system can contribute significantly to the overall success, competitiveness, and sustainability of Groove Monkey Pinoy Korean Bistro.

The proposed web-based management system aims to establish a more efficient operational framework that supports the daily activities of Groove Monkey Pinoy Korean Bistro. By integrating essential business processes into a single platform, the system can help reduce redundancies, improve coordination, and ensure smoother execution of restaurant operations.

Another objective of the study is to enhance the accuracy and consistency of business transactions. Manual processes are often susceptible to human error, which may affect order processing, inventory records, and sales reporting. Through automation, the system can improve data reliability and reduce the occurrence of costly mistakes.

The system is designed to provide management with better visibility into business operations. Access to real-time information regarding sales, inventory levels, and customer orders allows managers to monitor performance more effectively and make informed decisions that contribute to operational success.

A further goal is to improve the restaurant’s capability to manage customer demand efficiently. During peak hours, handling large volumes of orders manually can lead to delays and service disruptions. The proposed system can streamline order processing and help ensure that customer requests are handled accurately and promptly.

The study also seeks to promote a more organized approach to inventory management. By maintaining accurate digital records of stock movement and inventory levels, the system can assist management in preventing shortages, avoiding overstocking, and maintaining adequate supplies for daily operations.

In addition, the system aims to facilitate better financial management through accurate transaction recording and reporting. Automated sales tracking and report generation can provide valuable insights into revenue performance, operational expenses, and overall business profitability. These features can support more effective budgeting and financial planning.

Another objective is to improve employee productivity by minimizing time spent on repetitive administrative tasks. Automated workflows can simplify routine activities such as order recording, inventory updates, and report preparation, enabling employees to focus on customer service and operational excellence.

The proposed system also intends to support business continuity by providing secure and organized data storage. Digital records can be protected from loss, damage, or misplacement, ensuring that important business information remains accessible for future reference and decision-making purposes.

Furthermore, the study aims to strengthen the restaurant's competitiveness by incorporating technological solutions that align with current industry standards. Modern customers increasingly prefer businesses that offer efficient and technology-supported services, making digital transformation an important factor in maintaining market relevance.

Ultimately, the development of the web-based management system seeks to create a sustainable foundation for future business growth. By improving operational efficiency, enhancing customer satisfaction, supporting data-driven decision-making, and optimizing resource management, the system can help Groove Monkey Pinoy Korean Bistro achieve its long-term goals and maintain success within the competitive food service industry.

The web-based management system is intended to serve as a comprehensive technological solution that addresses the operational needs of Groove Monkey Pinoy Korean Bistro. By automating essential business functions, the system can help create a more efficient and structured operational environment that supports the restaurant's long-term objectives.

One of the goals of the study is to simplify the management of customer orders by providing a centralized digital platform. This feature can reduce the complexity of handling multiple transactions simultaneously and improve the overall accuracy of order processing. As a result, customers can receive faster and more reliable service.

The proposed system also aims to provide better control over inventory management activities. Through automated stock monitoring and inventory updates, management can easily track available supplies and identify items that require replenishment. This capability helps ensure uninterrupted operations and minimizes losses caused by inventory-related issues.

Another objective is to establish a reliable database for storing and managing business information. The system will maintain organized records of transactions, inventory movements, and operational activities, making it easier for management to retrieve information and monitor business performance over time.

The study further seeks to improve operational efficiency by reducing dependence on paper-based documentation. Digital recordkeeping can streamline administrative processes, reduce paperwork, and enhance the speed of information processing. This transition can contribute to a more environmentally friendly and costeffective business operation.

An additional goal is to support effective coordination among employees and departments. By enabling real-time information sharing, the system can facilitate smoother communication between front-ofhouse staff, kitchen personnel, and management. Improved coordination helps reduce misunderstandings and promotes efficient workflow management.

The proposed system is also designed to enhance customer satisfaction by providing more accurate and timely service. Customers benefit from improved order accuracy, shorter waiting times, and a more organized service process. These improvements can strengthen customer trust and encourage repeat patronage.

Another objective of the study is to provide management with valuable analytical tools for evaluating business performance. The system can generate reports that highlight sales trends, inventory usage, and operational efficiency, enabling managers to make strategic decisions based on accurate and up-to-date information.

The implementation of the web-based system also aims to improve the restaurant's ability to adapt to future technological advancements. As digital solutions continue to evolve, having an established technological infrastructure can make it easier for the business to integrate additional features and innovations that further enhance operations.

Ultimately, the general objective of this study is to develop a modern management system that contributes to the continuous improvement of Groove Monkey Pinoy Korean Bistro. Through enhanced operational efficiency, accurate data management, improved customer service, and stronger managerial control, the system can support sustainable growth and help the business maintain a competitive advantage within the food service industry.

1.3.2 Specific Objectives o To create a user-friendly web-based management system for the bistro o To automate the ordering and billing process o To improve inventory tracking and management o To generate real-time sales and inventory reports o To provide better customer experience through faster and more accurate service o To offer a centralized database for storing business transactions

The objective of creating a user-friendly web-based management system focuses on ensuring that all intended users, regardless of their technical background, can easily navigate and operate the system. A simple and intuitive interface reduces the learning curve and encourages efficient system adoption within the bistro environment.

In relation to automation of ordering and billing processes, the system aims to significantly reduce manual encoding of orders and computations. By automating these tasks, the system minimizes human errors such as incorrect pricing, missing items, or miscalculations, thereby improving transaction accuracy and speed.

Improving inventory tracking and management is another critical objective of the system. The system is designed to provide real-time updates of stock levels whenever items are sold or restocked. This helps prevent issues such as overstocking or stock shortages, which can negatively affect daily operations.

The generation of real-time sales and inventory reports supports better decision-making for management. With immediate access to updated data, administrators can analyze business performance, identify best-selling products, and monitor inventory usage without delays caused by manual report preparation.

Enhancing customer experience through faster and more accurate service is also a key goal of the system. By reducing waiting times and ensuring order accuracy, customers are more likely to receive satisfactory service, which can lead to increased customer loyalty and repeat visits to the bistro.

Providing a centralized database for storing business transactions ensures that all records are securely stored in one unified system. This eliminates data fragmentation and allows users to access consistent and updated information whenever needed, improving operational coordination.

The system also aims to streamline communication between different user roles such as administrators, cashiers, and staff. With a centralized system, all users can access relevant information based on their permissions, reducing miscommunication and improving workflow efficiency.

Another important objective is to improve data accuracy across all business processes. By eliminating manual record-keeping, the system reduces the likelihood of errors in transactions, inventory records, and reporting, ensuring that all business data remains reliable and consistent.

The system further aims to support business scalability and future growth. As the bistro expands, the system can accommodate additional features or increased transaction volumes without requiring major structural changes, making it a sustainable long-term solution.

The system also aims to reduce operational workload among employees by minimizing repetitive manual tasks. Through automation, staff members can allocate more time to customer service and other essential responsibilities, which contributes to a more efficient and productive working environment within the bistro.

Another objective is to enhance the accuracy of financial tracking and sales computation. By integrating billing and reporting modules, the system ensures that all transactions are properly recorded and calculated, reducing discrepancies that commonly occur in manual financial processing.

The system is also designed to improve order management efficiency by ensuring that all customer orders are properly recorded, updated, and tracked from entry to completion. This helps prevent issues such as lost orders, incorrect items, or delays in order fulfillment.

Strengthening internal operational control is another key objective of the system. With defined user roles and access levels, the system ensures that each employee can only perform tasks appropriate to their responsibilities, reducing the risk of unauthorized actions or system misuse.

The system also seeks to improve data organization within the business. By storing all information in a structured database, records such as transactions, inventory, and sales reports can be easily retrieved, searched, and analyzed when needed.

Another objective is to increase the speed of business transactions. By digitizing and streamlining processes, the system reduces the time required to process orders and generate receipts, leading to faster service delivery and improved customer satisfaction.

The system also aims to support better inventory decision-making by providing accurate and timely data on stock levels. This allows management to make informed decisions on purchasing, restocking, and product availability based on actual consumption patterns.

Improving consistency in business operations is another important objective. By standardizing procedures through a centralized system, the bistro can ensure that all transactions and processes follow uniform steps, resulting in more predictable and reliable outcomes.

The system further aims to reduce dependency on paper-based records and manual documentation. This shift to digital storage not only improves efficiency but also helps in preserving data, reducing physical storage requirements, and supporting environmentally friendly practices.

The system also aims to improve the traceability of all business transactions within Groove Monkey Pinoy Korean Bistro. By recording every action performed in the system, management can easily track the history of orders, inventory changes, and billing activities, which supports transparency and accountability.

Another objective is to enhance operational decision-making through data-driven insights. The system provides summarized and detailed reports that allow administrators to identify trends, compare sales performance, and evaluate inventory movement, leading to more informed business strategies.

The system is also designed to minimize operational downtime caused by manual errors or inefficient processes. By automating critical tasks, the likelihood of delays due to human mistakes is reduced, ensuring smoother and more continuous business operations.

Improving the accuracy of inventory replenishment is another key objective. The system helps management determine when items need to be restocked based on real-time usage data, reducing the risk of running out of essential ingredients during business hours.

The system also aims to improve coordination between kitchen operations and front-line staff. With a centralized ordering system, kitchen personnel receive accurate and timely order details, reducing miscommunication and ensuring correct order preparation.

Another objective is to support faster training of new employees. Since the system is designed to be user-friendly and intuitive, new staff members can quickly learn how to use it, reducing training time and increasing workforce efficiency.

The system also seeks to improve consistency in pricing and billing processes. By storing standardized menu prices within the system, it ensures that all transactions follow the same pricing structure, eliminating inconsistencies in manual price handling.

Enhancing operational transparency for management is also an important objective. The system allows administrators to monitor real-time activities, ensuring that all transactions and inventory changes are properly recorded and supervised.

The system further aims to reduce the risk of data loss commonly associated with manual record-keeping. By storing all information in a centralized database, the system ensures that records are securely maintained and can be retrieved whenever needed.

Another important objective of the system is to reduce the dependency on manual documentation within the bistro. By digitizing records such as orders, inventory logs, and sales transactions, the system ensures that information is stored accurately and can be retrieved easily whenever needed, minimizing paperwork and human error.

The system also aims to improve workflow efficiency among employees by integrating different business processes into a single platform. This integration reduces the need to switch between multiple manual records and allows staff to perform tasks in a more organized and systematic manner.

Another objective is to enhance data accuracy across all operational activities. By automating computations and record-keeping, the system minimizes errors commonly associated with manual entry, such as incorrect totals, missing items, or duplicated records.

The system is also designed to support better time management for both staff and management. With faster processing of orders and automated billing, employees can serve customers more quickly, while administrators can focus on decision-making rather than manual checking of records.

Improving communication between different roles within the bistro is another key objective. The system ensures that information such as orders and inventory updates is instantly shared across relevant modules, reducing delays caused by miscommunication or lack of coordination.

The system further aims to enhance financial tracking and accountability. By recording all transactions in a structured database, the system allows administrators to monitor daily sales, track revenue trends, and ensure that all financial data is properly accounted for.

Another objective is to minimize operational errors caused by repetitive manual tasks. Automation reduces the likelihood of mistakes in ordering, billing, and inventory updates, thereby improving overall service quality and consistency.

The system also aims to provide a scalable solution that can accommodate future business growth. As the bistro expands, the system can be upgraded with additional features or increased capacity without requiring a complete redesign of the platform.

In addition, the system is intended to improve customer satisfaction by ensuring that orders are processed accurately and delivered promptly. Faster service and fewer errors contribute to a better dining experience and increased customer loyalty.

Overall, these additional specific objectives reinforce the goal of modernizing the operations of Groove Monkey Pinoy Korean Bistro. They emphasize efficiency, accuracy, automation, and scalability, ensuring that the system provides long-term benefits for both the business and its customers.

1.4 Scope and Delimitation

This study focuses on developing a web-based management system for Groove Monkey Pinoy Korean Bistro. The system will handle ordering, billing, inventory management, and report generation.

The system will be limited to the internal operations of the bistro. It will not include advanced features like online delivery integration, mobile app development, or external payment gateways.

The target users of the system are the administrators, staff, and cashiers of the bistro.

The system will be developed as a web-based platform that can be accessed through computers or tablets within the establishment.

The scope of the study includes the design, development, and implementation of a centralized web-based management system that supports the daily operational activities of Groove Monkey Pinoy Korean Bistro. The system is intended to improve the efficiency of business processes by providing digital solutions for order management, billing, inventory monitoring, and report generation.

The ordering module of the system will allow staff and cashiers to record customer orders electronically, reducing the need for handwritten records and minimizing order-related errors. This feature is expected to improve the accuracy of transactions and facilitate faster communication between service personnel and kitchen staff.

The billing component of the system will focus on calculating customer purchases and generating accurate transaction records. It will assist cashiers in processing orders efficiently while maintaining organized records of sales activities. This functionality will help reduce calculation errors and improve transaction management.

The inventory management module will be designed to track available stock levels, monitor inventory movement, and provide updated information regarding ingredients and supplies. Through this feature, management can maintain better control over inventory resources and make informed purchasing decisions based on actual stock requirements.

The report generation feature will enable administrators to access operational and sales reports that summarize business activities over a specified period. These reports will support management in evaluating performance, monitoring trends, and identifying areas that require improvement or further attention.

The study is limited to the operational needs of Groove Monkey Pinoy Korean Bistro and is specifically tailored to its business processes and requirements. The system will not be designed for use by other businesses without further modifications and customization to accommodate different operational structures and workflows.

In terms of user access, the system will incorporate different access levels based on user roles and responsibilities. Administrators will have broader control over system settings and reports, while staff and cashiers will be granted access only to functions necessary for performing their assigned tasks. This limitation helps ensure data security and proper system usage.

The proposed system will operate within the local network or internet environment available within the establishment. While the platform is web-based, its primary purpose is to support internal operations rather than provide services directly to customers through public online access. Customer-facing functionalities are beyond the scope of this study.

The study does not include advanced technologies such as artificial intelligence, predictive analytics, automated supplier ordering, customer loyalty programs, or integration with third-party delivery services. These features may be considered for future enhancements but are not included within the scope of the current system development project.

Furthermore, the evaluation of the system will focus on its effectiveness in improving operational efficiency, inventory accuracy, order processing, and report generation within the bistro. Other aspects such as large-scale deployment, multi-branch management, and enterprise-level integrations are beyond the scope and limitations established for this study.

The proposed system will primarily focus on automating the essential operational processes of Groove Monkey Pinoy Korean Bistro to improve efficiency and accuracy. By concentrating on the restaurant's core business functions, the study aims to provide a practical and manageable solution that directly addresses the identified operational challenges.

The system will maintain a database for storing customer orders, inventory records, sales transactions, and generated reports. This centralized database will serve as the primary repository of information, allowing authorized users to access and manage data in a more organized and systematic manner.

Another aspect within the scope of the study is the management of menu items. Administrators will be able to add, update, and remove menu information as needed. This functionality ensures that the system reflects the restaurant's current offerings and supports accurate order processing and billing activities.

The study will also include user authentication and account management features to ensure that only authorized personnel can access the system. Login credentials and user roles will be implemented to provide appropriate access controls and protect sensitive business information from unauthorized use.

The web-based platform will be designed with a user-friendly interface to facilitate ease of use among administrators, staff, and cashiers.

The system will prioritize functionality and usability to ensure that users can perform their tasks efficiently without requiring extensive technical knowledge or training.

The system will support the recording and monitoring of daily sales transactions, enabling management to maintain accurate financial records. These transaction records can be used to generate summaries and reports that assist in evaluating business performance and operational effectiveness.

While the system can be accessed through computers and tablets within the establishment, it is not intended to function as a dedicated mobile application. The development of Android, iOS, or cross-platform mobile applications is beyond the scope of this study and may be considered as a future enhancement.

The study is also limited to the testing and implementation of the system within the operational environment of Groove Monkey Pinoy Korean Bistro. The performance of the system in other restaurant settings, industries, or business environments will not be evaluated as part of this research.

Additionally, the system will not include advanced customer relationship management features such as customer profiling, rewards programs, marketing automation, or personalized recommendations. The primary focus remains on improving internal operations rather than managing customer engagement activities.

Finally, the scope of this study is confined to the development, implementation, and evaluation of the proposed web-based management system. Factors beyond the control of the system, such as employee performance, market competition, customer preferences, economic conditions, and external business risks, are recognized but are not directly addressed within the boundaries of this research.

The proposed web-based management system will focus on providing a structured and efficient platform for managing the daily operations of Groove Monkey Pinoy Korean Bistro. The system is intended to support operational activities by replacing manual procedures with digital processes that improve accuracy, efficiency, and data accessibility.

Within the scope of the study, the system will allow authorized users to manage and monitor customer orders from order placement to transaction completion. This functionality will help ensure that order information is accurately recorded and easily accessible for operational and reporting purposes.

The system will also provide inventory monitoring capabilities that enable users to track stock availability and inventory movement. These features are designed to support better inventory control and reduce the occurrence of stock shortages, overstocking, and inventory-related inefficiencies.

Another component covered by the study is the generation of operational reports. The system will be capable of producing reports related to sales transactions, inventory status, and business activities. These reports will assist management in reviewing performance and making informed operational decisions.

The study includes the implementation of role-based access control to ensure that users can only access system functions relevant to their responsibilities. This feature enhances data security and promotes accountability by restricting unauthorized access to sensitive business information.

The system will be designed to operate using commonly available web technologies and standard internet browsers. Users will not be required to install specialized software to access the platform, making it more convenient to deploy and maintain within the establishment.

Although the system is web-based, its functionality is limited to supporting the internal operations of the restaurant. Customer-facing services such as online table reservations, public online ordering, and direct customer account management are not included within the scope of the current study.

Furthermore, the proposed system will not integrate with external accounting software, banking systems, or third-party payment processors. Financial transactions will be recorded within the system for monitoring and reporting purposes, but advanced financial integrations are beyond the scope of this research.

The study also excludes the implementation of advanced analytical technologies such as machine learning, predictive inventory forecasting, and business intelligence dashboards. While these technologies may provide additional benefits, they are not necessary for achieving the primary objectives of the proposed system.

Ultimately, the scope and limitations established in this study are intended to maintain a clear focus on the development of a practical and effective web-based management system for Groove Monkey Pinoy Korean Bistro. By concentrating on essential operational functions, the study seeks to provide a solution that is feasible, relevant, and capable of addressing the restaurant’s immediate management needs while remaining within the project's available resources and timeframe.

The scope of the proposed system includes the automation of routine operational tasks that are currently performed manually within the restaurant. By digitizing these processes, the system aims to reduce paperwork, improve efficiency, and provide a more organized approach to managing daily business activities.

The system will provide a centralized environment where authorized users can access relevant operational information. This centralized approach will help ensure consistency in data management and reduce the likelihood of information duplication or loss that may occur with manual record-keeping practices.

Another area covered by the study is the management of transaction histories. The system will maintain records of completed orders and sales transactions, allowing management to review historical data when necessary. These records can serve as valuable references for evaluating business performance and monitoring operational trends.

The proposed system will support inventory monitoring by recording stock additions, deductions, and updates. This functionality will provide users with better visibility into inventory movements and help management maintain adequate stock levels to support uninterrupted restaurant operations.

The study also includes the development of reporting features that can generate summaries of daily, weekly, or monthly business activities. These reports are intended to assist management in tracking operational performance and identifying areas where improvements may be required.

To ensure usability, the system will be designed with simple navigation and intuitive interfaces that accommodate users with varying levels of technical expertise. The focus will be on creating a practical and accessible platform that can be easily adopted by restaurant personnel.

The system's implementation is limited to a single establishment, specifically Groove Monkey Pinoy Korean Bistro. The study will not address the requirements of multi-branch operations, franchise management, or centralized control of multiple restaurant locations, as these features fall outside the project's intended scope.

In terms of hardware requirements, the system is expected to operate using existing computers, laptops, or tablets available within the establishment. The study does not include the acquisition, installation, or evaluation of specialized hardware such as selfservice kiosks, biometric devices, or advanced point-of-sale terminals.

The project is also limited to the management of restaurant operations and does not include functionalities related to human resource management, payroll processing, employee attendance tracking, or recruitment activities. These areas are considered separate business functions and are beyond the scope of the proposed system.

Finally, while the proposed web-based management system is expected to improve operational efficiency and business management, the study acknowledges that its effectiveness depends on proper implementation, user adoption, and regular system maintenance. Factors such as user training, organizational policies, and management practices may influence the overall success of the system but are not directly addressed within the scope of this research.

1.5 Significance of the Study

This study will benefit several groups:

- o Administrators – It will help them monitor sales, inventory, and overall business performance more efficiently.
- o End Users (Staff and Cashiers) – It will simplify their tasks by automating ordering, billing, and inventory processes.
- o Riders - Updates the specified order assigned to them.
- o Organization (Groove Monkey Pinoy Korean Bistro) – It will improve operational efficiency, reduce errors, and boost customer satisfaction.

- Researchers – This study will serve as a reference for future research on restaurant management systems.
- Future Developers – It can guide the development of similar systems with better features and functions.

The significance of this study extends beyond the immediate improvement of operational processes within Groove Monkey Pinoy Korean Bistro, as it also contributes to the broader understanding of how digital systems can transform small and medium-sized food service businesses. By implementing a web-based management system, the study demonstrates how technology can be effectively utilized to address common operational challenges.

For administrators, the system provides a centralized platform where they can easily monitor daily transactions, track inventory levels, and evaluate overall business performance. This real-time access to information enables more informed decision-making and allows management to respond quickly to operational issues such as stock shortages or sales fluctuations.

The system also benefits end users, particularly staff and cashiers, by reducing the complexity of manual processes. Automated ordering and billing functions minimize the need for handwritten records, thereby reducing human error and improving service speed. This allows employees to focus more on customer service rather than administrative tasks.

In addition, riders play an important role in the delivery and order fulfillment process. Through the system, they are provided with updated order assignments, which helps ensure accurate and timely delivery of customer orders. This improves coordination between kitchen staff and delivery personnel, leading to better service efficiency.

For the organization as a whole, Groove Monkey Pinoy Korean Bistro benefits significantly from improved operational efficiency and data accuracy. The integration of key business processes into a single system reduces redundancy, minimizes errors, and enhances workflow coordination. As a result, the business is able to operate more smoothly and effectively.

Customer satisfaction is also indirectly improved through the implementation of the system. Faster order processing, accurate billing, and improved inventory management contribute to better service quality. Satisfied customers are more likely to return, which can lead to increased sales and business growth.

From an academic perspective, researchers benefit from this study as it adds to the growing body of knowledge on web-based restaurant management systems. It provides a detailed example of system development, implementation, and evaluation, which can be used as a reference for similar future studies.

Future developers can also utilize this study as a foundation for creating more advanced systems. The methodologies, features, and evaluation approaches presented in this research can serve as a guide for improving system design and incorporating additional functionalities such as mobile integration or advanced analytics.

The study also contributes to the understanding of how automation can improve small business operations. By demonstrating the practical application of a web-based system, it highlights the importance of digital transformation in enhancing productivity and competitiveness in the food service industry.

The significance of the study also extends to improving data accuracy within the organization. By shifting from manual processes to a centralized web-based system, the likelihood of human error in recording sales, inventory, and transactions is significantly reduced. This ensures that the business operates using reliable and up-to-date information.

Another important benefit is improved decision-making for management. With access to real-time reports and consolidated data, administrators can analyze sales trends, monitor product performance, and identify operational issues more effectively. This supports faster and more strategic business decisions.

The system also enhances communication within the organization. Since all operational data is stored in a centralized platform, staff members can easily access updated information regarding orders, inventory status, and billing details. This reduces miscommunication and ensures smoother coordination between different roles.

In terms of efficiency, the study highlights how automation reduces workload for employees. Tasks that were previously done manually, such as computing totals or updating inventory records, are now handled automatically by the system. This allows employees to focus more on customer service and other important responsibilities.

The study is also significant in promoting digital transformation among small and medium-sized food businesses. It demonstrates how even simple web-based systems can greatly improve operational workflows. This may encourage other similar establishments to adopt technologybased solutions in their operations.

For customers, the system indirectly enhances the overall dining experience. Faster service, accurate order processing, and improved billing systems contribute to higher satisfaction levels. This can lead to increased customer loyalty and positive word-of-mouth promotion for the business.

The research also supports the development of standardized processes within the bistro. By digitizing operations, the system ensures that tasks are performed consistently and according to defined procedures. This reduces variability in service quality and improves overall business reliability.

In addition, the study contributes to better inventory control and waste reduction. By accurately tracking stock levels, the system helps prevent overstocking and understocking issues. This leads to more efficient resource utilization and cost savings for the organization.

The significance of this study also lies in its contribution to sustainability in business operations. By minimizing paper-based records and manual documentation, the system supports more environmentally friendly practices. This aligns with modern efforts toward digital and sustainable business solutions.

The study is also significant in strengthening accountability within the organization. Since all transactions are recorded in the system, each action performed by users such as staff, cashiers, and administrators can be traced. This promotes responsibility and reduces the likelihood of unauthorized or incorrect operations.

Another contribution of the system is improved transparency in business operations. Managers can easily review transaction histories, inventory movements, and sales activities at any time. This transparency helps ensure that all processes are properly monitored and verified.

The system also supports faster audit and reporting processes. Instead of manually compiling records, administrators can generate reports instantly through the system. This saves time and ensures that reports are more accurate and less prone to human error.

In addition, the study helps reduce operational delays commonly caused by manual processing. By automating key tasks such as ordering and billing, the system minimizes waiting time for both staff and customers, resulting in smoother workflow execution.

The research is also significant in promoting better resource management. With accurate inventory tracking and sales monitoring, the business can plan purchases more effectively. This reduces unnecessary expenses and improves overall profitability.

Another important aspect is the enhancement of staff productivity. Since repetitive and time-consuming tasks are automated, employees can allocate more time to customer interaction and service improvement. This leads to a more efficient and service-oriented workforce.

The system also plays a role in improving data security awareness within the organization. By implementing user authentication and access control, staff become more conscious of proper system usage and the importance of protecting sensitive information.

Furthermore, the study provides a scalable foundation for future system enhancements. As the business grows, additional features such as online ordering or mobile integration can be added without redesigning the entire system. This makes the solution adaptable to future needs.

The significance of the study also includes its contribution to reducing operational stress among employees. With simplified processes and automated workflows, staff experience less workload pressure, allowing them to perform their tasks more efficiently and with fewer errors.

Overall, this study serves as a comprehensive improvement tool for the organization, enhancing efficiency, accuracy, and operational control. It not only addresses current challenges but also prepares Groove Monkey Pinoy Korean Bistro for future technological and business advancements.

1.6 Definition of Terms o Bistro – A small restaurant serving moderately priced meals in a casual setting. o Web-Based System – A system that is accessed through a web browser over a network. o Inventory Management – The process of tracking and managing stock levels of products. o Point of Sale (POS) – A system used to process customer orders and payments.

- Database – An organized collection of data stored electronically.
- Real-Time Reports – Reports generated instantly as data is updated.
- Automation – The use of technology to perform tasks with minimal human involvement.
- User Interface – The visual layout through which users interact with the system.
- Transaction – A recorded business activity such as a sale or order.
- System Administrator – A person responsible for managing and maintaining the system.
- Cashier – An employee responsible for processing customer orders, handling payments, and issuing receipts.
- Staff – Employees who perform various operational tasks within the restaurant, including order-taking, food serving, and inventory handling.
- Order Management – The process of receiving, recording, tracking, and completing customer orders efficiently.
- Billing System – A component of the system that calculates customer purchases and generates transaction records or receipts.
- Stock Level – The quantity of products, ingredients, or supplies currently available in inventory.
- Stock-Out – A situation in which a product or ingredient is unavailable due to insufficient inventory.
- Overstocking – The condition of maintaining excessive inventory beyond the required amount, which may result in spoilage or unnecessary storage costs.
- Sales Report – A document or system-generated summary that contains information about sales transactions within a specific period.
- Inventory Record – A documented list of available stocks, including quantities, additions, and deductions.
- Data Management – The process of collecting, storing, organizing, and maintaining information within a system.

- Authentication – A security process used to verify the identity of a user before granting access to the system.
- Login Credentials – The username and password used by authorized users to access the system.
- Access Control – A security feature that restricts system access based on user roles and permissions. o Administrator – An authorized user with full access to manage system settings, user accounts, inventory records, and reports.
- Report Generation – The process of automatically creating reports based on data stored within the system.
- Digital Record – Information stored electronically within the system for future retrieval and analysis. o Data Accuracy – The correctness and reliability of information stored and processed within the system.
- System User – Any authorized individual who interacts with the system to perform specific tasks or access information.
- Menu Management – The process of adding, updating, organizing, and maintaining menu items within the system.
- Order Processing – The sequence of activities involved in receiving, recording, preparing, and completing customer orders.
- Operational Efficiency – The ability of a business to perform its activities effectively while minimizing time, effort, and resource consumption.
- Web Browser – A software application used to access and interact with web-based systems through the internet or a local network.
- Local Area Network (LAN) – A network that connects devices within a limited area, such as the restaurant premises, allowing access to the system.
- System Maintenance – The process of updating, monitoring, and improving the system to ensure continuous and reliable operation.
- Backup – A copy of system data created to prevent information loss in the event of technical failures or accidental deletion.

- o Dashboard – A visual interface that displays important system information, statistics, and summaries for quick monitoring and decision-making. o User Role – A classification that determines the permissions and responsibilities assigned to a particular system user.

- o Receipt – A document generated after a transaction that serves as proof of purchase and payment.

- o Food Waste – Ingredients or food items that are discarded due to spoilage, expiration, overproduction, or improper inventory management. o Business Performance – The measurement of how effectively a business achieves its operational, financial, and customer service objectives.

The Definition of Terms section is an important part of a research study because it provides clear and precise meanings of key concepts used throughout the paper. Its main purpose is to avoid misunderstanding and misinterpretation of terms that may have different meanings in other contexts or fields. By defining these terms, the researchers ensure that readers share a common understanding of the concepts being discussed.

In system development studies such as a web-based management system for a bistro, technical and operational terms are frequently used. These may include terms related to software, databases, user roles, and business processes. Without proper definitions, readers who are not familiar with these technical concepts may find the study difficult to understand. Therefore, defining these terms helps bridge the gap between technical language and reader comprehension.

Another importance of the Definition of Terms is that it establishes consistency throughout the study. Once a term is defined, it is used in the same meaning in all chapters, especially in the discussion of methodology, system design, and evaluation. This prevents confusion and ensures that the interpretation of results remains accurate and uniform.

The section also helps in improving the academic quality of the research. By clearly defining key terms such as "inventory management," "automation," or "user interface," the study becomes more structured and professional. It shows that the researchers have a clear understanding of the concepts being applied in their system.

Furthermore, the Definition of Terms supports better communication between researchers, readers, and future developers. It allows other researchers to replicate or build upon the study because they can clearly understand how each term was used and applied within the system.

In summary, the Definition of Terms serves as a foundational guide for understanding the entire research study. It ensures clarity, consistency, and accuracy in the presentation of ideas, making the study easier to read, interpret, and apply in real-world system development contexts.

CHAPTER 2: REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Foreign Studies

A research study conducted by Salakham and Talasee (2024) proposed a web-based online restaurant management system for Redbox Restaurant. In the study, a structured development process was used, and the proposed system was tested by the users, which included the employees and customers of the restaurant. The results indicated that the proposed system could enhance the performance of the business, improve the quality of service, and increase user satisfaction. This research study can be used for the current project because it presents a way of improving the performance of a restaurant through a web-based system.
The researchers utilized the System Development Life Cycle (SDLC) in developing the system and evaluated it through user satisfaction surveys involving restaurant owners, general users, and system experts. The findings revealed that the system effectively improved restaurant operations, enhanced service quality, and achieved high user satisfaction. The study demonstrates how web-based restaurant management systems can contribute to operational efficiency and customer satisfaction, making it highly relevant to the proposed system for Groove Monkey Pinoy Korean Bistro.
Another study published in the *International Journal of Gastronomy and Food Science* (2021) examined the operational effects of restaurant management systems. The results showed that restaurants using such systems experienced improved workflow efficiency and better service delivery compared to those using manual processes. This supports the idea that adopting a computerized system is beneficial for restaurant businesses like Groove Monkey Pinoy Korean Bistro.
The research involved 385 restaurant employees and assessed both the positive and negative aspects of RMS implementation. The results showed that restaurants using RMS experienced improvements in operations management, increased sales, and better service standardization. Employees working in restaurants without RMS perceived such systems as difficult due to unfamiliarity, while users of RMS acknowledged significant operational benefits. This study

supports the importance of implementing computerized management systems in restaurant operations.

In addition, Oghenekaro and Okafor (2023) developed a web-based integrated restaurant management system. The study found that manual systems often lead to errors such as inaccurate inventory records and untracked sales, while computerized systems ensure better data handling and faster transactions. This study is directly relevant as it highlights the importance of integrating multiple functions such as ordering, inventory, and reporting into one system.

The researchers implemented the system using PHP, HTML, CSS, and MySQL and adopted a structured system analysis and design methodology. The study concluded that the integrated system improved data management, streamlined restaurant operations, and enabled efficient order processing. This research is directly related to the current study because it integrates ordering, inventory management, and reporting functions into a single platform.

Zamora Saltos and Rodríguez Borges (2024) conducted a study entitled *"Design of an Inventory Management System for a Manabita Restaurant."* The researchers emphasized the importance of effective inventory management in reducing procurement costs, minimizing product spoilage, and preventing stock shortages. Their findings revealed that a properly designed inventory management system helps restaurants maintain adequate stock levels while reducing losses caused by expiration and inventory shrinkage. The study highlights the significance of inventory control in ensuring operational efficiency and customer satisfaction.

Erameh and Odoh (2021) developed a *Web-Based Inventory Control System* to address challenges associated with manual inventory tracking in small and medium-sized enterprises. The study found that manual inventory management often leads to difficulties in monitoring stock levels and meeting customer demand. The implementation of a web-based inventory control system improved stock monitoring, reduced overstocking, and enhanced inventory accuracy. These findings support the need for digital inventory management in restaurant businesses.

A study conducted by Alfiyan, Maulana, Izdihar, and Handayani (2024) focused on the implementation of a web-based food inventory management system using the Just-In-Time (JIT) method in a Korean restaurant. The researchers found that real-time inventory monitoring and automatic stock notifications significantly improved operational efficiency while reducing food waste and storage costs. The study demonstrated that web-based inventory systems help restaurant managers make timely decisions regarding inventory replenishment and resource allocation.

The study by Kocaman (2021) further revealed that restaurant employees perceived RMS as beneficial in standardizing production and service processes. The system helped improve consistency in restaurant operations and enhanced overall management effectiveness. Such findings suggest that the adoption of digital systems contributes not only to operational efficiency but also to the maintenance of service quality standards.

Salakham and Talasee (2024) also highlighted the importance of database-driven web applications in managing restaurant operations. Their system utilized PHP and MySQL technologies to facilitate information management, order processing, and user interactions. The successful implementation of these technologies demonstrates the practicality of web-based solutions for modern restaurant businesses.

According to Oghenekaro and Okafor (2023), one of the primary causes of inefficiency in restaurant operations is the continued reliance on manual processes. Their study found that automated systems reduce operational errors, improve transaction processing, and provide better access to business information. These benefits support the development of integrated management systems capable of addressing multiple operational challenges simultaneously.

Rosnan, Che Ahmat, and Md Nor (2023) conducted a review entitled *"An Overview of the Application of Restaurant Management Systems in the Foodservice Industry."* The study analyzed 31 empirical studies related to the implementation of restaurant management systems across different countries. The researchers found that restaurant management systems significantly improve operational efficiency, order processing, inventory management, customer service, and managerial decision-making. The study concluded that digital systems have become essential tools for modern food service establishments seeking to improve productivity and competitiveness. This study is relevant to the current research because it provides comprehensive evidence regarding the benefits of adopting restaurant management technologies in food service operations.

Setiawan and Nataliani (2025) developed and implemented a web-based inventory information system for Oemah Djari Restaurant. The researchers identified several issues associated with manual inventory management, including stock discrepancies, inefficient monitoring, and time-consuming inventory processes. The developed system improved inventory accuracy, streamlined stock monitoring, and enhanced operational efficiency. The study demonstrates how web-based inventory systems can effectively address common restaurant management challenges.

Sihombing (2024) conducted a study entitled *"Innovating Restaurant Inventory Management Application: Enhancing Quality and Efficiency Through Extreme Programming Approach."* The research focused on addressing inventory calculation errors, stock shortages, and material wastage in restaurants. Using the Extreme Programming (XP) methodology, the developed inventory application received positive evaluations from users, with the majority expressing satisfaction with its ability to improve efficiency and inventory control. The findings highlight the importance of digital inventory systems in restaurant management.

Jingga and Limantara developed a Restaurant Management System model using PHP and MySQL technologies. Their study emphasized the importance of real-time communication among waiters, kitchen personnel, and cashiers. The system facilitated efficient order transmission, billing, and food preparation coordination, reducing reliance on paper-based processes. The findings support the integration of ordering and billing functionalities within a centralized web-based platform.

Research on restaurant inventory management consistently identifies manual inventory monitoring as a major source of operational inefficiency. Studies show that manual systems often result in stock discrepancies, delayed replenishment, and inaccurate records. Automated inventory systems improve stock visibility, facilitate timely purchasing decisions, and reduce inventory-related losses. These findings provide strong support for implementing computerized inventory management solutions in restaurant businesses.

Overall, the reviewed foreign studies demonstrate that web-based restaurant management systems contribute significantly to operational efficiency, inventory accuracy, customer satisfaction, transaction management, and business performance. The consistent findings across multiple studies provide a strong theoretical and practical foundation for the development of a web-based management system for Groove Monkey Pinoy Korean Bistro, particularly in addressing challenges related to ordering, billing, inventory tracking, and report generation.

2.2 Local Studies

A study by Bari et al. (2024) emphasized that restaurant operations involve complex tasks such as staff management, transaction processing, and inventory tracking. The researchers concluded that implementing a restaurant management system significantly reduces workload and improves efficiency in daily operations. This aligns with the current study, which aims to simplify processes in Groove Monkey Pinoy Korean Bistro.

Another study by Galabi et al. (2023) introduced a smart restaurant management system that replaces traditional manual processes. The findings showed that digital systems reduce time consumption and improve service delivery. This supports the need for automation in restaurant environments.
Furthermore, Sudirman et al. (2024) developed a web-based ordering system using QR codes. The study found that the system was effective, user-friendly, and ready for implementation, proving that modern ordering systems enhance customer convenience and operational efficiency. This is relevant to the proposed system as it also aims to modernize ordering processes.

Baylen (2020) conducted a study entitled *"Analysis of Inventory Management Systems of Selected Small-Sized Restaurants in Quezon Province: Basis for an Inventory System Manual"* published in the *Journal of Business and Management Studies*. The study examined the inventory management practices of small restaurants in Quezon Province and found that ineffective inventory management contributed to reduced profitability and operational inefficiencies. The researcher recommended the implementation of a structured inventory management system to improve stock monitoring and business performance. This study is relevant to the current research because it highlights the importance of proper inventory control in restaurant operations.

Banas, Laguna, Paccial, Ramirez, and Susvilla (2017) developed a *WebBased Management System for Resona Catering Services and Event Planning* at the Central Philippine University. The system was designed to manage reservations, customer orders, inventory, payments, and reports through a centralized web platform. The researchers found that the system improved operational efficiency and provided faster access to business information. This study is related to the proposed system because it demonstrates the effectiveness of web-based management solutions in food-related service businesses.

Furtos (2026) conducted a study entitled *"Brewing Efficiency: An Assessment of Inventory Management Systems in Cebu City Cafés"*. The research examined inventory management practices among cafés in Cebu City and analyzed their relationship with organizational performance. The findings revealed that effective inventory management contributes to improved operational efficiency, cost control, and overall business performance. The study supports the inclusion of inventory management features in the proposed system.

Cejas (2025) conducted a study on the application of artificial intelligence technologies for food waste reduction in restaurant management in the Philippines. The research emphasized that proper inventory monitoring and technology-assisted management help minimize food waste and improve operational decision-making. Although focused on AI applications, the study highlights the importance of digital systems in managing restaurant inventory and resources effectively.

Grepon (2021) developed a *Web-Based Case Docket Information System* for a Regional Trial Court in Northern Mindanao. Although conducted in the judicial sector, the study demonstrated how web-based information systems improve efficiency, reliability, and data management compared to manual processes. The findings support the use of web technologies in improving organizational operations and information management.

Dela Rosa (2022) developed a *Web-Based Management Information System of Cases Filed with the National Labor Relations Commission*. The study found that manual record management often resulted in inconsistent and inaccurate information. After implementing the web-based system, operational processes became more efficient, and data management improved significantly. The system received high evaluation scores from both experts and end-users, demonstrating the effectiveness of web-based management systems in improving organizational efficiency.

Salvador, Botangen, Rabang, Salinas, Naagas, and Balagot (2024) developed a *Web-Based Research Consortium Database Management System* for the Central Luzon Agriculture, Aquatic and Natural Resources Research and Development Consortium (CLAARRDEC). The study highlighted the importance of centralized databases, real-time monitoring, and automated reporting in improving organizational performance. The researchers concluded that web-based systems enhance information sharing, support decision-making, and improve operational management. These findings are applicable to the proposed system because it also aims to provide centralized data management and real-time reporting capabilities.

A study by Peralta, Freo, Manalo, and Ocfemia (2015) entitled *"WebBased Purchasing Operations and Sales and Inventory System with Quality Evaluation and Various Modes of Transactions for Best Deals Laguna"* developed and evaluated a web-based sales and inventory system for a business establishment in Laguna. The researchers found that the system was functional, reliable, efficient, and beneficial to both customers and business owners. The study demonstrated how web-based technologies can improve inventory management and transaction processing, making it relevant to the proposed system for Groove Monkey Pinoy Korean Bistro.

Sison, Oreiro, Camalit, and Ng (2019) conducted a study entitled *"A Problem Oriented Approach to Implementing an Inventory and PointofSale System for Company KCP."* The study identified problems such as slow transaction processing, inventory inaccuracies, product shortages, overstocking, and data redundancy resulting from manual operations. The proposed automated POS and inventory management system successfully addressed these issues and improved overall business processes. The findings support the integration of POS and inventory modules in the proposed restaurant management system.

Bonecillo, Dorado, Mabalot, and Malaborbor (2016) developed *"EQUIPORT: An Online Equipment Requisition and Inventory System with Android App Support for the School of Tourism and Hospitality Management."* The study found that the automated system improved usability, accuracy, efficiency, and security compared to the previous manual process. The researchers concluded that online inventory systems contribute significantly to operational improvement and information management.

An earlier Philippine study entitled *"An Enhancement of the Existing Food Costing System of a Restaurant in Los Baños, Laguna"* examined inventory and food costing practices within a restaurant. The researchers found weaknesses in inventory recording, purchasing procedures, and material requisition processes. The study proposed a structured inventory and cost accounting system that improved monitoring and control of restaurant resources. This study is particularly relevant because it directly addresses restaurant inventory and operational management.

Research in Philippine information systems consistently demonstrates that organizations relying on manual record-keeping experience challenges in data retrieval, transaction monitoring, and report preparation. Studies involving web-based management systems show that automation improves data accuracy, reduces processing time, and enhances organizational productivity. These findings further justify the adoption of a web-based solution for Groove Monkey Pinoy Korean Bistro.

Overall, the reviewed local studies indicate that web-based management systems, inventory systems, and POS applications significantly improve operational efficiency, inventory accuracy, reporting capabilities, and business management. These studies provide strong local evidence supporting the development of a Web-Based Management System for Groove Monkey Pinoy Korean Bistro to address challenges related to ordering, billing, inventory management, and report generation.

2.3 Related Literature

According to Warlina and Noersidik (2018), web-based food ordering systems allow customers to place orders efficiently while enabling restaurants to manage transactions digitally. These systems improve communication between customers and staff and enhance service speed. Their study emphasized that such systems improve ordering efficiency, facilitate communication between customers and restaurants, and reduce service delays. The researchers concluded that web-based ordering applications provide convenience to customers while improving operational processes within restaurants.

Golovko, Borozdin, and Tokar (2021) highlighted the importance of automation and information systems in restaurant management. They argued that modern restaurants require information systems to support decision-making, improve operational efficiency, and maintain competitiveness in the hospitality industry. The authors emphasized that automation allows restaurants to streamline workflows and enhance overall business performance.

Kocaman (2021) stated that Restaurant Management Systems (RMS) play a vital role in improving operational efficiency, standardizing service delivery, and increasing productivity. The literature emphasized that digital restaurant systems provide advantages in transaction management, service coordination, and operational control, particularly in establishments with complex business operations.

Alt (2021) discussed the digital transformation occurring within the restaurant industry. According to the author, modern restaurants increasingly rely on digital technologies to improve customer experiences, automate business processes, and integrate front-ofhouse and back-of-house operations. The study emphasized that digital transformation enables restaurants to provide more personalized services while improving operational efficiency.

Nilsson, Pers, and Grubbström (2021) examined the use of self-service technologies in casual dining restaurants. Their findings suggested that technology-based ordering and service systems improve customer convenience, reduce waiting times, and enhance service quality. The authors noted that customers generally perceive technology-assisted restaurant services as efficient and user-friendly.

A systematic review conducted on information and communication technologies (ICTs) in food services and restaurants emphasized the growing role of digital technologies in hospitality operations. The review analyzed 165 scholarly articles and concluded that ICTs significantly contribute to customer service improvement, operational efficiency, inventory control, and business management. The literature further highlighted the importance of integrating technology into restaurant operations to remain competitive in a rapidly evolving industry.

Roy, Spiliotopoulou, and de Vries (2022) discussed the emergence of restaurant analytics as an important aspect of restaurant management. According to the authors, data-driven technologies provide valuable insights into customer behavior, sales performance, inventory utilization, and operational efficiency. The literature emphasized that access to accurate and timely information supports strategic decision-making and enhances business performance.

Santos and Guimarães (2023) conducted a systematic review on the use of information technology in restaurant management. Their findings revealed that information technology has become an essential component of restaurant operations, influencing inventory management, customer service, sales monitoring, and organizational decision-making. The study concluded that technology adoption contributes significantly to operational efficiency and long-term business sustainability.

Mussa, Yuliana, Ismiyar, Tjahjadi, and Munandar (2023) emphasized the importance of accounting and information systems in restaurant operations. Their literature review found that computerized systems improve financial reporting, transaction management, and managerial control. The authors noted that systems utilizing databases and web technologies help organizations maintain accurate records and support effective business management.

A recent systematic literature review on web technology in food ordering systems highlighted that modern food service establishments increasingly adopt web-based platforms due to their ability to provide real-time transactions, centralized information management, and enhanced customer convenience. The review concluded that web technologies improve service efficiency, support business scalability, and strengthen the relationship between customers and restaurant operators.

In the field of information systems, studies emphasize that digital systems are essential in managing, processing, and storing large volumes of business data. Restaurant management systems, in particular, help automate operations such as billing, inventory tracking, and reporting.
Moreover, modern restaurant systems are developed using web technologies and databases, allowing centralized access and real-time updates. These technologies ensure accuracy, accessibility, and scalability, making them suitable for small and medium-sized businesses.

2.4 Synthesis of the Reviewed Literature

The reviewed studies consistently show that restaurant management systems play a crucial role in improving operational efficiency, reducing human error, and enhancing customer satisfaction. Foreign and local studies both highlight the limitations of manual processes, such as inaccurate data recording, slow service, and difficulty in managing inventory.

The literature also emphasizes the importance of web-based systems in providing real-time data access, centralized databases, and improved service delivery. Despite these advancements, many small restaurants still rely on traditional methods, indicating a gap in the adoption of modern technology.

The gap identified is the lack of customized and affordable systems specifically designed for small-scale restaurants like Groove Monkey Pinoy Korean Bistro. Many existing systems are either too complex or not tailored to the needs of small businesses.
The current study addresses this gap by developing a user-friendly, web-based management system that integrates ordering, billing, inventory, and reporting. This system aims to improve efficiency, ensure data accuracy, and enhance overall business performance.

The reviewed literature and studies demonstrate that technological innovation has become an essential component of modern restaurant operations. As customer expectations continue to evolve, businesses are increasingly adopting digital solutions to improve service quality, operational efficiency, and overall customer satisfaction. This trend highlights the growing importance of information systems in supporting the sustainability and competitiveness of food service establishments.

Both foreign and local studies emphasize that manual operational processes often result in inefficiencies that negatively affect business performance. Common issues include inaccurate order processing, inventory discrepancies, delayed transactions, and difficulties in generating reports. These challenges can lead to customer dissatisfaction, increased operational costs, and reduced profitability.

The literature further reveals that integrated management systems provide a practical solution to many of these operational problems. By combining ordering, billing, inventory management, and reporting functions into a single platform, businesses can streamline workflows and reduce the likelihood of errors caused by fragmented or manual processes.

Several studies also highlight the importance of real-time information in business management. Access to updated sales records, inventory levels, and transaction data enables managers to make informed decisions quickly and accurately. This capability is particularly important in the food service industry, where timely decision-making directly affects customer service and operational efficiency.

Another significant finding from the reviewed literature is the role of centralized databases in improving information management. Centralized systems allow businesses to store, organize, and retrieve data more efficiently than traditional paper-based methods. As a result, organizations can maintain accurate records and reduce the risk of data loss or duplication.

The studies consistently indicate that inventory management is one of the most critical aspects of restaurant operations. Effective inventory monitoring helps prevent stock shortages, reduce food waste, and optimize purchasing decisions. The integration of inventory management features within a web-based system can therefore contribute significantly to improved resource utilization and cost control.

Customer satisfaction emerged as another recurring theme throughout the reviewed literature. Researchers found that digital systems improve service speed, order accuracy, and transaction convenience, all of which positively influence customer experiences. Satisfied customers are more likely to return and recommend the business to others, contributing to long-term business growth.

The reviewed studies also suggest that web-based systems offer greater flexibility and accessibility compared to traditional standalone applications. Authorized users can access system information through web browsers, allowing for more convenient management of business operations. This accessibility makes web-based solutions particularly suitable for small and medium-sized enterprises.

Furthermore, the literature indicates that adopting technology can enhance organizational transparency and accountability. Automated transaction records, inventory logs, and system-generated reports provide clear documentation of business activities, making it easier to monitor performance and identify areas for improvement.

Overall, the synthesis of the reviewed literature and studies establishes a strong foundation for the development of the proposed web-based management system for Groove Monkey Pinoy Korean Bistro. The findings consistently support the use of integrated digital solutions to address operational inefficiencies, improve customer service, strengthen inventory control, and facilitate better business decision-making. These insights validate the relevance and necessity of the current study in addressing the specific needs of the restaurant.

The reviewed literature indicates that the successful implementation of information systems depends not only on technological capabilities but also on their alignment with the operational needs of the organization. Systems that are specifically designed according to the requirements of a business tend to produce better results in terms of efficiency, usability, and user satisfaction. This highlights the importance of developing a customized solution that addresses the unique operational requirements of Groove Monkey Pinoy Korean Bistro.

The studies also reveal that small and medium-sized enterprises often face challenges in adopting advanced technologies due to financial constraints and limited technical expertise. As a result, many businesses continue to rely on manual processes despite the availability of modern digital solutions. This situation creates an opportunity for affordable and user-friendly systems that can facilitate digital transformation among smaller businesses.

Another common finding among the reviewed studies is the importance of system usability. Researchers emphasize that a management system should be easy to learn and operate to encourage user acceptance and maximize its effectiveness. A complex or difficult-to-use system may discourage employees from utilizing its features fully, thereby reducing its potential benefits.

The literature further suggests that automation significantly reduces the occurrence of repetitive administrative tasks. Activities such as recording transactions, updating inventory records, and generating reports can be performed more efficiently through automated processes. This allows employees to focus on customer service and other valueadding activities that contribute directly to business success.

Many researchers also highlight the role of reporting tools in supporting organizational decision-making. Automated reports provide managers with timely and accurate information regarding sales performance, inventory status, and operational activities. Such information serves as a valuable basis for strategic planning, performance evaluation, and continuous improvement efforts.

The reviewed studies consistently demonstrate that data accuracy is a critical factor in effective business management. Inaccurate records can lead to poor decision-making, financial losses, and operational disruptions. By maintaining a centralized database and automating data entry processes, organizations can improve the reliability and consistency of their information.

Security and access control were also identified as important considerations in the development of information systems. The literature emphasizes the need to protect business data from unauthorized access while ensuring that authorized users can perform their responsibilities efficiently. Role-based access controls and authentication mechanisms are commonly recommended to enhance system security.

Another important theme found in the literature is the scalability of web-based systems. As businesses grow and transaction volumes increase, management systems must be capable of accommodating additional users, data, and operational requirements. Web-based technologies provide a flexible foundation that can support future expansion and feature enhancements.

The reviewed studies further indicate that technology adoption contributes to a more competitive business environment. Restaurants that utilize modern management systems are better positioned to respond to customer demands, monitor operational performance, and adapt to changing market conditions. Consequently, digital transformation has become an important factor in achieving long-term business sustainability.

In summary, the collective findings from the reviewed literature and studies reinforce the significance of developing an integrated webbased management system for Groove Monkey Pinoy Korean Bistro. The proposed system aligns with established research findings regarding automation, inventory control, data management, customer service improvement, and decision support. By addressing the operational challenges identified in the literature, the system has the potential to enhance efficiency, productivity, and overall business performance.

The literature reviewed in this study demonstrates that information technology has become an indispensable tool in modern business operations. Across various industries, organizations have adopted computerized systems to improve efficiency, reduce operational costs, and enhance service delivery. The restaurant industry is no exception, as businesses increasingly recognize the value of digital solutions in managing daily activities and maintaining competitiveness.

A recurring theme in both foreign and local studies is the need for accurate and timely information. Business managers rely on data to make decisions regarding inventory replenishment, pricing strategies, staffing requirements, and service improvements. Manual methods often fail to provide real-time information, whereas web-based systems allow managers to access updated data whenever needed.

The reviewed literature also highlights the significance of process integration in improving organizational performance. Businesses that use separate systems or manual procedures for different operational functions often experience delays and inconsistencies. Integrated systems eliminate these issues by allowing information to flow seamlessly between ordering, billing, inventory management, and reporting modules.

Researchers have also emphasized the role of digital systems in reducing operational risks. Human errors in recording transactions, monitoring stock levels, or preparing reports can result in financial losses and customer dissatisfaction. Automated systems help minimize these risks by standardizing processes and ensuring greater consistency in data handling.

Several studies noted that restaurants operate in a fast-paced environment where service speed is a critical factor in customer satisfaction. Delays in order processing and billing can negatively affect the dining experience. By automating these functions, restaurants can improve response times, increase service efficiency, and better meet customer expectations.

The literature further indicates that inventory management directly influences a restaurant’s profitability. Poor inventory practices can lead to excessive waste, spoilage, and lost sales opportunities. Effective inventory monitoring systems help businesses maintain optimal stock levels, ensuring that resources are used efficiently while minimizing unnecessary expenses.

Another important observation from the reviewed studies is the growing expectation of customers for technology-enabled services. Modern consumers increasingly value convenience, accuracy, and speed when interacting with businesses. Restaurants that utilize digital systems are better able to meet these expectations and provide a more satisfying customer experience.

The studies also reveal that computerized management systems improve transparency and accountability within organizations. By maintaining detailed records of transactions and inventory activities, businesses can monitor performance more effectively and identify areas requiring improvement. This level of transparency supports stronger internal controls and better management practices.

Furthermore, the literature suggests that web-based systems offer long-term advantages because of their flexibility and accessibility. Unlike traditional desktop applications, web-based platforms can be accessed from multiple devices and locations, making them suitable for businesses seeking efficient and scalable solutions. This flexibility contributes to improved operational management and organizational adaptability.

Overall, the reviewed literature and studies provide substantial evidence that integrated web-based management systems are effective tools for addressing common operational challenges in the restaurant industry. The findings consistently support the implementation of digital solutions to improve efficiency, data accuracy, customer satisfaction, inventory control, and decision-making. These conclusions reinforce the rationale for developing a web-based management system tailored to the needs of Groove Monkey Pinoy Korean Bistro.

The reviewed literature further demonstrates that technological advancements continue to reshape the way businesses operate and interact with customers. As organizations increasingly embrace digital transformation, the use of computerized management systems has become a strategic necessity rather than merely an operational option. Restaurants that fail to adopt modern technologies may find it difficult to keep pace with competitors that utilize efficient and data-driven solutions.

Many studies emphasize that operational efficiency is closely linked to the quality of information available to management. Accurate and timely information enables restaurant owners and managers to identify trends, evaluate performance, and make informed decisions. In contrast, inaccurate or delayed information can result in poor planning, inefficient resource allocation, and reduced profitability.

The literature also highlights the importance of workflow optimization in restaurant operations. By automating repetitive tasks and reducing manual intervention, businesses can simplify procedures and improve coordination among employees. This contributes to smoother operations, reduced processing times, and enhanced productivity throughout the organization.

Another significant insight from the reviewed studies is the role of technology in improving communication within the workplace. Integrated systems facilitate the sharing of information between different departments, such as the kitchen, cashiering area, and management office. Improved communication helps prevent misunderstandings, minimizes operational delays, and promotes more efficient service delivery.

Researchers have also identified cost reduction as one of the primary benefits of implementing computerized management systems. Automated processes reduce the need for excessive paperwork, minimize inventory losses, and decrease the time spent on administrative tasks. These savings can contribute to improved financial performance and increased business sustainability.

The reviewed literature further suggests that businesses that adopt information systems are better equipped to monitor key performance indicators. Metrics such as sales volume, inventory turnover, customer demand patterns, and transaction frequency can be tracked more effectively through digital platforms. Access to these performance indicators enables managers to identify opportunities for growth and improvement.

Another important finding is that web-based systems promote consistency in business operations. Standardized procedures ensure that tasks are performed uniformly regardless of the employee handling them. This consistency enhances service quality, reduces operational errors, and contributes to a more professional customer experience.

Several studies also discuss the importance of adaptability in today's business environment. Market conditions, customer preferences, and operational requirements can change rapidly. Web-based management systems provide the flexibility needed to respond to these changes by allowing updates, modifications, and improvements without significant disruptions to daily operations.

The literature indicates that user acceptance is a critical factor in the successful implementation of any information system. Systems that are intuitive, user-friendly, and aligned with organizational needs are more likely to be embraced by employees. Therefore, the design of the proposed system should prioritize ease of use to encourage adoption and maximize its benefits.

In conclusion, the synthesis of the reviewed literature reveals a strong consensus regarding the benefits of web-based management systems in restaurant operations. The studies consistently support the use of integrated digital solutions to improve efficiency, accuracy, communication, inventory control, reporting, and customer service. These findings further validate the development of a webbased management system for Groove Monkey Pinoy Korean Bistro as an effective approach to addressing its operational challenges and supporting long-term business success.

CHAPTER 3: METHODOLOGY

3.1 Research Design

This study utilized a developmental research design as it focused on the design and development of a web-based management system for Groove Monkey Pinoy Korean Bistro. Developmental research is appropriate because it involves creating, testing, and improving a system to address existing operational problems.

Additionally, the study incorporated descriptive research methods to gather information about the current system used by the bistro. This includes identifying issues such as manual ordering, limited online presence, and inaccurate inventory tracking. These insights served as the basis for system development.

The developmental research design was selected because it provides a systematic approach to the creation of technological solutions intended to address specific organizational problems. Through this design, the researchers were able to identify operational challenges, develop appropriate system features, and evaluate the effectiveness of the proposed solution. This approach ensured that the resulting system was tailored to the actual needs of Groove Monkey Pinoy Korean Bistro.

Developmental research emphasizes the continuous improvement of a product or system through planning, design, implementation, testing, and evaluation. In this study, each stage of development was carefully carried out to ensure that the web-based management system would effectively address the identified issues within the bistro's operations. Feedback gathered during the development process was also considered in refining the system.

The descriptive aspect of the research played an important role in understanding the current operational practices of the business. By examining existing procedures and workflows, the researchers were able to identify weaknesses and inefficiencies that affect daily operations. This information provided a clear basis for determining the functional requirements of the proposed system.

To gather relevant data, the researchers conducted observations and interviews with the individuals directly involved in the business operations. These methods allowed the researchers to obtain firsthand information regarding the challenges encountered in ordering, billing, inventory monitoring, and report preparation. The collected data served as valuable input in the design and development process.

The use of a developmental research design also enabled the researchers to evaluate whether the proposed system could effectively improve operational efficiency. Through system testing and user feedback, the researchers assessed the performance, usability, and reliability of the developed application. This evaluation ensured that the system met the expectations and requirements of its intended users.

Furthermore, the study focused on developing a practical and userfriendly solution that could easily be adopted by the employees of the bistro. Since the target users may have varying levels of technological proficiency, the system was designed with simplicity and ease of use in mind. This consideration aimed to encourage user acceptance and minimize the learning curve associated with system implementation.

The research design also supported the integration of multiple business functions into a single platform. Instead of addressing individual operational issues separately, the study sought to create a comprehensive system that combines ordering, billing, inventory management, and report generation. This integrated approach promotes consistency, efficiency, and better information management.

Another important characteristic of developmental research is its focus on solving real-world problems. The study was not limited to theoretical analysis but extended to the actual development and testing of a functional web-based management system. As a result, the research generated a tangible output that can be utilized directly by the business to improve its operations.

The researchers also adopted a structured development process to ensure that the system was developed systematically and efficiently. Each phase of development was carefully documented to maintain consistency and provide a clear record of the procedures undertaken throughout the project. This approach contributed to the reliability and quality of the final system.

The research design adopted in this study provided a structured framework that guided the entire development process of the proposed system. By following a systematic approach, the researchers ensured that all stages of the project, from data gathering to system evaluation, were conducted in an organized and logical manner. This helped maintain the quality and reliability of the research outcomes.
In developing the system, the researchers focused on identifying the specific operational requirements of Groove Monkey Pinoy Korean Bistro. Understanding these requirements was essential to ensure that the features and functionalities of the system aligned with the actual needs of the business. This user-centered approach increased the relevance and practicality of the proposed solution.

The developmental research design also encouraged continuous assessment throughout the project. Instead of waiting until the completion of the system, the researchers evaluated each component during development. This allowed potential issues to be identified and corrected early, reducing the likelihood of major problems during implementation.

Another advantage of the chosen research design is its ability to bridge the gap between theory and practice. The concepts and principles derived from related literature and studies were applied directly to the development of the web-based management system. This ensured that the system was built upon established best practices and proven technological approaches.

The study also considered the importance of system functionality in addressing operational challenges. Through the developmental process, features were carefully designed to automate repetitive tasks, improve transaction accuracy, and simplify inventory monitoring. These functionalities were intended to enhance the efficiency and effectiveness of daily business operations.

In addition, the research design supported the collection of user feedback throughout the development process. Input from potential users helped identify areas that required improvement and ensured that the system remained aligned with user expectations. Incorporating feedback contributed to the development of a more effective and userfriendly application.

The descriptive component of the study enabled the researchers to document the current business processes and evaluate their strengths and weaknesses. This analysis provided valuable insights into the operational environment of the bistro and served as a basis for determining the system features necessary to address identified inefficiencies.

The developmental nature of the research also emphasized innovation and problem-solving. Rather than simply analyzing existing conditions, the study sought to introduce a practical technological solution capable of improving business operations. This approach reflects the growing role of information technology in addressing organizational challenges and enhancing service delivery.

Furthermore, the research design facilitated the evaluation of the system’s overall effectiveness after development. Various testing procedures were conducted to determine whether the system performed according to its intended functions. The results of these evaluations provided evidence regarding the system’s capability to improve operational processes within the establishment.

Ultimately, the selected research design ensured that the study achieved both its investigative and developmental objectives. By combining the strengths of descriptive and developmental research, the researchers were able to gain a comprehensive understanding of the existing problems while simultaneously developing a practical solution. This approach contributed significantly to the successful creation of the web-based management system for Groove Monkey Pinoy Korean Bistro.

The chosen research design also allowed the researchers to examine the relationship between existing operational challenges and the potential benefits of technological solutions. By analyzing current business practices and comparing them with modern management approaches, the study was able to identify opportunities for improvement that could be addressed through system development.

An important aspect of the developmental research design is its emphasis on iterative improvement. Throughout the development process, system components were reviewed, modified, and enhanced based on observations and user feedback. This iterative approach ensured that the final system met the functional requirements of the business and addressed the concerns raised during the initial assessment.

The research design further supported the establishment of clear objectives and measurable outcomes. By defining specific goals for the system, such as improving order processing, enhancing inventory accuracy, and simplifying report generation, the researchers were able to evaluate the success of the project based on tangible performance indicators.

In conducting the study, the researchers recognized the importance of understanding user needs and expectations. Employees and management personnel play a significant role in the successful implementation of any information system. Therefore, their insights and experiences were considered throughout the development process to ensure that the resulting system would be practical and beneficial in real-world operations.

The developmental research design also promoted the use of systematic problem-solving techniques. Identified issues were carefully analyzed to determine their root causes before appropriate system features were developed. This approach helped ensure that the proposed solution addressed underlying problems rather than merely treating their symptoms.

Another strength of the research design is its focus on producing a functional output that can be utilized by the organization. Unlike purely theoretical studies, developmental research aims to create a tangible product that generates practical benefits. In this study, the primary output is a web-based management system that can directly support the operational needs of Groove Monkey Pinoy Korean Bistro.

The study also considered the role of technology in enhancing organizational adaptability. As business environments continue to evolve, organizations must be capable of responding to changing customer demands and operational requirements. The proposed system was designed with flexibility in mind, allowing future modifications and enhancements as the business grows.

Moreover, the research design encouraged careful documentation of the development process. Maintaining detailed records of system requirements, design decisions, testing procedures, and evaluation results contributed to the transparency and credibility of the study. Such documentation also serves as a useful reference for future system maintenance and enhancement activities.

The combination of developmental and descriptive methods enabled the researchers to approach the study from both analytical and practical perspectives. While descriptive research provided a comprehensive understanding of the current operational environment, developmental research facilitated the creation of a technology-based solution that addresses identified challenges.

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The combination of developmental and descriptive methods enabled the researchers to approach the study from both analytical and practical perspectives. While descriptive research provided a comprehensive understanding of the current operational environment, developmental research facilitated the creation of a technology-based solution that addresses identified challenges.

Overall, the selected research design provided a comprehensive and effective framework for achieving the objectives of the study. It supported the systematic investigation of existing business processes, guided the development of a customized web-based management system, and ensured that the final product was capable of improving efficiency, accuracy, and overall business performance within Groove Monkey Pinoy Korean Bistro.

3.2 System Development Methodology

The study used the Agile Model as the system development methodology. Agile is an iterative and flexible approach that allows continuous improvement of the system through collaboration and feedback.

Phases of the Agile Model Used:

1. **Planning**

   In this phase, the researchers identified the problems of the current system and defined the project objectives. Requirements such as system features (ordering, billing, inventory, and reporting) were gathered through data collection methods.

2. **Design**

   The system design was created based on the identified requirements. This includes designing the user interface, system flow, and database structure. Mock-ups and diagrams were prepared to visualize how the system will function.

3. **Development**

In this phase, the actual coding of the system was carried out. The researchers developed the core features of the system, including ordering, billing, and inventory management modules.

The Agile Model was selected for this study because of its flexibility and adaptability in handling system development projects. Unlike traditional development methodologies that follow a rigid sequence of activities, Agile allows developers to continuously improve the system based on user feedback and changing requirements. This approach is particularly beneficial for projects that require frequent modifications and refinements throughout the development process.

One of the key advantages of Agile development is its emphasis on collaboration among stakeholders. Throughout the project, the researchers maintained communication with the intended users of the system to ensure that their needs and expectations were properly addressed. This collaborative approach helped minimize misunderstandings and contributed to the development of a system that effectively supports business operations.

The planning phase served as the foundation of the entire project. During this stage, the researchers conducted a detailed assessment of the existing processes within Groove Monkey Pinoy Korean Bistro. Information gathered from interviews, observations, and document reviews enabled the researchers to identify operational inefficiencies and determine the requirements necessary for the proposed system.

In addition to identifying system requirements, the planning phase also involved establishing project goals, defining the scope of development, and determining the resources needed for implementation. Careful planning ensured that the project remained focused on addressing the most critical operational challenges faced by the business.

During the design phase, considerable attention was given to creating a user-friendly interface that would be easy to understand and navigate. The researchers recognized that system usability is a critical factor in successful implementation. As a result, interface designs were developed with simplicity, accessibility, and functionality in mind to accommodate users with varying levels of technical expertise.

The design phase also included the development of system architecture and database structures. Proper database design was necessary to ensure efficient storage, retrieval, and management of business information. Relationships between system entities such as products, orders, inventory records, and reports were carefully defined to maintain data consistency and integrity.

Mock-ups, wireframes, and process flow diagrams were utilized during the design stage to provide a visual representation of the proposed system. These design tools helped stakeholders better understand how the system would function and allowed potential issues to be identified before actual development began. Early visualization contributed to more efficient development and reduced the likelihood of major design revisions.

The development phase focused on transforming the approved system designs into a fully functional web-based application. The researchers utilized appropriate programming languages, development frameworks, and database technologies to build the various modules of the system.

Each component was developed according to the requirements identified during the planning phase.

Throughout development, individual system modules were implemented incrementally rather than all at once. This iterative approach allowed the researchers to test and evaluate features as they were completed. By identifying and resolving issues early in the development process, the overall quality and reliability of the system were improved.

The Agile methodology also encouraged continuous refinement and enhancement throughout the project lifecycle. As new insights emerged during development and testing, adjustments were made to improve system functionality and user experience. This flexibility ensured that the final product remained aligned with the operational needs of Groove Monkey Pinoy Korean Bistro and provided an effective solution to its existing challenges.

Another important characteristic of the Agile Model is its focus on delivering functional components of the system within short development cycles. Instead of waiting until the entire system is completed, specific features can be developed, tested, and evaluated incrementally. This approach allows users to provide feedback on completed functionalities and helps ensure that the final system meets their expectations.

The Agile methodology also promotes adaptability to changing requirements. During the course of system development, new needs or improvements may be identified by stakeholders. Agile enables the researchers to incorporate these changes without significantly disrupting the overall development process. This flexibility is particularly valuable in projects where operational requirements may evolve over time.

Communication played a crucial role throughout the Agile development process. Regular discussions between the researchers and system users facilitated the exchange of ideas, concerns, and suggestions regarding system functionality. Continuous communication helped ensure that the developed features aligned closely with the operational needs of the business.

The iterative nature of Agile development supports continuous quality improvement. At the end of each development cycle, system components were reviewed and evaluated to determine whether they met established requirements. Necessary adjustments were then implemented before proceeding to subsequent phases, resulting in a more refined and reliable system.

Another advantage of using the Agile Model is its ability to reduce project risks. By testing features as they are developed, potential issues can be identified and addressed at an early stage. This proactive approach minimizes the likelihood of major problems emerging during the later stages of development and implementation.

The development process also emphasized modular design principles. Each major function of the system, such as ordering, billing, inventory management, and reporting, was developed as a separate module while remaining integrated within the overall system. This modular structure simplifies maintenance, troubleshooting, and future enhancements.

In the development phase, attention was given to ensuring that the system performed efficiently under typical operational conditions. The researchers optimized database queries, interface responsiveness, and system workflows to improve performance and provide users with a smooth and reliable experience during daily operations.

Security considerations were also incorporated into the development process. Appropriate measures were implemented to protect sensitive business information and prevent unauthorized access to system resources. User authentication, password protection, and access controls were included to enhance data security and maintain the confidentiality of business records.

The Agile methodology further supports future scalability of the system. Because the system was developed using an iterative and flexible framework, additional features and functionalities can be incorporated as business requirements change. This capability ensures that the system can continue to support the growth and development of the bistro over time.

Overall, the use of the Agile Model provided a structured yet flexible framework for the successful development of the web-based management system. Through continuous collaboration, iterative improvement, and regular evaluation, the methodology enabled the researchers to develop a reliable, user-friendly, and effective solution that addresses the operational needs of Groove Monkey Pinoy Korean Bistro.

3.3 System Architecture

The proposed system follows a client-server architecture, where users interact with the system through a web browser (client), while data is processed and stored in a centralized server. System Overview:

- Users (admin and staff) access the system via a web interface
- Requests are sent to the server for processing
- The server communicates with the database to store and retrieve data
- Results are displayed back to the user in real time

User Interaction Flow:

1. User logs into the system
2. User selects an operation (e.g., order, inventory, reports)
3. System processes the request
4. Data is stored or retrieved from the database
5. Output is displayed to the user

The client-server architecture was chosen for the proposed system because it provides a reliable and efficient framework for managing business operations. This architecture separates the user interface from the data processing and storage components, allowing the system to operate more effectively. By centralizing data management on a server, the system ensures consistency, security, and accessibility of information across all authorized users.

In a client-server environment, users interact with the system through web browsers installed on computers or tablets within the establishment. This eliminates the need to install specialized software on every device, making the system easier to deploy and maintain. Users can access the system using standard web technologies, which enhances convenience and usability.

The server serves as the core component of the system architecture. It is responsible for receiving user requests, processing business logic, managing transactions, and communicating with the database. By handling these functions centrally, the server ensures that all users work with the same set of accurate and up-to-date information.

The database acts as the repository of all system information, including user accounts, menu items, customer orders, billing records, inventory data, and generated reports. Storing data in a centralized database helps maintain data integrity and prevents inconsistencies that may occur when information is stored in multiple locations.

One of the major advantages of the proposed architecture is real-time data processing. Whenever a user submits a transaction or updates a record, the information is immediately processed and reflected within the system. This capability allows staff and administrators to access the most current information available, improving decision-making and operational efficiency.

The architecture also supports multiple users accessing the system simultaneously. For example, while a cashier processes customer orders, an administrator can monitor inventory levels and generate reports without disrupting ongoing transactions. This multi-user capability enhances productivity and supports the daily operations of the business.

To ensure efficient communication between system components, requests generated by users are transmitted through secure network connections to the server. The server then processes the requests, retrieves or updates the necessary data, and returns the results to the user's browser. This communication process occurs seamlessly and transparently from the user's perspective.

Security is an important consideration within the system architecture. User authentication mechanisms are implemented to verify the identity of individuals accessing the system. Access privileges are also assigned according to user roles, ensuring that sensitive information and administrative functions are only available to authorized personnel.

The proposed architecture is designed to support future scalability and expansion. As the business grows and operational requirements change, additional modules and functionalities can be integrated into the existing framework without requiring major modifications to the entire system. This flexibility makes the architecture suitable for long-term use.

The system architecture was designed to ensure that information flows efficiently between users and system components. Every transaction initiated by a user follows a structured process that minimizes delays and ensures accurate data handling. This organized flow of information contributes to the overall reliability and effectiveness of the system.

Another advantage of the client-server architecture is its centralized management capability. Since all data and processing activities are handled by a single server, administrators can easily monitor system performance, manage user accounts, and maintain database integrity. Centralized management simplifies system administration and reduces the complexity associated with maintaining multiple independent applications.

The architecture also supports data consistency across all modules of the system. Information entered in one module is automatically reflected in related modules. For example, when an order is processed, the corresponding inventory records can be updated automatically, ensuring that all users have access to accurate and synchronized information.

The proposed system incorporates structured database relationships to facilitate efficient data retrieval and storage. Tables within the database are organized according to established database design principles, reducing redundancy and improving data integrity. This structure enables the system to process transactions quickly while maintaining accurate records.

To enhance operational efficiency, the architecture supports automated processing of routine tasks. Activities such as calculating order totals, updating inventory quantities, generating reports, and recording transactions are performed automatically by the system. This reduces manual effort and minimizes the possibility of human error.

The user interface acts as the communication bridge between users and the underlying system components. It is designed to provide intuitive navigation and clear presentation of information. By simplifying user interactions, the system allows employees to perform their tasks more efficiently and with greater confidence.

Another important feature of the architecture is its ability to maintain transaction accuracy. Whenever users submit information, validation mechanisms are applied to ensure that the data entered meets predefined requirements. These validation procedures help prevent incomplete, duplicate, or incorrect records from being stored in the database.

The architecture also facilitates report generation by consolidating information from various system modules. Data related to orders, sales, inventory levels, and user activities can be processed and presented in report formats that support managerial decision-making. Access to comprehensive reports allows administrators to monitor business performance more effectively.

Backup and recovery considerations are incorporated into the overall architecture to safeguard critical business information. By maintaining secure backups of the database, the system can recover important records in the event of hardware failures, software issues, or accidental data loss. This capability contributes to the reliability and continuity of business operations.

The proposed architecture also promotes efficient resource utilization by centralizing processing activities within the server. Instead of requiring each client device to perform complex computations, most processing tasks are handled by the server. This approach reduces the hardware requirements of client devices and ensures consistent system performance across different workstations.

Another benefit of the architecture is its support for continuous data availability. Authorized users can access information whenever needed, provided that the system and network are operational. This accessibility allows employees and administrators to retrieve records, process transactions, and monitor business activities without unnecessary delays.

The architecture is designed to accommodate various operational scenarios within the restaurant. Whether processing customer orders, updating inventory records, generating reports, or managing user accounts, the system follows a standardized workflow that ensures consistency and accuracy across all functions. This standardization contributes to smoother and more organized business operations.

Data security remains a fundamental component of the system architecture. In addition to user authentication, security measures such as password encryption, session management, and controlled access permissions are implemented to protect sensitive business information. These safeguards help prevent unauthorized access and maintain the confidentiality of organizational data.

The client-server architecture also supports efficient maintenance and system updates. Since the application is centrally managed through the server, updates and enhancements can be deployed without requiring manual installation on each client device. This simplifies system maintenance and ensures that all users have access to the latest version of the application.

Furthermore, the architecture enhances system reliability by reducing the likelihood of data duplication and inconsistencies. Since all transactions are processed through a centralized database, information is maintained in a single authoritative source. This ensures that users access the same accurate and updated records regardless of their location within the establishment.

The architecture also facilitates better monitoring and auditing of system activities. Every transaction, modification, and user action can be recorded within system logs. These records provide valuable information for tracking user activities, identifying operational issues, and ensuring accountability among system users.

Performance optimization is another important consideration within the system architecture. Efficient database queries, structured data storage, and optimized communication processes help minimize response times and improve overall system speed. This enables users to complete tasks quickly, which is particularly important in a fast-paced restaurant environment.

The proposed architecture supports future integration with additional technologies should the business decide to expand its operations. Features such as online ordering, customer management, digital payment processing, and mobile accessibility can potentially be incorporated into the existing framework without requiring a complete system redesign. This scalability ensures that the system remains adaptable to future business needs.

The architecture of the proposed system is designed to support the efficient handling of large amounts of operational data generated on a daily basis. As transactions accumulate over time, the centralized database ensures that information remains organized and easily accessible. This capability allows the business to maintain historical records that can be used for future analysis and decision-making.

An additional advantage of the system architecture is its ability to provide a consistent user experience across different devices. Since the application is web-based, users can access the system through compatible web browsers without significant differences in functionality or appearance. This consistency helps reduce user confusion and improves overall system usability.

The architecture also enables efficient management of user roles and responsibilities. Different access levels can be assigned to administrators, cashiers, and staff members based on their specific duties within the organization. This role-based access structure helps maintain security while ensuring that users can perform their assigned tasks effectively.

Another important feature of the architecture is its support for data validation and error prevention. Before information is stored in the database, the system verifies that the data entered by users meets predefined requirements. These validation mechanisms help maintain data quality and reduce the occurrence of inaccurate or incomplete records.

The proposed architecture encourages efficient coordination among various operational functions. Activities such as order processing, inventory updates, billing transactions, and report generation are interconnected within the system. This integration allows information to flow seamlessly between modules, minimizing delays and improving operational efficiency.

In terms of performance, the architecture is designed to respond quickly to user requests. Optimized communication between the client, server, and database ensures that transactions are processed efficiently. Faster response times contribute to a smoother user experience and help employees complete tasks more effectively during busy business hours.

The system architecture also supports data monitoring and performance evaluation. Administrators can review transaction histories, inventory movements, and operational reports generated by the system. These monitoring capabilities provide valuable insights into business activities and help management identify opportunities for improvement.

Another key consideration in the architectural design is system reliability. The centralized structure minimizes the risk of conflicting information and ensures that all users access the same data source. Consistent access to accurate information supports informed decision-making and enhances the overall effectiveness of business operations.

The architecture further promotes operational transparency by maintaining detailed records of system activities. User actions, transaction updates, and inventory modifications can be tracked and reviewed when necessary. This transparency improves accountability and provides a reliable audit trail for management purposes.

In summary, the client-server architecture provides a comprehensive and dependable framework for the proposed web-based management system.

Through centralized control, secure data management, efficient communication, scalability, and enhanced operational support, the architecture enables Groove Monkey Pinoy Korean Bistro to manage its business processes more effectively while preparing for future technological advancements and organizational growth.

Overall, the client-server architecture provides a stable, secure, and efficient foundation for the web-based management system. Through centralized data management, real-time processing, multi-user accessibility, and enhanced security, the architecture supports the system's objective of improving operational efficiency and business performance within Groove Monkey Pinoy Korean Bistro.

3.4 Tools and Technologies Used

The system was developed using the following tools and technologies:

- **Programming Languages:**
  - o HTML (HyperText Markup Language) o CSS (Cascading Stylesheet) o JavaScript (Vanilla) o PHP (HyperText Preprocessor)
- **Frameworks:**
  - o Bootstrap 5.3
- **Database:**
  - o MySQL

- **Development Tools:**
    - o Visual Studio Code

The selection of appropriate tools and technologies was a critical aspect of the system development process. These technologies were chosen based on their reliability, compatibility, ease of use, and ability to support the functional requirements of the proposed webbased management system. Together, they provide a stable and efficient platform for developing, deploying, and maintaining the application.

The selected technologies were carefully chosen to ensure compatibility and seamless integration throughout the development process. Since all components of the technology stack are widely used in web development, they provide a stable environment for creating a reliable and maintainable system. Their compatibility minimizes technical issues and facilitates efficient communication between the front-end and back-end components of the application.

HTML (HyperText Markup Language) served as the foundation of the system's web pages. It was used to structure and organize the content displayed to users, including forms, tables, navigation menus, and other interface components. HTML provides the basic framework that allows web browsers to properly interpret and display application content.

HTML plays a significant role in establishing the structural framework of the system. Every page within the application, including login forms, dashboards, inventory pages, billing interfaces, and report modules, relies on HTML elements to organize and present information logically. This structure ensures that users can navigate the system efficiently and access the necessary features with ease.

CSS (Cascading Style Sheets) was utilized to enhance the visual appearance of the system. Through CSS, the researchers were able to control the layout, typography, colors, spacing, and overall design of the user interface. The use of CSS contributed to a more professional and user-friendly appearance, improving the overall user experience.

The use of CSS contributes not only to visual appeal but also to usability and accessibility. Through proper styling techniques, the researchers designed interfaces that are clear, organized, and easy to read. Well-designed layouts help users locate information quickly and perform tasks more effectively, thereby improving overall user satisfaction.

JavaScript was implemented to provide interactivity and dynamic functionality within the application. It enabled features such as form validation, real-time calculations, responsive user interactions, and dynamic content updates. By reducing the need for constant page reloads, JavaScript improved system responsiveness and usability.

JavaScript enhances the functionality of the system by enabling interactive features that improve user engagement and operational efficiency. For example, form validation can be performed instantly before data is submitted to the server, reducing errors and preventing invalid information from being stored in the database. Such features contribute to a smoother and more efficient user experience.

PHP (Hypertext Preprocessor) was used as the primary server-side programming language for the system. PHP handled the processing of user requests, execution of business logic, communication with the database, and generation of dynamic web content. Its compatibility with MySQL and web-based environments made it a suitable choice for the development of the proposed application.

PHP serves as the backbone of the application by handling server-side operations. It processes requests submitted by users, executes necessary computations, manages authentication procedures, and interacts with the database. Because PHP is specifically designed for web development, it provides a flexible and efficient solution for implementing the system's core functionalities.

Bootstrap 5.3 was employed as the front-end framework to simplify the design and development of responsive user interfaces. Bootstrap provides a collection of pre-built components, layouts, and styling utilities that help ensure consistency throughout the application. The framework also supports responsive design, allowing the system to function properly across different screen sizes and devices.

Bootstrap 5.3 significantly accelerated the development process by providing ready-made interface components such as navigation bars, buttons, forms, tables, and modal windows. These components helped maintain consistency throughout the application while reducing the amount of custom code required. As a result, development time was optimized without compromising interface quality.

MySQL was selected as the database management system due to its reliability, efficiency, and widespread use in web application development. The database stores essential business information, including user accounts, product records, inventory data, transaction histories, and generated reports. MySQL enables fast data retrieval and secure storage, ensuring the integrity of business records.

MySQL provides robust database management capabilities that support the storage and retrieval of large volumes of information. As the system accumulates transaction records, inventory updates, and user activities over time, MySQL ensures that data remains organized, searchable, and accessible. Its performance and reliability make it suitable for supporting the long-term operational needs of the business.

The use of a relational database structure within MySQL allowed the researchers to establish logical relationships among various data entities. This approach minimized data redundancy, improved data consistency, and enhanced the efficiency of information retrieval. Proper database design also contributed to the overall performance of the system.

Visual Studio Code served as the primary integrated development environment (IDE) for the project. It provided the researchers with tools for code editing, debugging, syntax highlighting, extension support, and project management. The flexibility and user-friendly interface of Visual Studio Code helped streamline the development process and improve productivity.

Visual Studio Code offered several development features that improved coding efficiency and project organization. Built-in tools such as syntax highlighting, code completion, integrated terminal access, and debugging support enabled the researchers to write, test, and maintain code more effectively. These features contributed to the overall quality and consistency of the developed system.

The use of widely adopted technologies also provides long-term benefits in terms of maintenance and future enhancements. Since HTML, CSS, JavaScript, PHP, Bootstrap, and MySQL are supported by extensive documentation and large developer communities, future developers can easily understand, modify, and expand the system when necessary.

Overall, the combination of HTML, CSS, JavaScript, PHP, Bootstrap 5.3, MySQL, and Visual Studio Code provided a comprehensive technology stack for the development of the web-based management system. These tools and technologies worked together to support the creation of a secure, responsive, efficient, and maintainable application capable of addressing the operational needs of Groove Monkey Pinoy Korean Bistro.

3.5 Data Gathering Techniques

The researchers used the following methods to collect data:

- **Interviews** – Conducted with the bistro staff and management to understand current processes and challenges
- **Surveys** – Used to gather feedback on user needs and preferences
- **Observation** – Direct observation of daily operations such as ordering and inventory handling
- **Document Analysis** – Review of existing records such as receipts and inventory logs

Data gathering is a critical component of the research process because it provides the information necessary for understanding the existing operational environment and identifying system requirements. Through the use of multiple data collection methods, the researchers were able to obtain comprehensive and reliable information regarding the current practices and challenges of Groove Monkey Pinoy Korean Bistro.

The interview method allowed the researchers to gather detailed insights directly from individuals involved in the daily operations of the business. By engaging with management and staff members, the researchers obtained firsthand information regarding the strengths and weaknesses of the current operational procedures. These discussions provided valuable perspectives that contributed to the identification of system requirements.

Interviews also enabled the researchers to clarify issues that could not be fully understood through observation alone. Participants were given the opportunity to explain specific challenges, provide suggestions for improvement, and share their experiences regarding existing processes. This qualitative information served as an important basis for designing system functionalities that address actual user needs.

Surveys were conducted to collect structured feedback from potential users of the proposed system. The use of survey questionnaires allowed the researchers to gather information from multiple respondents in a consistent manner. Survey results helped identify user preferences, expectations, and desired system features that could improve operational efficiency and usability.

The survey method also provided quantitative data that could be analyzed to identify common trends and concerns among respondents. By examining the responses systematically, the researchers gained a broader understanding of the operational challenges faced by the business and the specific areas where technological solutions could provide the greatest benefits.

Observation was another essential data gathering technique utilized in the study. Through direct observation of daily activities, the researchers were able to examine how orders were processed, how inventory was monitored, and how transactions were recorded. This method allowed the researchers to witness actual workflows and identify inefficiencies that might not be apparent through interviews or surveys alone.

Direct observation provided objective information regarding the current operational practices of the bistro. By observing employees as they performed their duties, the researchers gained a clearer understanding of task sequences, communication patterns, and procedural bottlenecks. These observations contributed to the development of system features that align with existing workflows while addressing identified inefficiencies.

Document analysis was conducted to review existing business records and operational documents. Materials such as receipts, sales records, inventory logs, and transaction reports were examined to understand how information was currently recorded and managed. This analysis provided valuable insights into the structure and content of the data that the proposed system would need to process and store.

The review of existing documents also helped verify information obtained through interviews, surveys, and observations. Comparing data from multiple sources improved the accuracy and reliability of the findings. This process allowed the researchers to identify inconsistencies, validate operational procedures, and gain a more comprehensive understanding of the organization's information management practices.

The use of multiple data gathering techniques allowed the researchers to obtain a comprehensive understanding of the current operational environment of Groove Monkey Pinoy Korean Bistro. By collecting information from different sources, the researchers were able to validate findings and ensure that the identified system requirements accurately reflected the needs of the business.

Data triangulation was achieved through the combination of interviews, surveys, observations, and document analysis. This approach strengthened the credibility of the collected information by allowing the researchers to compare and verify data obtained from various methods. The use of multiple sources minimized bias and increased the reliability of the research findings.

The information collected through interviews provided valuable insights into the operational challenges encountered by employees and management. These challenges included delays in order processing, difficulties in inventory monitoring, and limitations in generating accurate reports. Understanding these concerns enabled the researchers to design system features that directly address existing operational problems.

Survey responses played an important role in identifying user expectations regarding the proposed system. Respondents were able to express their preferences for system functionality, interface design, and ease of use. The feedback gathered helped guide the development process and ensured that the final system would be aligned with user needs and preferences.

Observation sessions allowed the researchers to evaluate actual work practices within the establishment. By monitoring routine activities, the researchers gained firsthand knowledge of how information flows through the organization. This understanding was essential in designing workflows and system processes that complement existing business operations.

The observation method also helped identify inefficiencies that might otherwise remain unnoticed. For example, delays caused by manual record-keeping, repetitive tasks, and communication gaps between employees became evident during the observation process. These findings contributed to the formulation of system solutions intended to streamline operations and improve productivity.

Document analysis provided a historical perspective on business activities by examining existing records and reports. Through the review of receipts, inventory logs, and transaction documents, the researchers were able to assess the accuracy, completeness, and organization of current record-keeping practices. This information served as a reference for designing the database structure of the proposed system.

Another benefit of document analysis was the identification of data elements that needed to be incorporated into the system. Information such as product details, transaction records, inventory quantities, and user information were examined to determine how they should be organized and managed within the database. This ensured that the system would effectively support the business's information management requirements.

The gathered data also assisted the researchers in establishing priorities during system development. By understanding which operational issues had the greatest impact on efficiency and service quality, the researchers were able to focus their efforts on developing features that would provide the most significant benefits to the organization.

The researchers ensured that the data gathering process was conducted systematically to obtain accurate and relevant information. Each data collection activity was carefully planned and executed to maximize the quality of the information gathered. This systematic approach contributed to the reliability of the findings and provided a strong basis for the development of the proposed system.

Prior to conducting interviews and surveys, the researchers prepared data collection instruments designed to capture the information necessary for the study. Interview guides and survey questionnaires were developed based on the objectives of the research and the operational requirements of the business. These instruments helped ensure consistency in data collection and facilitated effective analysis of the responses.

The participation of management personnel was particularly important in understanding the strategic and administrative aspects of the business. Through interviews, managers were able to provide insights regarding operational policies, reporting requirements, inventory management practices, and overall business objectives. This information helped define the administrative functionalities required in the proposed system.

Employees also contributed valuable information regarding daily operational procedures and challenges. Their firsthand experiences provided practical insights into how orders are processed, inventory is monitored, and transactions are recorded. These perspectives enabled the researchers to design system features that support actual workplace activities and improve operational efficiency.

During observation activities, the researchers paid close attention to the sequence of tasks performed by employees. This process allowed them to identify redundant activities, workflow interruptions, and areas where automation could be beneficial. The findings from these observations played a significant role in improving the design of the system's operational processes.

The survey process allowed respondents to provide feedback in a structured and measurable format. Responses were analyzed to identify recurring concerns and common suggestions. By evaluating these patterns, the researchers gained a clearer understanding of user expectations and priorities, which guided the selection of system functionalities.

Document analysis further enhanced the depth of the study by providing access to actual business records. Existing documents served as evidence of current operational practices and helped verify information obtained through interviews and observations. This method ensured that the system requirements were based on documented business activities rather than assumptions.

The integration of different data gathering techniques allowed the researchers to develop a comprehensive view of the business environment. Information obtained from one method often complemented and reinforced findings from another. This integration strengthened the overall validity of the research and reduced the likelihood of incomplete or inaccurate conclusions.

Another advantage of the data gathering process was the identification of opportunities for process improvement beyond the immediate scope of the system. By thoroughly examining current operations, the researchers gained insights into potential enhancements in workflow management, communication practices, and resource utilization. These observations contributed to the overall understanding of the organization's operational needs.

In summary, the data gathering techniques employed in this study provided extensive and reliable information necessary for the successful development of the web-based management system. Through interviews, surveys, observations, and document analysis, the researchers obtained a detailed understanding of the business environment, operational challenges, and user requirements. These findings served as the foundation for designing a system that effectively addresses the needs of Groove Monkey Pinoy Korean Bistro and supports its operational improvement goals.

3.6 Testing Procedures

Testing procedures were conducted to ensure that the proposed webbased management system functioned correctly, efficiently, and according to the specified requirements. The testing process aimed to identify errors, evaluate system performance, and verify that all modules operated as intended before implementation.

The researchers performed a series of tests throughout the development process to assess the functionality, reliability, usability, and overall quality of the system. Testing was conducted incrementally as each module was developed, allowing issues to be identified and corrected at an early stage. This approach helped improve system stability and minimize potential operational problems.

In addition to identifying software defects, the testing procedures were designed to ensure that the system met the operational requirements of Groove Monkey Pinoy Korean Bistro. Each feature was evaluated to determine whether it effectively addressed the problems identified during the data gathering phase. This process helped verify that the developed system provided practical solutions to the organization's operational challenges.

The researchers adopted a systematic testing approach to ensure that all components of the system were thoroughly evaluated. Test cases were prepared for each module to assess whether expected outputs were produced when specific inputs were provided. These test cases served as a guide for validating system functionality and identifying areas requiring improvement.

During the testing process, special attention was given to the accuracy of data processing and storage. Since the system handles critical business information such as customer orders, inventory records, and sales transactions, it was important to ensure that all data entered into the system was processed correctly and stored securely within the database.

The testing procedures also evaluated the responsiveness of the system under normal operating conditions. Users performed common tasks such as placing orders, updating inventory records, generating reports, and processing transactions to determine whether the system responded efficiently. This assessment helped ensure that users could complete tasks without experiencing significant delays.

Error handling mechanisms were tested to verify the system's ability to respond appropriately to invalid inputs and unexpected situations. The researchers intentionally entered incorrect or incomplete data to determine whether the system could detect errors and provide meaningful feedback to users. Effective error handling contributes to improved usability and data integrity.

Another important aspect of testing involved validating user authentication and access control functions. Different user accounts were used to confirm that administrators, cashiers, and staff members could only access features appropriate to their assigned roles. This process helped ensure the security and confidentiality of business information.

The researchers also tested the consistency of data across different modules of the system. Transactions processed through the ordering module were verified against billing records, inventory updates, and generated reports to ensure that all related information remained synchronized. This testing helped prevent discrepancies and maintained data accuracy throughout the system.

Performance testing was conducted to evaluate the system's ability to handle multiple transactions and user requests. Although the system was designed primarily for a small-scale business environment, it was important to verify that it could maintain acceptable performance levels during periods of increased activity. This testing contributed to the overall reliability of the application.

Feedback obtained during testing was documented and analyzed to identify opportunities for further refinement. Any issues encountered by users were reviewed by the researchers, and necessary modifications were implemented to improve system functionality, usability, and performance. This iterative improvement process enhanced the quality of the final product.

The testing phase served as a crucial step in validating the effectiveness of the developed web-based management system. By conducting thorough evaluations, the researchers ensured that the system not only met technical specifications but also supported the practical requirements of daily business operations. This validation process increased confidence in the system's readiness for actual use within the establishment.

To maintain consistency during testing, predefined testing criteria were established before the evaluation process began. These criteria served as benchmarks for measuring system performance, functionality, usability, and reliability. By comparing actual outcomes with expected results, the researchers were able to determine whether each module met the required standards.

The researchers also performed repeated testing procedures to confirm the consistency of system outputs. Tasks such as order entry, inventory updates, and report generation were executed multiple times under similar conditions. Repeated testing helped verify that the system produced accurate and dependable results regardless of how often the functions were used.

Special consideration was given to the validation of database transactions during testing. The researchers examined whether records were correctly saved, updated, retrieved, and deleted when necessary. Ensuring the integrity of database operations was essential because inaccurate data handling could negatively affect inventory management, billing processes, and report generation.

Compatibility testing was conducted to determine whether the system functioned properly across different web browsers and supported devices. Since the application was designed as a web-based platform, it was important to ensure that users could access and utilize the system effectively regardless of the browser or device being used within the establishment.

The researchers also assessed the navigation structure of the application during usability testing. Menus, buttons, forms, and links were evaluated to ensure that users could move between different system modules without confusion. A well-organized navigation structure contributes to user efficiency and improves the overall user experience.

Another area of evaluation involved testing the system's ability to recover from common user mistakes. Users occasionally enter incorrect information or perform unintended actions during normal operations. The system was therefore tested to determine whether it could provide warnings, error messages, or recovery options that help users correct mistakes without compromising stored data.

The testing process further examined the clarity and accuracy of system-generated outputs. Reports, receipts, inventory summaries, and transaction records were reviewed to ensure that information was presented in an understandable and professional manner. Accurate outputs are important for supporting business decisions and maintaining reliable operational records.

Throughout the testing phase, detailed records of identified issues, corrective actions, and testing outcomes were maintained. This documentation provided a systematic record of system improvements and served as evidence that appropriate quality assurance procedures had been followed. Proper documentation also supports future maintenance and enhancement activities.

In conclusion, the extensive testing procedures conducted during the study contributed significantly to the quality and reliability of the proposed web-based management system. Through continuous evaluation, correction of identified issues, and validation of system performance, the researchers ensured that the application was capable of meeting the operational needs of Groove Monkey Pinoy Korean Bistro while providing a stable, secure, and user-friendly environment for its intended users.

Functional Testing

Functional testing was conducted to verify that each feature of the system performed according to its intended purpose. Individual modules such as user authentication, ordering, billing, inventory management, and report generation were tested to ensure that inputs, processes, and outputs functioned correctly. Any errors identified during testing were documented and corrected before proceeding to the next phase.

The login module was tested to ensure that only authorized users could access the system using valid credentials. Different user roles were also evaluated to confirm that access permissions were correctly enforced. This testing ensured that system security and user management functions operated properly.

The ordering module was tested to determine whether customer orders could be accurately recorded, processed, and stored within the database. Test transactions were performed to verify that order details were correctly displayed and that the system generated accurate transaction records.

The billing module underwent testing to ensure that order totals, prices, and payment information were calculated correctly. Various transaction scenarios were evaluated to verify the accuracy of billing computations and receipt generation. This process helped ensure reliable financial transaction processing.

Inventory management functions were tested to confirm that stock quantities were updated accurately whenever products were added, modified, or deducted through sales transactions. The researchers verified that inventory records reflected actual stock movements and that inventory reports were generated correctly.

Report generation features were tested to determine whether sales reports, inventory reports, and transaction summaries could be produced accurately. The generated reports were reviewed to ensure completeness, consistency, and correctness of the information presented.

The ordering module underwent extensive testing to ensure that customer orders could be accurately recorded and processed. Various order scenarios were simulated to verify that menu items, quantities, prices, and order details were correctly captured by the system. The results confirmed that the module was capable of handling transactions accurately and efficiently.

Testing of the ordering functionality also involved verifying that orders were properly transmitted to the appropriate system records.

The researchers ensured that all submitted orders were correctly stored in the database and could be retrieved when needed. This process helped maintain the accuracy and reliability of transaction records.

The billing module was tested to determine whether the system could correctly calculate order totals based on selected menu items and quantities. Different transaction combinations were entered to validate the accuracy of pricing calculations, subtotal computations, and final billing amounts. The testing results indicated that the billing module generated accurate payment information.

In addition, receipt generation functions were evaluated to confirm that transaction details were displayed correctly. Information such as purchased items, quantities, prices, total amounts, and transaction dates were reviewed to ensure completeness and correctness. Accurate receipt generation is important for maintaining reliable sales records and customer documentation.

The inventory management module was tested to verify its ability to accurately monitor stock levels. Inventory records were updated through simulated stock additions, deductions, and adjustments. The researchers observed whether inventory quantities changed appropriately after each transaction and confirmed that the module maintained accurate stock information.

Inventory testing also focused on ensuring that sales transactions automatically affected inventory records. Whenever products were sold through the ordering system, corresponding inventory quantities were deducted accordingly. This integration helped eliminate the need for manual inventory updates and reduced the possibility of record discrepancies.

The report generation module was evaluated to determine its capability to produce accurate and organized reports. Various report types, including sales summaries, inventory reports, and transaction histories, were generated and reviewed. The researchers verified that the reports contained complete and up-to-date information derived from the database.

Testing of the report module further examined the accuracy of calculations and summaries presented within generated reports. Totals, averages, and inventory counts were compared against source data to ensure consistency. The successful generation of accurate reports supports managerial decision-making and operational monitoring.

Data validation features were also subjected to functional testing. The system was tested using invalid, incomplete, and duplicate data entries to determine whether appropriate validation mechanisms were triggered. The results showed that the system effectively prevented erroneous information from being stored and provided clear feedback messages to users.

Overall, the functional testing process demonstrated that the major modules of the system operated according to their intended specifications. The successful execution of tests involving ordering, billing, inventory management, reporting, and user authentication provided evidence that the system was capable of supporting the operational requirements of Groove Monkey Pinoy Korean Bistro in a reliable and efficient manner.

Integration Testing

Integration testing was conducted to evaluate how different modules interacted with one another. The researchers verified that information flowed correctly between the ordering, billing, inventory, and reporting modules. This testing ensured that updates in one module were accurately reflected in related modules without causing inconsistencies or data errors.

Integration testing was performed after the successful completion of functional testing for individual modules. The primary objective of this phase was to determine whether the various components of the system could operate together as a unified application. This process helped identify issues that might not be apparent when modules were tested independently.

The researchers carefully examined the interaction between the ordering and billing modules. Whenever a customer order was entered into the system, the billing module was expected to automatically retrieve the relevant order details and calculate the corresponding payment amount. Testing confirmed that transaction information was transferred accurately between these modules.

Particular attention was given to the integration between the ordering and inventory management modules. The researchers verified that completed sales transactions automatically reduced inventory quantities based on the products sold. This automated update process ensured that stock records remained accurate and up to date without requiring manual intervention.

The integration between inventory management and report generation modules was also evaluated. Changes made to inventory records were expected to appear correctly in inventory reports and stock summaries.

Testing results showed that updated inventory information was accurately reflected in generated reports, supporting effective inventory monitoring.

Another important aspect of integration testing involved the synchronization of billing records with reporting functions. Sales transactions processed through the billing module were examined to ensure that corresponding information was correctly included in sales reports. This integration enabled management to obtain accurate financial summaries and transaction histories.

The researchers also tested the interaction between user authentication and other system modules. Access permissions assigned to different user roles were evaluated to verify that users could only access functions appropriate to their responsibilities. This testing ensured that module interactions complied with established security and access control requirements.

Data consistency was a major focus throughout the integration testing process. The researchers monitored how information moved between modules and verified that records remained synchronized across the entire system. This helped prevent discrepancies that could result from incomplete updates or communication failures between components.

Various transaction scenarios were simulated to evaluate the system's ability to handle multiple operations simultaneously. Orders were processed, inventory levels were updated, bills were generated, and reports were created to observe how the modules interacted under normal operational conditions. The successful completion of these tests demonstrated the effectiveness of module integration.

Error handling between interconnected modules was also examined during testing. The researchers intentionally introduced invalid transactions and incomplete data to determine whether errors could be detected and managed appropriately. The system successfully prevented incorrect information from affecting related modules, thereby maintaining data integrity.

The integration testing phase also evaluated the accuracy of data transfer between the system interface and the database. Whenever users performed transactions, the researchers verified that information entered through the user interface was correctly stored in the database and subsequently retrieved by other modules when needed. This ensured reliable communication between the front-end and back-end components of the system.

The researchers tested whether updates made in one module were immediately reflected in other related modules. For example, modifications to product information within the inventory management module were expected to appear automatically in the ordering module.

Successful synchronization confirmed that users always accessed current and consistent information throughout the system.

Special attention was given to the integration of transaction records with reporting functionalities. Every completed order and billing transaction was monitored to ensure that the corresponding information was accurately incorporated into daily sales reports and transaction summaries. This verification helped maintain the reliability of business reports generated by the system.

The testing process also examined the handling of simultaneous operations performed by multiple users. Scenarios involving concurrent transactions were simulated to determine whether the system could maintain data consistency while processing multiple requests. The results indicated that the integrated modules functioned effectively without causing conflicts or data corruption.

Integration testing further evaluated the efficiency of communication between system components. The researchers observed how quickly information moved between modules and whether any delays occurred during processing. Efficient communication was necessary to support real-time updates and provide users with accurate information during daily operations.

The relationship between user management functions and operational modules was also reviewed. Different user roles were assigned specific permissions, and testing was conducted to verify that these permissions were consistently enforced across all integrated components. This process helped ensure both security and proper system operation.

The researchers also assessed the system’s ability to maintain accurate audit trails across integrated modules. Actions performed within one component were tracked and recorded appropriately within related records. This capability supports transparency, accountability, and effective monitoring of business activities.

Data recovery and error prevention mechanisms were evaluated during integration testing as well. The researchers simulated unexpected interruptions and invalid transactions to determine how the integrated system responded. The results demonstrated that appropriate safeguards were in place to prevent data inconsistencies and maintain operational continuity.

Another aspect of integration testing involved verifying the consistency of calculations across different modules. Inventory deductions, sales totals, billing computations, and report summaries were compared to ensure that all calculations produced consistent and accurate results throughout the application. This validation contributed to the overall reliability of the system.

In conclusion, the integration testing procedures demonstrated that all modules of the web-based management system were successfully interconnected and capable of working together as a unified platform. The seamless exchange of information, accurate synchronization of records, effective handling of transactions, and maintenance of data integrity confirmed that the system was well-prepared to support the operational requirements of Groove Monkey Pinoy Korean Bistro.

Overall, the integration testing process confirmed that the individual modules of the web-based management system functioned cohesively and effectively as a single application. The successful exchange of information between ordering, billing, inventory management, reporting, and user management modules demonstrated the system's ability to support seamless business operations while maintaining data accuracy, consistency, and reliability.

Usability Testing

Usability testing was performed to assess the ease of use, accessibility, and user-friendliness of the system. Selected users, including staff and management personnel, were asked to perform typical tasks using the system. Their feedback regarding navigation, interface design, system responsiveness, and overall user experience was collected and analyzed. Suggestions provided by users were considered in refining the system.

Usability testing was conducted to determine how effectively users could interact with the system while performing their daily responsibilities. The researchers focused on evaluating whether users could complete assigned tasks efficiently, accurately, and without requiring extensive assistance. This assessment helped ensure that the system would be practical for actual business operations.

The selected participants represented the intended users of the system, including administrators, cashiers, and staff members. Their involvement provided valuable insights regarding the system's usability from different operational perspectives. Since these individuals would be the primary users of the application, their feedback was essential in evaluating the effectiveness of the interface design.

During the testing process, participants were asked to perform common activities such as logging into the system, processing customer orders, updating inventory records, generating reports, and managing user accounts. These tasks allowed the researchers to observe how users interacted with various system features under realistic conditions.

The researchers carefully monitored the time required for users to complete assigned tasks. Completion times were analyzed to determine whether the system supported efficient workflow execution. Faster task completion indicated that users could navigate the application effectively and utilize its features without unnecessary delays.

Navigation usability was also evaluated during testing. Users explored menus, buttons, links, and interface components to determine whether the system structure was intuitive and easy to understand. Feedback regarding navigation helped identify areas where interface organization could be improved to enhance user convenience.

Another important aspect of usability testing involved assessing the clarity of instructions, labels, and messages displayed by the system. The researchers examined whether users could easily understand prompts, notifications, and error messages. Clear communication within the interface contributes to a more positive user experience and reduces the likelihood of operational mistakes.

System responsiveness was also considered during usability evaluation. Users interacted with different modules to determine whether pages loaded efficiently and whether actions were processed within a reasonable time. Responsive system performance is important in maintaining productivity, particularly during busy operational periods.

The testing process further examined the visual design and layout of the application. Participants provided feedback regarding readability, color schemes, spacing, and overall presentation. A welldesigned interface improves accessibility, reduces user fatigue, and supports efficient task execution.

User feedback collected during usability testing was documented and analyzed by the researchers. Suggestions regarding interface improvements, feature enhancements, and workflow adjustments were carefully reviewed. Where appropriate, modifications were implemented to improve the overall usability and effectiveness of the system.

The usability testing process also focused on measuring user satisfaction when interacting with the system. Participants were encouraged to share their opinions regarding the overall experience, including the convenience of accessing features and the ease of performing routine tasks. Positive user feedback indicated that the system was capable of meeting the expectations of its intended users.

The researchers observed how quickly users became familiar with the system's interface. Since the application was designed for individuals with varying levels of technical knowledge, it was important to determine whether new users could learn the system without extensive training. The results showed that most participants were able to understand the basic functions and navigation structure within a short period.

Particular attention was given to the organization of menus and system modules. Users evaluated whether features were logically grouped and easily accessible. A well-organized interface reduces confusion and allows users to locate functions more efficiently, thereby improving productivity during daily operations.

The readability of information displayed on the screen was also assessed during usability testing. Participants reviewed forms, reports, tables, and transaction records to determine whether information was presented clearly and effectively. Proper formatting and presentation contribute significantly to user understanding and decision-making.

The researchers also evaluated the consistency of interface elements throughout the application. Buttons, icons, menus, and form layouts were reviewed to ensure that similar actions followed consistent design patterns. Consistency helps users develop familiarity with the system and reduces the likelihood of operational errors.

Another aspect examined during usability testing was the effectiveness of system feedback mechanisms. Users interacted with the application to determine whether confirmation messages, alerts, and notifications were displayed appropriately after performing specific actions. Effective feedback helps users understand the outcomes of their activities and improves overall confidence in using the system.

The accessibility of frequently used functions was also considered. Features such as order entry, inventory updates, billing operations, and report generation were evaluated to determine whether they could be reached quickly and conveniently. Easy access to essential functions enhances workflow efficiency and supports faster task completion.

Participants were encouraged to identify any difficulties encountered while using the system. These observations provided valuable insights into potential usability issues that may not have been detected during development. Addressing such concerns contributed to the refinement of the system and improved the overall user experience.

The results of usability testing demonstrated that the system's design effectively supported user interaction and operational requirements. Most participants were able to complete assigned tasks successfully with minimal assistance, indicating that the interface was intuitive and aligned with user expectations. This outcome suggested that the system could be adopted with minimal disruption to existing business processes.

In conclusion, the usability testing phase confirmed that the proposed web-based management system was easy to learn, simple to navigate, and effective in supporting daily business activities. Through continuous evaluation and incorporation of user feedback, the researchers were able to enhance the system's usability and ensure that it provided a positive and productive experience for administrators, staff, and management personnel of Groove Monkey Pinoy Korean Bistro.

Overall, the usability testing phase demonstrated that the web-based management system was accessible, user-friendly, and capable of supporting the operational needs of Groove Monkey Pinoy Korean Bistro. The positive feedback received from participants and the successful completion of assigned tasks indicated that the system could be adopted effectively by its intended users while promoting efficiency, accuracy, and ease of use.

User Acceptance Testing (UAT)

User Acceptance Testing was conducted to determine whether the developed system met the operational requirements and expectations of Groove Monkey Pinoy Korean Bistro. Representatives from the target users evaluated the system by performing actual business-related tasks. The results of this evaluation served as the basis for determining the system's readiness for deployment and implementation.

User Acceptance Testing (UAT) represented the final stage of evaluation before the proposed system was considered ready for implementation. This phase focused on determining whether the developed application met the actual needs of the organization and functioned effectively within a realistic operational environment. The participation of intended users ensured that the evaluation reflected real-world business requirements.

The primary objective of UAT was to validate that the system fulfilled all functional and operational requirements identified during the planning and analysis stages of the project. Users assessed whether the features provided by the system aligned with their daily responsibilities and whether the application could effectively support business operations.

Representatives from management, cashiers, and staff members participated in the evaluation process. These individuals were selected because of their direct involvement in the activities that the system was designed to support. Their feedback provided valuable insights into the practicality, efficiency, and usefulness of the application in actual workplace scenarios.

During the testing period, participants were instructed to perform tasks that closely resembled their regular operational duties. These tasks included processing customer orders, updating inventory records, generating reports, managing transactions, and accessing administrative functions. Observing users perform these activities allowed the researchers to evaluate the system under realistic conditions.

The researchers carefully monitored user interactions and documented observations throughout the testing process. Particular attention was given to the ease of task completion, the accuracy of system outputs, and the overall effectiveness of system workflows. Any issues encountered during testing were recorded for further analysis and possible refinement.

User feedback was collected through evaluation forms, interviews, and discussions conducted after testing sessions. Participants were encouraged to provide comments regarding system functionality, interface design, performance, and overall satisfaction. This feedback served as an important source of information for assessing system quality and identifying potential improvements.

The evaluation also focused on determining whether the system improved existing operational processes. Users compared the developed application with previous manual procedures and assessed whether the system provided benefits such as increased efficiency, improved accuracy, faster transaction processing, and better information management.

Another important aspect of UAT involved evaluating user confidence in utilizing the system. Participants assessed whether they felt comfortable performing their assigned tasks using the application and whether additional training would be required. Positive responses indicated that the system was intuitive and suitable for everyday use within the organization.

The results of User Acceptance Testing demonstrated that the majority of participants were satisfied with the performance and functionality of the system. Users reported that the application successfully addressed many of the operational challenges associated with manual processes, including record management, inventory tracking, and transaction processing. These findings supported the system's readiness for deployment.

The User Acceptance Testing phase also provided an opportunity to evaluate the system from the perspective of actual end users rather than developers. While technical testing focused on functionality and performance, UAT emphasized user satisfaction and practical applicability. This distinction ensured that the system not only operated correctly but also met the expectations of those who would use it on a daily basis.

During the evaluation process, users assessed whether the system's features were relevant to their operational needs. They examined the effectiveness of the ordering, billing, inventory management, and reporting modules in supporting their responsibilities. The feedback received indicated that the integrated nature of the system simplified many routine tasks previously performed manually.

The researchers encouraged participants to use the system under conditions similar to normal business operations. By simulating actual workplace scenarios, users were able to evaluate how well the application supported transaction processing, inventory monitoring, and information retrieval. This approach provided a realistic assessment of system performance in a practical environment.

Another objective of UAT was to determine whether the system reduced the complexity of existing procedures. Participants compared the new system with traditional manual methods and assessed whether tasks could be completed more efficiently. Many users observed that the system streamlined workflows and reduced the amount of time required to perform administrative activities.

The testing process also measured the accuracy of information generated by the system. Users reviewed transaction records, inventory reports, sales summaries, and other outputs to verify that the displayed information was complete and correct. Accurate reporting is essential for supporting business decisions and maintaining reliable organizational records.

Participants were given the opportunity to identify any remaining issues or areas requiring improvement. Their observations provided valuable insights into aspects of the system that could be enhanced further. This feedback allowed the researchers to implement final refinements before considering the system fully ready for deployment.

The researchers also evaluated the consistency of system performance throughout the testing period. Multiple users accessed different modules and performed various operations to determine whether the application maintained stable performance under typical usage conditions. The successful completion of these activities demonstrated the reliability of the system.

An important aspect of UAT involved assessing user confidence in adopting the new technology. Since successful implementation depends on user acceptance, participants were asked whether they felt comfortable utilizing the system as part of their regular duties. Positive responses suggested that the system's design and functionality supported smooth adoption within the organization.

The results of the evaluation further indicated that the system achieved its intended objectives of improving operational efficiency, reducing manual workload, and enhancing information management. Users reported that the application simplified business processes and provided easier access to important operational data. These benefits highlighted the practical value of the developed system.

In conclusion, the extended User Acceptance Testing process provided strong evidence of the system's readiness for implementation. Through direct user involvement, practical task execution, and comprehensive feedback collection, the researchers were able to confirm that the web-based management system effectively met organizational requirements and was capable of supporting the operational goals of Groove Monkey Pinoy Korean Bistro.

User Acceptance Testing provided confirmation that the web-based management system met the expectations and requirements of Groove Monkey Pinoy Korean Bistro. The positive evaluation results indicated that the application was capable of supporting business operations effectively, enhancing productivity, and improving overall management efficiency. Consequently, the system was considered suitable for implementation and operational use within the establishment.

3.7 Evaluation Criteria

The evaluation criteria were established to assess the quality, functionality, usability, and overall performance of the proposed web-based management system for Groove Monkey Pinoy Korean Bistro. These criteria served as the basis for determining whether the developed system successfully met its intended objectives and satisfied the requirements of its target users.

The evaluation was conducted through a structured assessment involving selected respondents, including administrators, staff members, and management personnel. Participants were asked to evaluate the system after using its various features and functionalities. Their responses were collected using an evaluation questionnaire designed to measure different aspects of system quality.

The researchers adopted the ISO/IEC 25010 Software Quality Model as the framework for evaluating the developed system. This internationally recognized standard provides a comprehensive set of quality characteristics used to assess software products. The selected evaluation criteria included Functional Suitability, Performance Efficiency, Usability, Reliability, Security, and Maintainability.

The evaluation criteria also ensured that the assessment of the system was objective and standardized. By using clearly defined measures, the researchers were able to reduce subjectivity in interpreting user feedback. This approach helped produce more reliable and consistent evaluation results that accurately reflected the system's performance.

Each criterion was carefully aligned with the specific objectives of the study to ensure relevance. The researchers made sure that every aspect of the system being evaluated directly corresponded to its intended functions, such as ordering, billing, inventory tracking, and reporting. This alignment strengthened the validity of the evaluation process.

In applying the evaluation criteria, the researchers considered both technical performance and user experience. While system efficiency and functionality were assessed from a technical perspective, usability and satisfaction were evaluated based on user interaction. This balanced approach ensured a comprehensive assessment of the system.

The evaluation process also emphasized the importance of real-world usage conditions. Respondents were encouraged to assess the system based on actual tasks they would normally perform in the bistro. This helped ensure that the evaluation results accurately reflected how the system would perform in a real operational environment.

To further strengthen the evaluation process, respondents were given sufficient time to explore and use the system before answering the questionnaire. This allowed them to fully experience the system's features and functionalities, resulting in more informed and meaningful feedback.

The researchers ensured that the evaluation instrument used clear and understandable statements to avoid confusion among respondents. Each question in the questionnaire was designed to focus on a specific aspect of system quality, making it easier for participants to provide accurate and honest responses.

Data collected from the evaluation were systematically organized and analyzed using statistical tools. Weighted mean was used to determine the average rating for each criterion, allowing the researchers to interpret the overall performance level of the system. This quantitative approach contributed to the objectivity of the evaluation results.

The ISO/IEC 25010 model provided a structured framework that enhanced the credibility of the evaluation process. By adhering to internationally recognized standards, the researchers ensured that the system was assessed using well-established software quality metrics. This made the findings more reliable and academically acceptable.

The evaluation criteria also served as a guide for identifying areas of improvement in the system. Lower-rated aspects were carefully analyzed to determine possible enhancements that could be implemented in future updates. This continuous improvement approach ensured that the system could evolve based on user feedback.

Overall, the use of well-defined evaluation criteria allowed the researchers to comprehensively assess the effectiveness of the webbased management system. The structured evaluation process confirmed that the system met the required standards of quality, usability, and functionality, making it suitable for implementation in Groove Monkey Pinoy Korean Bistro.

Functional Suitability

Functional Suitability refers to the degree to which the system provides functions that meet specified user requirements. This criterion evaluates whether the system can accurately perform essential tasks such as order processing, billing, inventory management, report generation, and user account management. A high rating in this area indicates that the system effectively supports business operations and fulfills user needs.

Functional Suitability also measures how completely the system's features cover all identified business requirements. In the context of Groove Monkey Pinoy Korean Bistro, this means ensuring that every operational need identified during the requirements analysis phase is properly addressed by a corresponding system function. The completeness of these features is essential in determining the overall effectiveness of the system.

In evaluating functional suitability, the researchers also considered the correctness of system outputs. Each module was examined to ensure that it produces accurate and reliable results when given valid inputs. For example, order totals, inventory deductions, and generated reports were checked against expected values to confirm that the system performs computations correctly.

Another important aspect of functional suitability is the appropriateness of system functions for actual business processes. The system was evaluated to determine whether its features align with how the bistro operates on a daily basis. This includes assessing whether workflows in ordering, billing, and inventory management reflect real-world restaurant procedures.

The researchers also assessed the consistency of system functions across different modules. Functional consistency ensures that similar operations behave in a uniform manner throughout the system. This helps users develop familiarity with the system and reduces the likelihood of confusion or operational errors.

Functional suitability further examines the system's ability to handle both routine and exceptional business scenarios. The system was tested not only under normal conditions but also under unusual or edge cases such as cancelled orders, modified transactions, and inventory adjustments. This ensured that the system could function properly under varying operational conditions.

The degree of automation provided by the system was also considered under this criterion. By automating tasks such as inventory updates, billing calculations, and report generation, the system reduces the need for manual intervention. This automation enhances accuracy and efficiency in daily operations.

Another factor evaluated was the system's ability to support user roles and responsibilities effectively. Different users such as administrators, cashiers, and staff members require specific functionalities based on their tasks. Functional suitability ensures that each user role is provided with appropriate access and tools to perform their duties efficiently.

The researchers also examined whether the system supports error-free transaction processing. This includes verifying that invalid inputs are properly handled and that the system prevents incorrect data from being processed. Effective error handling contributes to the overall reliability and correctness of system functions.

Functional suitability also includes evaluating how well the system supports decision-making processes within the business. By providing accurate reports and real-time data, the system enables management to make informed decisions regarding inventory control, sales performance, and operational planning.

Overall, the assessment of functional suitability confirms that the web-based management system is capable of performing all required functions effectively and accurately. The system's ability to meet user requirements, support business processes, and produce correct outputs demonstrates its effectiveness in improving the operational efficiency of Groove Monkey Pinoy Korean Bistro.

Performance Efficiency

Performance Efficiency measures how well the system performs under specified conditions. This includes evaluating system responsiveness, processing speed, and efficient utilization of resources. The criterion determines whether users can complete transactions and access information without experiencing significant delays or interruptions.

Performance Efficiency also evaluates the system's ability to maintain acceptable response times during routine operations. In the context of the web-based management system for Groove Monkey Pinoy Korean Bistro, this means ensuring that actions such as logging in, processing orders, and generating reports are executed within a reasonable time frame without causing delays to users.

The researchers assessed how quickly the system responds when multiple modules are accessed consecutively. Since restaurant operations often require fast decision-making and continuous transaction processing, the system must be able to handle rapid user interactions without performance degradation.

Another important consideration under performance efficiency is system throughput, which refers to the number of transactions the system can process within a given period. The system was evaluated to determine whether it could handle multiple orders, inventory updates, and billing processes efficiently during peak operational hours.

Resource utilization was also examined to ensure that the system does not excessively consume hardware or network resources. Efficient use of CPU, memory, and storage contributes to smoother system performance and ensures that the application remains stable even when multiple users are active simultaneously.

The researchers also evaluated the system's performance under different workload conditions. This included testing both light and heavy usage scenarios to determine whether the system could maintain consistent performance regardless of the volume of transactions being processed.

Database performance played a significant role in evaluating efficiency. The speed of data retrieval and storage operations was assessed to ensure that queries related to orders, inventory, and reports could be executed quickly and accurately without causing system delays.

Another aspect considered was the optimization of system processes. The researchers examined whether redundant operations were minimized and whether system workflows were structured in a way that supports faster processing. Efficient system logic contributes significantly to overall performance.

The responsiveness of the user interface was also evaluated under this criterion. The system was tested to determine whether pages load quickly and whether user actions such as button clicks and form submissions provide immediate feedback. A responsive interface enhances user experience and operational efficiency.

Performance efficiency also includes evaluating system stability during continuous usage. The system was observed over extended periods to determine whether it could maintain consistent performance without crashing, slowing down, or experiencing errors during prolonged operation.

Overall, the assessment of performance efficiency indicates that the system is capable of delivering fast, stable, and reliable performance under expected operational conditions. The ability to process transactions quickly, manage resources efficiently, and maintain responsiveness ensures that the system effectively supports the daily operations of Groove Monkey Pinoy Korean Bistro.

Usability

Usability assesses the ease with which users can learn, operate, and interact with the system. This criterion focuses on factors such as interface design, navigation, accessibility, user-friendliness, and overall user experience. A highly usable system enables users to perform tasks efficiently with minimal training and assistance.

Usability also evaluates how quickly new users can become familiar with the system without extensive technical training. In the case of Groove Monkey Pinoy Korean Bistro, the system was designed so that staff and administrators could understand its basic functions within a short period of use. This reduces the learning curve and supports faster adaptation to the new system.

The clarity of the system interface is another important factor in usability. Clear labeling of buttons, menus, and input fields helps users easily understand the purpose of each function. A wellorganized interface reduces confusion and allows users to complete tasks with greater confidence and accuracy.

Navigation structure plays a significant role in determining usability. The system was evaluated based on how easily users can move between modules such as ordering, billing, inventory, and reporting. Intuitive navigation ensures that users do not waste time searching for features and can focus more on operational tasks.

Consistency in design is also considered under usability. Similar elements across different pages of the system were designed to behave in a uniform manner. This consistency helps users predict system behavior, which improves efficiency and reduces the likelihood of errors during operation.

Another aspect of usability is the accessibility of system features. The system was assessed to ensure that important functions are easy to locate and use. Frequently used operations, such as order entry and inventory updates, were made readily accessible to improve workflow efficiency.

The system’s ability to provide helpful feedback to users was also evaluated. Notifications, alerts, and error messages were designed to clearly inform users about the results of their actions. This feedback helps users understand system responses and correct mistakes when necessary.

Usability also considers the visual appeal of the system interface. A clean and organized layout contributes to a more pleasant user experience and reduces cognitive load. Proper spacing, readable fonts, and balanced design elements help users interact with the system more comfortably.

Error prevention and correction features are also important usability factors. The system was evaluated based on its ability to minimize user mistakes through input validation and confirmation prompts. When errors occur, the system provides clear instructions to help users resolve them efficiently.

The efficiency of task completion was also assessed under usability criteria. Users were observed performing typical operations to determine whether tasks could be completed in fewer steps compared to manual processes. A more efficient system improves productivity and reduces operational workload.

Overall, the usability evaluation indicates that the system is userfriendly, intuitive, and suitable for daily business operations. Its clear interface design, logical navigation, and supportive feedback mechanisms enable users to perform tasks effectively, making it highly appropriate for implementation in Groove Monkey Pinoy Korean Bistro.

Reliability

Reliability refers to the system's ability to perform its intended functions consistently and accurately over time. This criterion evaluates system stability, error handling, data consistency, and operational dependability. Reliable systems minimize disruptions and ensure continuous support for business activities.

Reliability also evaluates the system’s ability to operate continuously without unexpected failures or interruptions. In the context of Groove Monkey Pinoy Korean Bistro, the system is expected to function consistently during business hours, ensuring that critical operations such as ordering, billing, and inventory management are not disrupted.

A key aspect of reliability is system stability under normal and extended usage conditions. The system was designed and tested to ensure that it does not crash or slow down significantly even when multiple transactions are being processed consecutively. Stable performance ensures smooth daily operations in a fast-paced restaurant environment.

Error handling is another important component of reliability. The system was evaluated based on its ability to detect, manage, and respond appropriately to errors. When invalid inputs or unexpected situations occur, the system provides clear error messages and prevents system breakdowns, ensuring continued operation.

Data consistency plays a vital role in ensuring reliability. The system was assessed to confirm that all data stored in the database remains accurate and synchronized across all modules. This ensures that updates in ordering, billing, and inventory are reflected correctly throughout the system.

The system's ability to recover from failures was also considered under this criterion. In cases of system interruptions, such as unexpected shutdowns or connection issues, the system is designed to preserve data integrity and allow recovery without significant data loss. This capability enhances operational dependability.

Another factor evaluated is transaction reliability, which ensures that all completed operations are properly recorded in the system. Each order, payment, and inventory update is stored accurately, preventing duplication or missing records that could affect business operations.

The researchers also examined the system's consistency in handling repeated operations. Tasks performed multiple times under similar conditions should produce the same results. This consistency ensures that users can trust the system's outputs for daily decision-making and business monitoring.

System availability is also an important aspect of reliability. The system was evaluated to ensure that it remains accessible to authorized users whenever needed during operational hours. High availability reduces downtime and ensures uninterrupted support for business activities.

The reliability of the database was also assessed, particularly in terms of data integrity and secure storage. The system ensures that records are properly saved, updated, and retrieved without corruption or loss. This strengthens the overall trustworthiness of the system.

Overall, the reliability evaluation confirms that the system is stable, consistent, and dependable in supporting the operations of Groove Monkey Pinoy Korean Bistro. Its ability to handle errors, maintain data integrity, and operate continuously ensures that it can be trusted for daily business activities and long-term use.

Security

Security measures the system's ability to protect information and prevent unauthorized access. This criterion evaluates user authentication, access control mechanisms, data protection measures, and the confidentiality of business records. Strong security helps safeguard sensitive information and maintain system integrity.

Security also evaluates how effectively the system safeguards sensitive business data from both internal and external threats. In the context of Groove Monkey Pinoy Korean Bistro, this includes protecting customer orders, financial records, inventory data, and user credentials from unauthorized access or manipulation. Ensuring data protection is essential for maintaining trust and operational integrity.

User authentication is a primary component of the system’s security framework. The system requires users to log in using valid credentials before accessing any features. This ensures that only registered and verified users can interact with the system, reducing the risk of unauthorized entry.

Access control mechanisms were also implemented and evaluated to ensure that each user role is granted appropriate permissions. Administrators, cashiers, and staff members have different access levels depending on their responsibilities. This role-based access control prevents users from accessing or modifying data that is beyond their authority.

Data confidentiality is another critical aspect of system security. The system was designed to ensure that sensitive information such as sales records and inventory data is only visible to authorized personnel. This helps prevent information leakage and ensures that business operations remain confidential.

The system also incorporates measures to protect data during transmission between the client and server. Secure communication protocols help reduce the risk of data interception or unauthorized access while information is being transferred. This ensures that all transactions remain secure within the system environment.

Database security was evaluated to ensure that stored data is protected from unauthorized modifications, deletions, or corruption. Proper database management practices were applied to maintain data integrity and ensure that all records remain accurate and consistent over time.

Another important aspect of security is session management. The system was assessed to ensure that user sessions are properly managed, including automatic logout after periods of inactivity.

This prevents unauthorized users from accessing active sessions left unattended by legitimate users.

The system also includes validation mechanisms to prevent malicious or incorrect data input. Input validation helps protect the system from common security threats such as SQL injection and invalid data entries, thereby enhancing overall system protection.

Audit trails and logging features were also considered in the security evaluation. The system records user activities such as logins, transactions, and data modifications. These logs provide accountability and allow administrators to monitor system usage and detect any suspicious activities.

Overall, the security evaluation confirms that the system provides strong protection for business data and user information. Through authentication, access control, data confidentiality, secure communication, and activity monitoring, the system ensures a secure operational environment for Groove Monkey Pinoy Korean Bistro.

Maintainability

Maintainability assesses the ease with which the system can be modified, updated, corrected, or enhanced. This criterion evaluates the structure, organization, and flexibility of the system. A maintainable system allows future improvements to be implemented efficiently as business requirements evolve.

To measure user perceptions, a **five-point Likert Scale** was utilized in the evaluation questionnaire. Respondents were asked to indicate their level of agreement with statements related to each evaluation criterion.

Maintainability also refers to how easily the system can be understood by developers or future maintainers. A well-structured system with clear code organization allows future programmers to quickly interpret how each module functions, reducing the time needed for troubleshooting and enhancements.

The system was designed using modular development principles to improve maintainability. Each functional component, such as ordering, billing, inventory, and reporting, was separated into distinct modules. This separation allows developers to modify one part of the system without significantly affecting other components.

Another important aspect of maintainability is code readability. The system was developed using clear naming conventions, proper indentation, and logical structuring of code. This ensures that future developers can easily identify functions and understand system logic without unnecessary complexity.

Documentation also plays a critical role in system maintainability. The researchers provided system documentation that includes module descriptions, database structure, and system workflows. This documentation serves as a guide for future updates, debugging, and system enhancements.

The flexibility of the system was also evaluated under maintainability. The system was designed to accommodate future changes such as additional features, updated business rules, or expanded operational requirements. This flexibility ensures that the system remains useful even as the business grows or evolves.

Error correction and debugging capabilities were also considered. The system was evaluated based on how easily issues can be identified and resolved. A maintainable system allows errors to be traced efficiently, reducing downtime and improving system reliability during updates or fixes.

Database structure also contributes significantly to maintainability. The system uses an organized and normalized database design that minimizes redundancy and ensures data consistency. This structure makes it easier to update, expand, or modify data without affecting the entire system.

The researchers also considered the ease of implementing system upgrades. Maintainable systems allow new features or improvements to be integrated without requiring a complete redesign. This ensures that the system can continuously evolve alongside technological advancements and business needs.

Version control practices were also acknowledged as part of maintainability. Proper tracking of system changes allows developers to monitor updates, revert to previous versions if necessary, and maintain stability during development. This ensures controlled and organized system evolution.

Overall, the maintainability evaluation confirms that the system is structured, flexible, and easy to update or modify. Its modular design, clear documentation, and organized codebase ensure that future improvements can be implemented efficiently, supporting the long-term sustainability of the web-based management system for Groove Monkey Pinoy Korean Bistro.

Scale Range		**Verbal Interpretation**
5	4.21 – 5.00	Excellent
4	3.41 – 4.20	Very Good
3	2.61 – 3.40	Good
2	1.81 – 2.60	Fair
1	1.00 – 1.80	Poor

The weighted mean was computed to determine the overall evaluation score for each criterion. The results were analyzed and interpreted to assess the acceptability and effectiveness of the developed webbased management system.

The evaluation criteria provided a systematic method for measuring the quality and performance of the proposed application. Through these criteria, the researchers were able to determine whether the system successfully achieved its objectives of improving operational efficiency, enhancing customer service, ensuring data accuracy, and supporting the overall management needs of Groove Monkey Pinoy Korean Bistro.

Furthermore, the evaluation results served as the basis for identifying potential improvements and future enhancements. Feedback obtained from respondents contributed valuable insights regarding system strengths and areas that may require additional refinement. This process ensured that the final system was aligned with user expectations and capable of delivering meaningful benefits to the organization.

The computation of the weighted mean provided a quantitative basis for interpreting the evaluation results in a more objective manner. By assigning numerical values to respondents' ratings, the researchers were able to systematically analyze perceptions of system performance across different criteria. This method helped reduce subjectivity in interpreting user feedback.

Each criterion's weighted mean was carefully calculated to reflect the collective responses of all participants. This allowed the researchers to determine which aspects of the system performed strongly and which areas required improvement. The use of statistical analysis ensured that conclusions were based on measurable data rather than personal judgment.

The interpretation of the weighted mean followed a predefined scale, which allowed the researchers to categorize results into descriptive levels such as "Poor," "Fair," "Good," "Very Good," and "Excellent." This standardized interpretation made it easier to communicate findings in a clear and understandable manner.

The evaluation results also helped establish the overall acceptability of the system. High weighted mean scores across most criteria indicated that users found the system satisfactory and suitable for daily operations. This acceptance is essential for successful implementation within Groove Monkey Pinoy Korean Bistro.

In addition to measuring acceptability, the results also reflected the effectiveness of the system in addressing identified operational problems. Positive evaluations in areas such as usability, functionality, and performance efficiency demonstrated that the system successfully improved business processes.

The systematic evaluation process also allowed the researchers to compare different aspects of the system in relation to one another. By examining variations in weighted mean scores across criteria, they were able to identify which features performed exceptionally well and which required further refinement.

The results of the evaluation served as a foundation for decisionmaking regarding system deployment. A strong overall performance rating supported the conclusion that the system was ready for implementation. This ensured that the final product met both technical and user requirements.

User feedback gathered during the evaluation process provided qualitative support for the quantitative findings. Respondents' comments and suggestions helped explain the numerical results and provided deeper insight into user experiences. This combination of data strengthened the validity of the study.

The identification of potential improvements based on evaluation results contributed to the continuous enhancement of the system. By addressing areas that received lower ratings, the researchers were able to refine system features and improve overall performance and usability.

Overall, the use of weighted mean analysis and structured evaluation criteria ensured a comprehensive assessment of the web-based management system. The findings confirmed that the system effectively meets its intended objectives and provides significant benefits in improving operational efficiency, enhancing service delivery, and supporting the management needs of Groove Monkey Pinoy Korean Bistro.

Chapter 4

SYSTEM DEVELOPMENT, IMPLEMENTATION, TESTING, EVALUATION,RESULTS, AND DISCUSSION

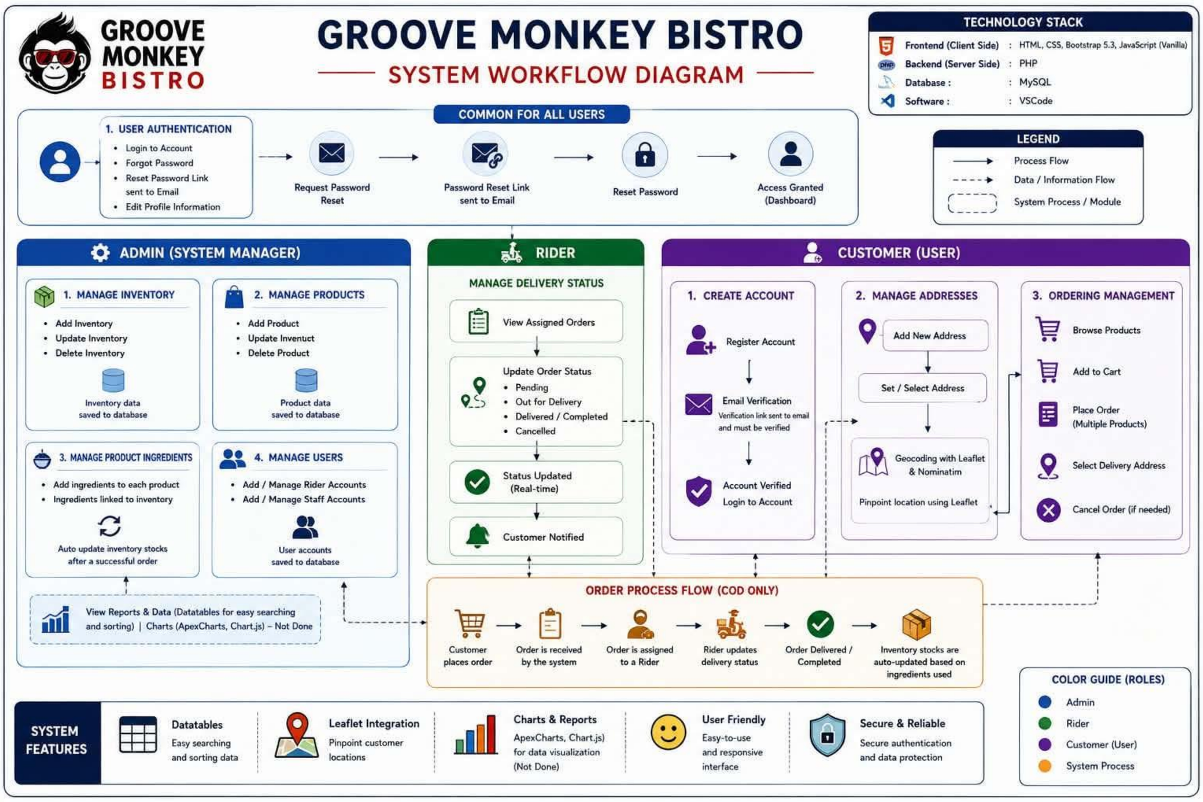
4.2 Overview of the developed system

The developed system is a web-based management system designed for Groove Monkey Pinoy Korean Bistro to improve its daily business operations by replacing manual processes with an integrated digital platform. The system was developed to address the problems identified in the study, particularly manual order processing, inaccurate inventory monitoring, and inefficient record management. It provides a centralized solution that enables administrators, cashiers, and staff to perform their responsibilities more efficiently through a web browser.

The system consists of four major modules: ordering, billing, inventory management, and report generation. The ordering module allows staff to record customer orders electronically, minimizing order errors and improving communication between the service staff and the kitchen. The billing module automatically computes customer purchases and generates accurate transaction records, reducing manual computation errors and speeding up the payment process. The inventory management module monitors stock levels and inventory movement in real time, enabling management to maintain accurate inventory records and make timely purchasing decisions. Meanwhile, the reporting module generates sales and operational reports that assist administrators in monitoring business performance and making informed decisions.

The developed system follows a client-server architecture, where users access the application through a web interface while all business data is processed and stored in a centralized database. This architecture ensures data consistency, security, and accessibility for authorized users. The typical workflow begins when a user logs into the system, selects a desired function such as order processing or inventory management, submits the request, and receives the processed results in real time. This approach allows multiple users to access accurate and updated information simultaneously.

To protect business information, the system implements user authentication and role-based access control. Administrators are granted full access to system settings, inventory records, reports, and user management, while cashiers and staff are limited to the functions necessary for their assigned responsibilities. This security mechanism helps maintain data integrity and prevents unauthorized access to sensitive information.



4.3 System Development Methodology

The development of the Groove Monkey Pinoy Korean Bistro Web-Based Ordering and Inventory Management System followed the Agile Software Development Methodology. Agile was selected because it emphasizes continuous planning, development, testing, and user feedback throughout the project. This methodology enabled the developers to build the system incrementally while ensuring that the requirements of the bistro were met effectively. Throughout the development process, regular evaluations and revisions were conducted to improve the functionality, usability, and reliability of the system based on user feedback and testing results. The methodology also allowed the developers to respond quickly to changes in system requirements and operational needs identified during development.

The Agile methodology consists of the following phases:

1. Planning and Requirements Gathering

During this phase, the researchers identified the operational problems encountered by Groove Monkey Pinoy Korean Bistro through interviews, observations, and analysis of the existing manual system. The business requirements were gathered, including the need for digital order processing, inventory management, sales monitoring, and report generation. The system objectives and functional requirements were also established based on the identified problems.

2. System Design

After gathering the requirements, the researchers designed the overall architecture of the system. This included creating the database structure, system flow, user interfaces, and navigation. The design focused on providing a user-friendly environment that allows administrators and staff to perform transactions efficiently while ensuring data accuracy and security.

3. Development

In this phase, the researchers developed the different modules of the system in small iterations. Core functionalities such as user authentication, order management, inventory management, billing, and report generation were implemented and integrated gradually. Each completed module was evaluated before proceeding to the next iteration to ensure that it satisfied the required specifications.

4. Testing

Each module underwent functional testing to verify that it performed according to the specified requirements. Errors, bugs, and system issues identified during testing were corrected immediately before proceeding with further development. User testing was also conducted to evaluate the system's usability, reliability, and performance in actual business operations.

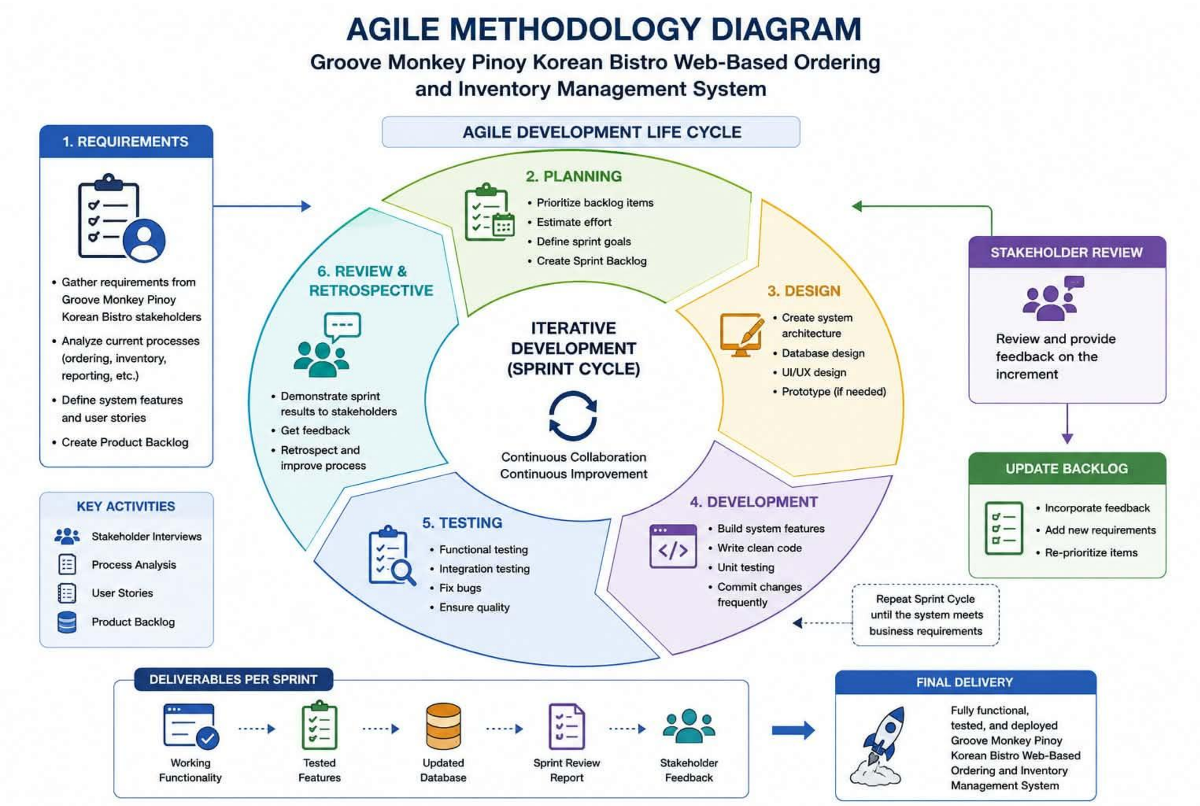
5. Deployment

After successful testing, the completed system was deployed for implementation at Groove Monkey Pinoy Korean Bistro. The system became available for use by authorized personnel, replacing the manual processes previously used in order processing, inventory monitoring, and report generation.

6. Maintenance and Continuous Improvement

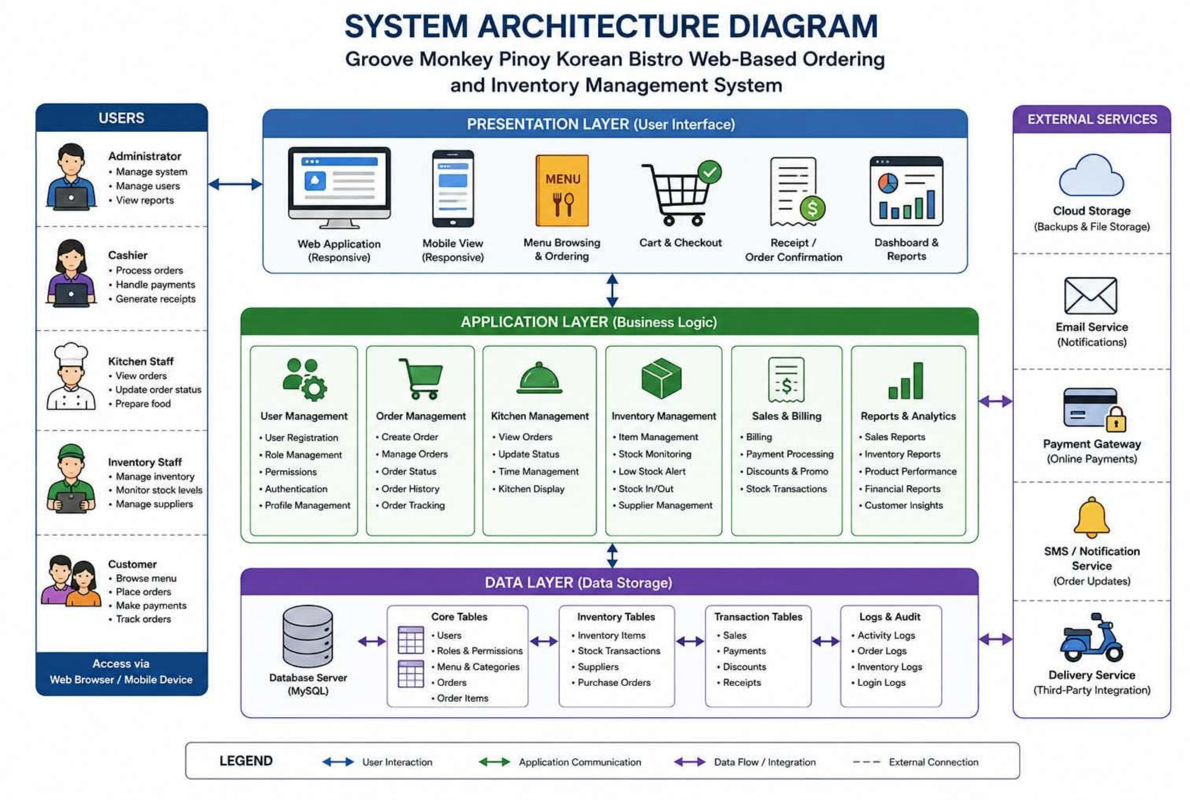
Following deployment, the system continues to be monitored to ensure its stability and effectiveness. User feedback, identified issues, and future business requirements may be incorporated into subsequent updates and enhancements. This continuous improvement process allows the system to remain responsive to the changing operational needs of the business while maintaining high performance and reliability.

The Agile Software Development Methodology enabled the researchers to develop a reliable and efficient web-based ordering and inventory management system through continuous planning, iterative development, testing, and user evaluation. By following this methodology, the resulting system effectively addressed the operational problems identified in the study and provided a scalable solution that improves order processing, inventory control, reporting, and overall business management.



4.4 System Architecture

The Groove Monkey Pinoy Korean Bistro Web-Based Ordering and Inventory Management System adopts a three-tier client-server architecture consisting of the presentation layer, application layer, and database layer. This architecture enables efficient communication between users and the system while ensuring centralized data storage, security, and reliability. The system allows authorized users to access the application through a web browser, where all requests are processed by the application server before being stored or retrieved from the database. This architecture supports real-time order processing, inventory updates, sales recording, and report generation, addressing the operational problems identified in the study.



Presentation Layer

The presentation layer serves as the user interface where administrators, cashiers, and staff interact with the system. Through this layer, users can log in, manage customer orders, process payments, update inventory records, and generate reports. The interface is designed to be simple and user-friendly to minimize errors and improve operational efficiency.

Major interface components include:

- Login Page
- Dashboard
- Product Management
- Customer Management ● Sales Transactions
- Analytics Dashboard
- Reports

Application Layer

The application layer contains the business logic of the system. It processes user requests, validates data, computes customer bills, updates inventory quantities after every transaction, manages user authentication, and generates reports. This layer serves as the bridge between the user interface and the database, ensuring that all business rules are properly executed before data is stored or displayed.

Its primary responsibilities include:

- User Authentication
- Transaction Processing
- Customer Profiling
- Sales Analysis
- Product Trend Analysis
- Dashboard Generation
- Report Generation

Database Layer

The database layer functions as the centralized repository of all system information. It stores user accounts, menu items, customer orders, inventory records, sales transactions, and generated reports. Because all information is maintained in a single database, authorized users can access accurate and up-to-date data whenever needed, improving decision-making and reducing data redundancy.

Stored information includes:

- User Accounts
- Products
- Categories
- Customers
- Sales Transactions
- Transaction Details
- Behavioral Analytics
- Reports

The overall flow of the system begins when an authorized user logs into the application. The user's request is sent to the application server, where it is validated and processed. If database access is required, the application communicates with the database server to retrieve or update the necessary information. The processed results are then returned to the user interface and displayed immediately. This workflow enables real-time transaction processing while maintaining data integrity and security.

The three-tier architecture provides several advantages for the developed system. It improves system performance by separating the user interface, business processes, and database functions into independent layers. It also enhances scalability, making future enhancements such as online ordering, customer accounts, or mobile application integration easier to implement. Furthermore, centralized data management improves record accuracy, minimizes duplicate information, strengthens data security, and supports efficient business operations.

4.5 Hardware and Software Requirements

The successful implementation of the Groove Monkey Pinoy Korean Bistro Web-Based Ordering and Inventory Management System requires appropriate hardware and software resources to ensure reliable performance, efficient processing, and secure data management. The specified requirements provide the minimum and recommended specifications needed for the development, deployment, and daily operation of the system.

4.5.1 Hardware Requirements

The following hardware components are required to run the developed system efficiently:

Hardware Component	Minimum Specification	Recommended Specification
Processor	Intel Core i3 or AMD Ryzen 3	Intel Core i5 or AMD Ryzen 5
Memory (RAM)	4 GB	8 GB or higher
Storage	128 GB SSD or 500 GB HDD	256 GB SSD or higher
Monitor	1366 × 768 Resolution	1920 × 1080 Full HD
Keyboard and Mouse	Standard USB Keyboard and Mouse	Standard USB Keyboard and Mouse
Internet Connection	10 Mbps	20 Mbps or higher
Printer (Optional)	Thermal Receipt Printer	Thermal Receipt Printer

The hardware requirements ensure that users can efficiently perform order processing, inventory monitoring, billing, and report generation without significant delays.

4.5.2 Software Requirements

The developed system requires the following software environment for development and deployment:

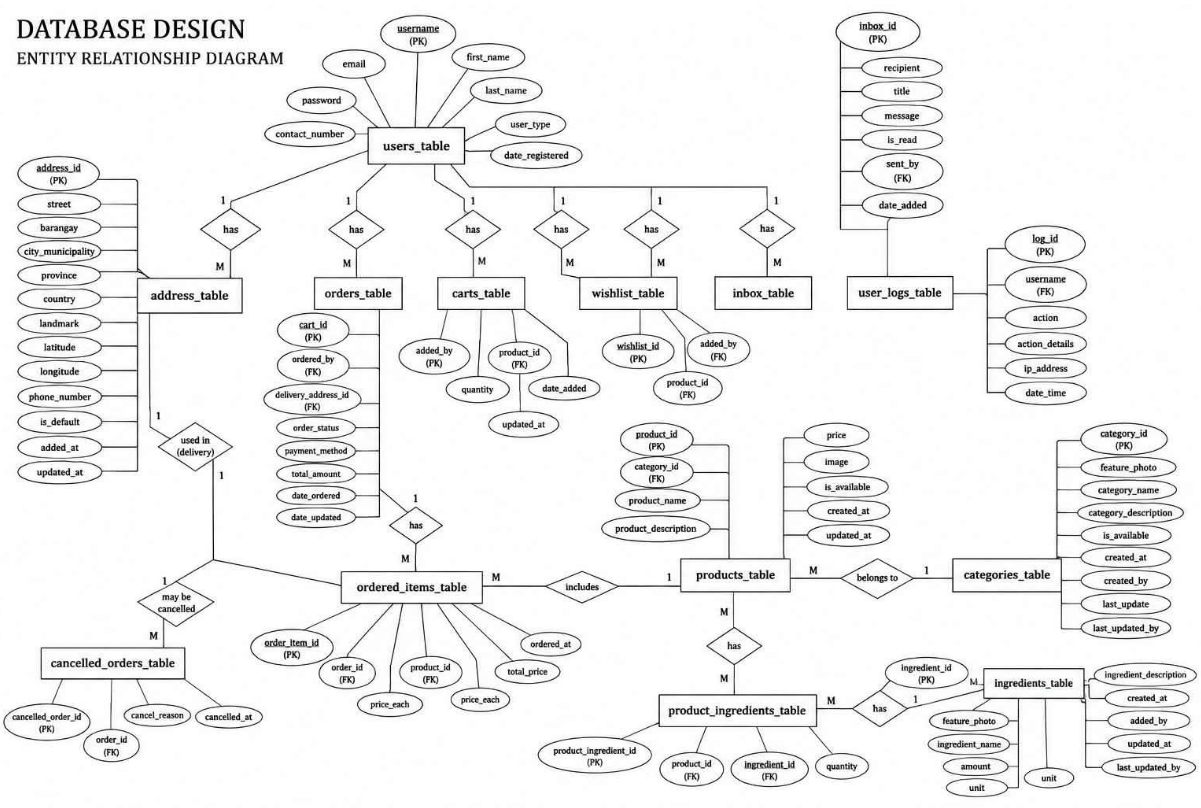
Software	Purpose
Operating System	Windows 10 or Windows 11
Web Browser	Google Chrome, Microsoft Edge, or Mozilla Firefox
Web Server	XAMPP (Apache Server)
XAMPP (Apache Server)	PHP
Database Management System	MySQL
Front-end Technologies	HTML5, CSS3, JavaScript, Bootstrap
Code Editor	Visual Studio Code
Database Administration	phpMyAdmin

The software requirements provide the necessary environment for executing the web application, managing the database, and maintaining the system. The use of PHP and MySQL enables efficient processing of transactions and secure storage of business information, while HTML, CSS, JavaScript, and Bootstrap provide a responsive and user-friendly interface.
The selection of these software tools was based on their stability, community support, compatibility, and widespread adoption in modern web application development.

4.6 Database Design

The database of the Groove Monkey Pinoy Korean Bistro Web-Based Ordering and Inventory Management System was designed to provide an organized, secure, and centralized repository for all business information. It supports the core functionalities of the system, including user authentication, order processing, inventory management, billing, and report generation. The database utilizes a relational database model implemented in MySQL, where related information is stored in separate tables connected through primary and foreign keys. This structure minimizes data redundancy, improves data integrity, and enables efficient retrieval of information during daily operations.

The database consists of several interconnected tables that store operational data required by the system. Each table serves a specific purpose while maintaining relationships with other tables to ensure consistency and accuracy of information.



Users Table

The Users table stores the login credentials and account information of authorized system users. It is responsible for user authentication and role-based access control.

Field name	Data Type	Description
User_id	INT(primary Key)	Unique user identification number
Full_name	VARCHAR(100)	Full name of the user
username	VARCHAR(50)	Username use for login
password	VARCHAR(255)	Encrypted user password
role	VARCHAR(20)	User role(Administrator,Cashier,Staff)

Menu Table

The Menu table contains all food and beverage items offered by the bistro.

Field Name	Data Type	Description
Menu_id	INT(Primary Key)	Unique menu item ID
Item_name	VARCHAR(100)	Name of the menu item
categorty	VARCHAR(50)	Food or beverage category
Price	VARCHAR(10,2)	Selling price

Status	VARCHAR(20)	Available or Unavilable

Orders Table

The Orders table records every customer transaction processed through the system.

Field Name	Data Type	Description
Order_id	INT (Primary Key)	Unique order number
Order_date	DATETIME	Date and time of transaction
Total_amount	DECIMAL(10,2)	Total customer payment
Payment_method	VARCHAR(30)	Cash or digital payment
User_id	INT (Foreign Key)	Cashier who processed the order

Order Details Table

This table stores the individual items included in each customer order.

Field Name	Data Type	Description
Detail_id	INT (Primary Key)	Unique detail ID
Order_id	INT (Foreign Key)	Associated order
Menu_id	INT (Foreign Key)	Ordered menu item
Quantity	INT	Quantity ordered subtotal
Subtotal	DECIMAL(10,2)	Total price of the item

Inventory Table

The Inventory table maintains records of ingredients and available stock.

Field Name	Data Type	Description
inventory_id	INT (Primary Key)	Inventory record ID
ingredient_name	VARCHAR(100)	Name of ingredient
quantity	INT	Current stocks quality
unit	VARCHAR(20)	Unit of measurement
reorder_level	INT	Minimum stock level

Entity Relationship Overview

The database tables are related as follows:

One User can process many Orders.

One Order can contain multiple Order Details.

One Menu Item can appear in many Order Details.

The Inventory table maintains stock information that supports menu preparation and inventory monitoring.

The relational database design ensures that the system can efficiently manage transactions, maintain accurate inventory records, generate reliable reports, and support informed decisionmaking. By organizing data into related tables, the system minimizes duplication, improves data consistency, enhances security, and provides a scalable foundation for future enhancements of the Groove Monkey Pinoy Korean Bistro Web-Based Ordering and Inventory Management System.

CHAPTER 5
CONCLUSION AND RECOMMENDATIONS

5.1 Introduction

This chapter presents the co conclusion and evaluation of the Groove Monkey Pinoy Korean Bistro Web-Based Ordering and Inventory Management System. It discusses the procedures used to verify whether the developed system meets its intended objectives and functional requirements. The chapter also presents the evaluation conducted by selected respondents to assess the system's performance in terms of functionality, usability, reliability, efficiency, and overall user satisfaction.

System testing was conducted to ensure that each module of the system, including user authentication, order processing, billing, inventory management, and report generation, functions correctly and performs according to the specified requirements. The testing process was carried out to identify and correct errors, verify data accuracy, and ensure the proper integration of all system components before implementation.

Following the testing phase, the developed system was evaluated by its intended users through a structured evaluation instrument. The respondents assessed the system based on established software quality criteria, and their responses were analyzed to determine the overall acceptability and effectiveness of the developed application.

The results presented in this chapter demonstrate whether the Groove Monkey Pinoy Korean Bistro Web-Based Ordering and Inventory Management System successfully addressed the operational problems identified in the study. The findings also provide evidence of the system's capability to improve order processing, inventory management, business reporting, and overall operational efficiency while enhancing customer service and supporting informed decisionmaking.

5.2 Summary of Findings

The study focused on the development of the Groove Monkey Pinoy Korean Bistro Web-Based Ordering and Inventory Management System to address the operational challenges associated with manual order processing, limited online presence, and inefficient inventory management. Based on the system development, testing, and evaluation, the following findings were obtained.
The developed system successfully automated the restaurant's order processing, allowing staff to record customer orders electronically, thereby minimizing manual errors and improving the speed and accuracy of service. The billing module accurately calculated customer transactions and generated reliable sales records, reducing the time required to complete payment transactions. The inventory management module effectively monitored stock levels and inventory movement in real time. This enabled management to maintain accurate inventory records, identify items requiring replenishment, and reduce the occurrence of stock shortages and excessive inventory. As a result, the system contributed to more efficient inventory control and improved resource utilization.
The reporting module generated organized sales and inventory reports that provided management with timely and accurate operational information. These reports supported better decision-making in areas such as inventory replenishment, sales monitoring, and business planning.

System testing confirmed that all major modules, including user authentication, order processing, billing, inventory management, and report generation, functioned according to the specified requirements. The integration of these modules allowed the system to operate efficiently while maintaining data accuracy, consistency, and security.

The evaluation conducted among the intended users indicated that the developed system was acceptable in terms of functionality, usability, reliability, efficiency, and overall performance. Respondents found the system easy to use, responsive, and capable of improving daily business operations.

The findings demonstrate that the Groove Monkey Pinoy Korean Bistro Web-Based Ordering and Inventory Management System successfully addressed the operational problems identified in the study. The system improved operational efficiency, enhanced inventory accuracy, streamlined business processes, supported informed managerial decision-making, and contributed to better customer service and overall business performance.

5.3 Conclusion

Based on the findings of the study, it can be concluded that the Groove Monkey Pinoy Korean Bistro Web-Based Ordering and Inventory Management System successfully achieved the objectives of the research. The developed system effectively addressed the operational problems associated with manual order processing, inaccurate inventory tracking, and inefficient record management by providing an integrated digital solution for the bistro. The implementation of the system significantly improved the efficiency of daily business operations through the automation of order processing, billing, inventory monitoring, and report generation. By reducing manual tasks and minimizing human errors, the system enabled staff to perform their responsibilities more accurately and efficiently while enhancing the overall quality of customer service.

The centralized database and automated reporting features provided management with accurate and timely operational information, allowing better monitoring of sales, inventory levels, and business performance. These capabilities supported informed decision-making and contributed to more effective resource management and operational planning.

Furthermore, the results of system testing and user evaluation indicated that the developed system was functional, reliable, userfriendly, and suitable for the operational requirements of Groove Monkey Pinoy Korean Bistro. The positive evaluation of the intended users demonstrates that the system is capable of supporting the business in improving productivity, maintaining inventory accuracy, and enhancing customer satisfaction.
Therefore, the researchers conclude that the Groove Monkey Pinoy Korean Bistro Web-Based Ordering and Inventory Management System is an effective solution for modernizing the restaurant's operations. The system provides a reliable foundation for efficient business management and has the potential to support the continued growth, competitiveness, and long-term sustainability of the establishment.

5.4 Recommendations

Based on the findings and conclusions of this study, the following recommendations are proposed to improve the Groove Monkey Pinoy

Korean Bistro Web-Based Ordering and Inventory Management System, guide future researchers, and support its successful implementation.

System Improvements

To further enhance the functionality and performance of the developed system, the following improvements are recommended:

1. Integrate online ordering and delivery management to allow customers to place orders remotely and track their order status.

2. Add online payment options such as GCash, Maya, credit/debit cards, and other digital payment gateways to provide customers with more convenient payment methods.

3. Implement a customer loyalty and rewards program that allows customers to earn points and receive discounts or promotional offers.

4. Enhance the inventory management module by incorporating automatic low-stock notifications, supplier management, and inventory forecasting to minimize stock shortages and food wastage.

5. Improve system security by implementing stronger password encryption, role-based permissions, audit logs, and regular database backup and recovery mechanisms.

6. Develop a mobile-responsive version or dedicated mobile application to enable management and staff to access the system using smartphones or tablets.

7. Include an analytics dashboard that provides graphical reports on daily sales, inventory trends, best-selling menu items, and customer purchasing patterns to support better business decisionmaking.

Future Research Directions

Future researchers are encouraged to extend the present study by exploring additional features and technologies that can further improve restaurant management systems. Future studies may:

1. Develop a mobile application version of the system for Android and iOS platforms.

2. Integrate Artificial Intelligence (AI) or machine learning techniques to forecast customer demand, optimize inventory levels, and recommend menu items.

3. Conduct comparative studies to evaluate the effectiveness of the system in different restaurants or food service establishments.

4. Expand the system to support multi-branch management, enabling centralized monitoring of sales, inventory, and reports across multiple business locations.

5. Evaluate the long-term impact of the system on customer satisfaction, employee productivity, operational efficiency, and business profitability after full implementation.

Implementation Strategies

To ensure the successful adoption and long-term sustainability of the developed system, the following implementation strategies are recommended:

1. Conduct comprehensive user training for administrators, cashiers, and staff before deploying the system to ensure proper utilization of all system features.

2. Perform pilot testing in actual business operations before full implementation to identify and resolve any remaining issues.

3. Schedule regular system maintenance, software updates, and database backups to maintain optimal performance, reliability, and data security.

4. Assign a designated system administrator responsible for user account management, data maintenance, troubleshooting, and monitoring system performance.

5. Gather regular user feedback from management, employees, and customers to identify opportunities for continuous improvement.

6. Develop standard operating procedures (SOPs) for system usage, data management, and security to ensure consistency and minimize operational errors.

7. Allocate sufficient resources for future upgrades and technical support to ensure that the system continues to meet the changing operational needs of Groove Monkey Pinoy Korean Bistro.

By implementing these recommendations, the Groove Monkey Pinoy Korean Bistro Web-Based Ordering and Inventory Management System can be further enhanced to provide more efficient restaurant operations, improved customer service, better inventory control, and stronger decision-making capabilities. Continuous system improvement, regular evaluation, and future research will help ensure that the system remains effective, scalable, and responsive to the evolving needs of the food service industry.

5.5 Impact of the Study

The development of the Groove Monkey Pinoy Korean Bistro Web-Based Ordering and Inventory Management System has a significant impact on the business, its employees, customers, and future researchers. By replacing manual processes with an integrated digital system, the study contributes to improving operational efficiency, enhancing service quality, and supporting informed business decision-making.

Impact on Groove Monkey Pinoy Korean Bistro

The developed system provides the management with an efficient platform for handling daily business operations. It automates order processing, inventory monitoring, billing, and report generation, reducing manual work and minimizing human errors. The availability of accurate and real-time information enables management to make informed decisions regarding inventory replenishment, sales monitoring, and business planning. As a result, the system improves productivity, reduces operational costs, and strengthens the bistro's competitiveness in the food service industry.

Impact on Employees

The system simplifies the daily tasks of administrators, cashiers, and staff by reducing repetitive manual processes. Employees can process customer orders more quickly and accurately, monitor inventory efficiently, and generate reports with minimal effort. This reduces workload, improves productivity, minimizes errors, and allows employees to focus more on delivering quality customer service.

Impact on Customers

Customers benefit from faster and more accurate order processing, resulting in shorter waiting times and improved dining experiences. The system enhances service reliability by minimizing order errors and ensuring that menu availability is properly managed through accurate inventory monitoring. These improvements contribute to higher customer satisfaction and encourage repeat patronage.

Impact on Business Operations

The implementation of the developed system promotes more organized and efficient business operations. Centralized data management improves record accuracy, facilitates report generation, and supports better monitoring of sales and inventory. Automated processes help reduce food wastage, optimize resource utilization, and improve overall operational efficiency, contributing to the long-term sustainability of the business.

Impact on Future Researchers

This study may serve as a valuable reference for future researchers interested in developing information systems for restaurants and other food service establishments. The methodology, system design, implementation process, and evaluation results presented in this study can provide guidance for future research involving digital ordering systems, inventory management, and restaurant automation. Future researchers may also expand the system by integrating advanced technologies such as mobile applications, online ordering, artificial intelligence, and business analytics.

5.6 Summary

This chapter presented the testing, results, evaluation, conclusions, recommendations, impact, and overall outcomes of the Groove Monkey Pinoy Korean Bistro Web-Based Ordering and Inventory Management System. The developed system was designed to address the operational challenges identified in the study, particularly those related to manual order processing, inaccurate inventory management, inefficient record-keeping, and limited operational efficiency.
The findings demonstrated that the system successfully automated essential business processes, including order processing, billing, inventory monitoring, and report generation. System testing verified that all modules functioned according to their intended requirements, while the evaluation results indicated that the system

was functional, reliable, user-friendly, and acceptable to its intended users. The developed system contributed to improved operational efficiency, enhanced inventory accuracy, reduced manual errors, and supported informed managerial decision-making. The chapter also concluded that the implementation of the system provides significant benefits to Groove Monkey Pinoy Korean Bistro by improving service quality, streamlining daily operations, and increasing overall business productivity. Recommendations were presented to guide future system enhancements, implementation strategies, and further research, including the integration of online ordering, digital payment options, mobile applications, advanced analytics, and additional security features. Furthermore, the study highlighted the positive impact of the developed system on business management, employees, customers, and future researchers. The system promotes a more efficient, organized, and technology-driven approach to restaurant operations while providing a foundation for future expansion and continuous improvement.

In conclusion, the Groove Monkey Pinoy Korean Bistro Web-Based Ordering and Inventory Management System achieved the objectives of the study by providing an effective and reliable solution to the operational challenges faced by the business. The successful development and evaluation of the system demonstrate its potential to enhance restaurant operations, improve customer satisfaction, strengthen business performance, and support the long-term growth and sustainability of Groove Monkey Pinoy Korean Bistro.

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